WELL DRILLING PROGRAM

AZADEGAN OIL FIELD

WELL DRILLING PROGRAM

F – 20 (PAD #93), TYPE: IIIF

Drilling Department

OCTOBER 2024

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643 m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Asmari, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is located in Khuzestan province and is approximately located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in the Northwest of the YADAVARAN Field and in the West of JUFEYR Field. the average monthly Temperature is 45ºc in summer, and highest

temperature is 52ºc. The lowest temperature is -4.4ºc in winter. Maximum rainfall is 78 mm in February, winter Season.

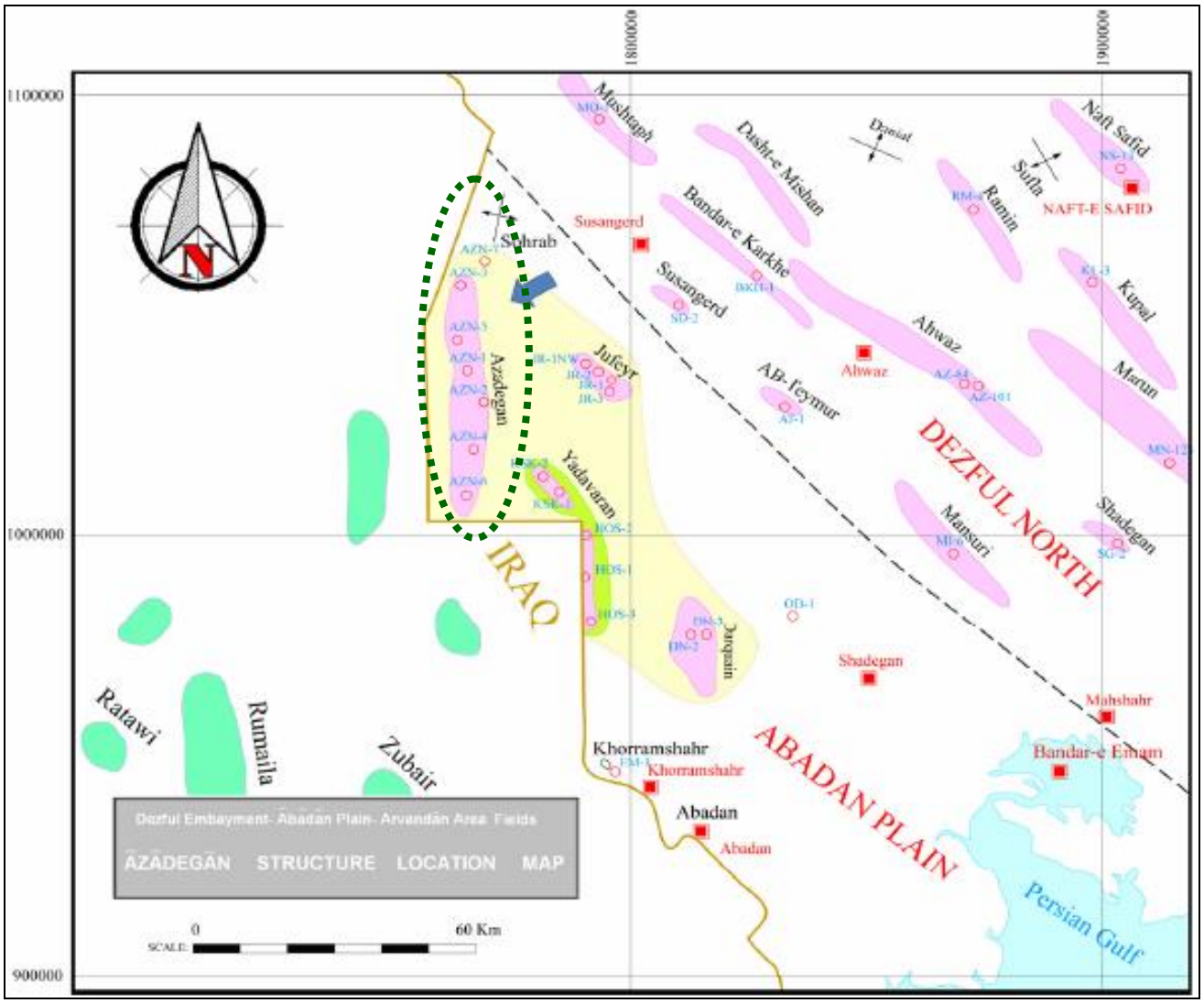


Figure 1: AZNS Oilfield Location Map

1.2 Well Information

F-20 Development Well Information				
Country / Area	Iran / Khuzestan			
Client:	PPL/PEDEC			
Contractor:	MSA/OEOC			
Field Name:	South Azadegan			
Well Name:	F-20			
Location:	PAD-93 (Area: III)			
Wellhead Co-ordinate (UTM39N, WGS84):	E: 192,581 m N: 3,441,012 m			
Well Type:	IIIF Vertical Well			
Well Objective:	Produce Oil from Fahliyan Reservoir, 7" Cemented Liner (Production through Perforation)			
RTE:	14.45 m	**Est. Moving Time:**	10 Days	
GLE:	3.95m	**Spud Date:**	TBA	
Rig Name:	OEOC 208	**Est. Well Duration:**	125.4 Days	
Rig Contractor:	OEOC	**Est. Rig Released:**	TBA	

Casing / Liner Size				
CSG/LNR Size:	20" CSG	13-3/8" CSG	9-5/8" CSG	7" Liner
Shoe Depth (MD):	**150**	**2182**	3915	4180

Other Information			
Total Depth:	4180 MD (-4166m TVDss)	**Entry Point in Fahliyan Fm.:**	E: 192,581m N: 3,441,012m
Producer Formation:	Fahliyan		
Reservoir Top:	3915 MD (-3900m TVDss)	**Target Coordinate:**	E: 192,581m N: 3,441,012m
KOP:	--		

Reservoir Data			
Item		**Measure** **Fahliyan**	**Unit**
Datum Depth		3915	**M, TVDss**
Datum Pressure (Initial)		8843	**Psi**
Press Gradient		0.912	**Psi/m**
Temp. @ Datum Depth		132	**°C**
Temperature Gradient		N/A	**°C/100m**
H2S (f/ PVT Separator Oil)		0.28	**Mol%**
CO2 (f/ PVT Separator Oil)		2.68	**Mol%**
Formation Water	**Salt**	143000	**PPM**
	Ca^{++}	8750	**Mg/lit**

Table 1: -F-20 (AZNS-207) information

1.3 F-20 Well Location Map

F-20, which is Fahliyan producer as a vertical well, is planned to be drilled in pad-93, located amongst AZNS-070-ST2 & AZNS-049 wells as Fahliyan wells & AZNS-173 as Gadvan well in the area-III. This well was supposed to be drilled in pad-72, then a decision was made by client to change the pad to a new location (pad-93), so the geoname was changed from G20 to F20.

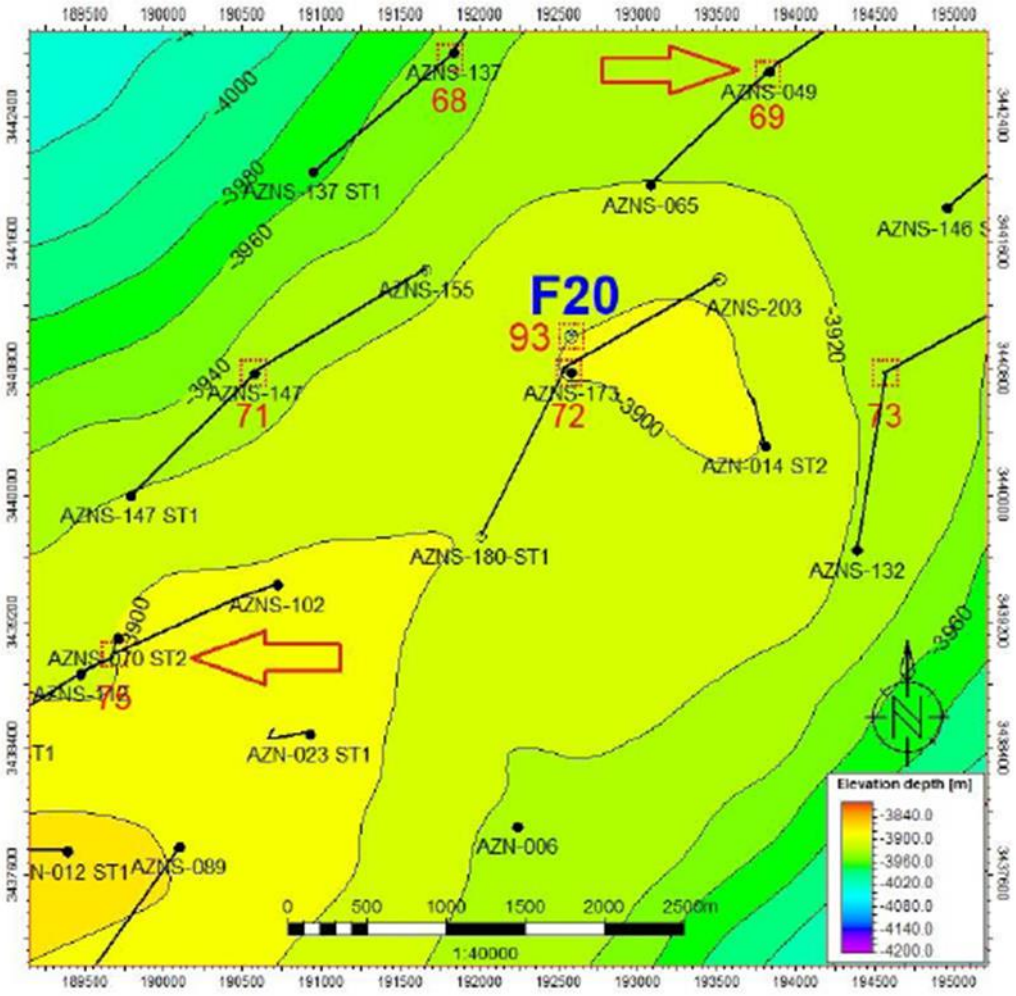


Figure 2: AZNS Area Distribution
Table 3: Well Pad Specification

1.4 Reservoir Information

1.4.1 Key Reservoir Descriptions

Fahliyan Formation: This carbonate to argillaceous carbonate Formation is limited by overlying thick argillaceous successions in the upper part (eaquivalent to the Iraqi Ratawi Formation) and the main part consists of different bioclastic limestones with some layers of shale and argillaceous facies that can be assumed equivalent with Iraqi Yamama Formation. However, in South Azadegan wells 1, 2 and 5 the Fahliyan Formation is characterized by fossiliferous limestone beds and interbedded shales. It should be noted that most of Majnoon wells penetrated into the Sarvak and Fahliyan carbonates as one of the main reservoir targets.

1.4.2 Formation Pore Pressure and Fracture Gradient

Formation	Interval	Pore Pressure	Collapse Pressure (Break-Out Pressure)	Fracture Pressure (Break-Down Pressure)
	m	SG	SG	SG
Aghajari	0~1024.4	0.98~1.18	0.06~1.39	1.01~1.89
Gachsaran	1024.4~1366.6	1.13~1.31	0.78~1.36	1.89~2.01
Asmari	1366.6~1581.6	1.14~1.26	0.6~1.28	1.43~1.79
Pabdeh	1581.6~2177.2	1.20~1.26	0.82~1.36	1.40~1.78
Gurpi	2177.2~2560.4	1.21~1.26	0.66~1.02	1.40~1.76
Ilam	2560.4~2632.8	1.18~1.24	0.2~0.9	1.41~1.72
Laffan	2632.8~2639.4	1.13~1.18	0.24~1.11	1.41~1.72
Sarvak	2639.4~3274.6	1.14~1.26	0.24~1.23	1.44~1.81
Kazhdumi	3274.6~3468.6	1.1~1.14	0.46~0.96	1.93~1.94
Dariyan	3468.6~3623.7	1.18~1.25	0.55~1.10	1.96~2.06
Gadvan	3623.7~3914.7	1.1~1.25	0.55~1.44	1.97~2.11
Fahliyan	3914.7-TD	1.60~1.3	1.20~1.75	2.08~2.12

1.4.3 Fahlian Reservoir Predicted Pressure, Temperature and OWC

Reservoir	Sub-layer	Mid of Sub-Layer Depth (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure	Top of Transition Zone (mss)	Basis for TTZ and FWL	Oil Gradient (psi/ft)
Fahliyan	Fah-1a	-3948	255.5	1.32	9000	Predicted from MDT 195	-4149	Dynamic Model	0.277
	Fah-1b	-3963	256.1		9000	Predicted from MDT 195		Dynamic Model	
	Fah-1c	-3985	257.1		9000	Predicted from MDT 195		Dynamic Model	
	Fah-2a	-4010	258.2		7769	Dynamic Model	-4046	Dynamic Model	
	Fah-3a	-4035	259.3		7855	Dynamic Model	-4151	Dynamic Model	
	Fah-3b	-4045	259.7		7868	Pressure Gradient	-	Dynamic Model	
	Fah-4a	-4057	260.2		7920	Dynamic Model	-4076	Dynamic Model	
	Fah-4b	-4076	261		8100	Predicted from MDT 195	-4086	Dynamic Model	
	Fah-5a	-4101	262.1		8014	Dynamic Model	-4109	Dynamic Model	

1.4.4 Corrosion Fluid

Reservoir	Sarvak	Kazhdumi	Gadvan	Fahliyan
H_2S Content (ppm)	2436	10	1650	30000
CO_2 Content (ppm)	142929	61107	40684	72769

Note: The corrosive gas concentration is subject to future well test sampling and lab test.

1.4.5 Formation Water Salinity

Reservoir	Sarvak	Kazhdumi	Gadvan	Fahliyan
Salinity (ppm as NaCl)	200000	180000	185000	143000
Calcium Content (mg/lit)	9500-11600	11400	9200-10000	8750

Note: Since there is high corrosive fluid in the reservoirs, during cementing operation, additives to anti-corrosion should be considered.

1.5 Geological Information

Province	Khuzestan
Area	Abadan Plain
Field	South Azadegan Oil Field
Main Client	PEDEC
Client	Petro Pars Ltd.
Contractor	Oil Exploration Operation Company (OEOC)
Drilling Sub-Contractor	Mahan Sanat Arya Company (MSA)
Location	Pad-93
Well Geological Name	F20
Coordination System	UTM_WGS84, Zone: 39N
Surface Coordinate	E: 192581.0 N: 3441012.0
Entry.P in Fahliyan	E: 192581.0 N: 3441012.0 (-3900 mTVDss)
TD Coordinates	E: 192581.0 N: 3441012.0 (-4166 mTVDss)
Planned Rig Days	125.4 days
Well Classification	Development Well
Well Type	Fahliyan Vertical Well
Drilling Rig Name	OEOC-208
Well Objective	Fahliyan Producer
Target Formation/Zone	Fahliyan Formation
G.L.E.	3.95 m.
R.T.E.	14.45 m.
Total Depth (T.D.)	4180.0 mMD (-4166 mss)

Type of Completion	7” Cemented Liner (Production through Perforation)
Nearby Wells	AZNS-070-ST2, AZNS-049, AZNS-173 (Gadvan target)

1.5.1 Expected Stratigraphy and Formation Top in F-20

The area is completely covered by the alluvium of Holocene age; the drilled will be Aghajari, Gachsaran, Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan formations. A brief stratigraphic sequence has been predicted as follows:

- ***Aghajari Formation (surface to 1021mMD) (Drilled Thickness: 1021 m)***

The Aghajari Formation consists of a repetition of brown to grey calcareous sandstones, chert debris and gypsum-veined red marl. A fauna of brackish to freshwater was observed, together with more marine fauna.. Aghajari overlays Gachsaran Formation with a conformable boundary.

- ***Gachsaran Formation (1021-1368 mMD) (Drilled Thickness: 347m)***

The Gachsaran Formation with a thickness ranges from 400 m to 500 m thickens towards the north. Deposition of the Gachsaran Formation is cyclic. The cycle began with a deposition of shallow-water marine carbonates, and then it was overlain by a thick evaporitic phase. The evaporates were capped by red to variegated marl, which represented the end of cycle. Gachsaran is laid on Asmari Formation with a comformable boundary.

- ***Asmari Formation (1368-1583 mMD) (Drilled Thickness: 215 m)***

The Asmari Formation in Azadegan Field developed a massive sandstone interval, with a few thin argillaceous limestones in the upper sections. Sandstone are clear, fine to medium-sorted, and poorly consolidated. On the other hand, the lower sections consist of intercalation of sandy, chalky, slightly dolomitic high-energy limestone with sandstone layers. Asmari is parallel to Pabdeh Formation at the bottom.

- ***Pabdeh Formation (1994-2178 mMD) (Drilled Thickness: 165 m)***

Pabdeh Formation is divided into two parts in the Azadegan field. The upper part is made of grey to dark-brown argillaceous limestones and marls. Planktonic fauna dates that this part was deposited in Oligocene. The lower part is composed of dark-brown, slightly argillaceous limestone. Pelagic fauna, mostly Globigerina and Globorotalia, indicates that this part was formed during late Paleocene to late Eocene in age. It seems that no early Paleocene Formation exists in the Azadegan area. Chert and cherty limestone layers at the bottom of Pabdeh Formation are very important for drilling parameters on the PDC bit (as a notification for drilling operation department).

- ***Gurpi Formation (2178-2560mMD) (Drilled Thickness: 382 m)***

Gurpi Formation consists of argillaceous limestone with higher clay content rather than Pabdeh Formation. About 50 meters of pure limestone will be observed in the middle part of the formation which divides it into 3 different parts as below:

- ***Upper Gurpi (2178-2343mMD) (Drilled Thickness: 165 m)***

The Gurpi Formation in the Azadegan Field consists of grey to dark-brownish, slightly argillaceous micritic limestone with Maestrichtian to Campanian aged fauna. At Well AZN-01, about 50 meters of chalky limestones were encountered in the middle of the Gurpi formation, which contains abundant fragments of rudists, bryozoa, corals, and Orbitoides.

- ***Tarbur Member (2343-2418 mMD) (Drilled Thickness: 75 m)***

In the western Abadan Plain about 50 meters thick chalky limestone was encountered in the middle part of the Gurpi Formation, which contains abundant fragments of bioclasts, suggesting very shallow depositional environment. This interval is equivalnt to the Tarbur Formation with reservoir quality in other regions.

- ***Lower Gurpi (2418-2560 mMD) (Drilled Thickness: 142 m)***

This member consists of gray, light gray, medium hard argillaceous Limestone. The Gurpi formation is overlain directly on Ilam formation by conformity contact.

- ***Ilam Formation (2560-2629 mMD) (Drilled Thickness: 69 m)***

In the Azadegan Field, the Ilam Formation consists of grey micritic limestone, often argillaceous, rarely pyritic, limestones. Microfauna, mostly pelagic, including rotalids and oligostegina assemblage, a few algae, bryozoa, and rudist fragments exist.

- ***Laffan Member (2629-2635mMD) (Drilled Thickness: 6 m)***

The Laffan Formation developed a thin greenish-grey pyritic shale layer with ostracods fauna that capping the Sarvak Formation.

- ***Sarvak Formation (2635-3270 mMD) (Drilled Thickness: 635 m)***

The Sarvak Formation developed a carbonate unit, which was gradually overlying the Kazhdumi Formation, and it is abruptly terminated by an erosion phase, covered by a various late Cretaceous sediment. The lithology of Sarvak consists of the dark grey fine grained nodular argillaceous limestone interbedded with thin marls in the bottom. The Limestone graded upward into a white-to-buff massive chalky limestone with siliceous nodules; developed a very massive tan-to-brown limestone containing abundant rudist debris in the middle part. In the upper part is made up of rubbly limestone with red staining and leached zones of breccia and the top of Sarvak Formation is very ferruginous.

- ***Kazhdumi Formation (3270-3464 mMD) (Drilled Thickness: 194 m)***

Kazhdumi Formation begins with dark brown, soft to medium hard, some fissile and calcareous Shale. This formation is start with dark brown to dark green gray, medium hard to hard, soft to medium, high bituminous, calcareous Shale interbedded with argillaceous limestone, sandy claystone, claystone, marl, sandy limestone, limestone beds. The Argillaceous Limestone is dark brown to brown gray, medium hard to hard, micro bituminous. The Sandy claystone is light green gray, soft, high calcareous. The Claystone is gray to light gray, soft. The Marl is green gray, soft, sticky. The Sandy Limestone is dark green gray, hard, faint sandy. The Limestone is light gray to green gray, medium hard to soft, partly argillaceous. The Kazhdumi Formation at base is consisting of Azadegan Member.

- ***Dariyan Formation (3464-3622 mMD) (Drilled Thickness: 158 m)***

In Dariyan Formation a variety of lithological rock types such as carbonates, rich organic shales and sandstones were deposited in various environments. The lower part of the Gadvan Formation is composed of low-energy, but not very deep, micritic, often argillaceous limestone

with echinoids and algal fragments. Two sandstone units were defined in the Gadvan Formation lateral changes and especially discontinuity of the sandstone layers which are consist of fine clastics (shales and siltstones) may be expected

The Gadvan formation divided to three part:Dariyan formation starts with white to white cream grey clean limestone type I/II, I. Base of Dariyan formation, appears grey, green Pyritic shale interbedded with argillaceous limestone. This zone called transition zone and thickness is 16m. after last grey limestone indicates of Gadvan formation.

This carbonate formation contains three zones. D1 and D3 are non-reservoirs and D2 is a moderately poor reservoir.

- ***Gadvan Formation (3622-3915 mMD) (Drilled Thickness: 293 m)***

In This Formation a variety of lithological rock types such as carbonates, rich organic shales and sandstones were deposited in various environments. The lower part of the Gadvan Formation is composed of low-energy, but not very deep, micritic, often argillaceous limestone with echinoids and algal fragments. Two sandstone units were defined in the Gadvan Formation lateral changes and especially discontinuity of the sandstone layers which are consist of fine clastics (shales and siltstones) may be expected

The Gadvan Formation divided to three part;

Upper Gadvan; the upper part starts with occurrence of a terrigenous rock. The lithology is composed of grey claystone and some shale.

Khalij Mbr.; appears with pure limestone, cream, light brown and in some intervals oil stain.

Lower Gadvan; consisting of alternating of marl, argillaceous limestone, limestone, claystone, sandy claystone and sandstone beds. The Marl is green gray, soft, sticky, slightly calcareous. The Argillaceous Limestone is gray white gray medium hard to soft, partly pyritic. The Limestone is white, white to light gray, hard. The Claystone is light gray, soft, sticky. The Sandy Claystone is light gray, soft, sticky, fine sand grained. The Sandstone is brown to gray, partly white buff, gray, friable, sorted, sub rounded, medium quartz grained, partly calcareous. This sandstone which occur in Zone-3 and Zone-5 are the reservoir of this formation and in this part is equivalent with Kushk member.

- ***Fahliyan Formation (3915-4180.0 mMD)(TD) (Drilled Thickness: 265m +)***

This formation is start with light gray, light green gray, soft Limestone interbedded with green gray, soft, sticky Marl, green gray, light gray, soft, Argillaceous Limestone and dark green gray, soft, sticky Clay stone beds. This formation at continue is consisting of buff white light gray, light brown, brown, medium hard, hard, densely, slightly argillaceous Limestone and off white, white gray, medium hard Chalky Limestone beds.

1.5.2 Formation Top & Thickness

Formation/Member Name		Tops		Thickness
		mTVDSS (RTE: 14.45m)	mMD	mTVD
Aghajari		Surface	Surface	1021
Gachsaran		-1007	1021	347
Asmari		-1353	1368	215
Pabdeh or Pd./Ja.		-1568	1583	102
Jahrum		-1671	1685	309
Pabdeh		-1980	1994	184
Gurpi	**U. Gurpi**	-2164	2178	165
	Tarbur	-2328	2343	75
	L. Gurpi	-2404	2418	142
Ilam		-2546	2560	69
Laffan		-2614	2629	6
Sarvak		-2621	2635	635
Kazhdumi		-3256	3270	194
Dariyan		-3450	3464	158
Gadvan		-3607	3622	293
Fahliyan	**Fahliyan**	-3900	3915	40
	Fah-1a	-3940	3955	16
	Fah-1b	-3956	3971	11
	Fah-1c	-3968	3982	34
	Fah-2a	-4001	4016	17
	Fah-2b	-4018	4033	13
	Fah-3a	-4031	4046	7
	Fah-3b	-4039	4053	13
	Fah-4a	-4051	4066	10
	Fah-4b	-4062	4076	27
	Fah-5a	-4088	4103	24
	Fah-5b	-4112	4127	53mTVD inside of Fah-5b zone
T.D.		-4166	4180	

Formation	Lithology	Lithology Description	Depth (MD/TVDss) (m)
Aghajari		Red Marl, Grey Marl, Sandstone, Anhydrite	1021m (-1007mss)
Gachsaran		Marl, Anhydrite, Limestone	1368m (-1353mss)
Asmari		Sandstone, Limestone & Marl	1583m (-1568mss)
Pd. or Pd./Ja.		Argillaceous Limestone	1685m (-1671mss)
Jahrum		Dolomite & Chert	1994m (-1980mss)
Pabdeh		Limestone, Chert, Cherty Limestone, Argillaceous Limestone	2178m (-2164mss)
U. Gurpi		Argillaceous Limestone	2343m (-2328mss)
Tarbur		Limestone	2418m (-2404mss)
L. Gurpi		Argillaceous Limestone	2500m (-2486mss)
Ilam		Limestone, Marl	2629m (-2614mss)
Laffan		Shale	2635m (-2621mss)
Sarvak		Limestone, Argillaceous Limestone, Marl, Cherty Limestone	3270m (-3256mss)
Kazhdumi		Argillaceous Limestone, Shale, Marl, Sandstone	3464m (-3450mss)
Dariyan		Limestone	3622m (-3607mss)
Gadvan		Claystone, Shale, Limestone, Argillaceous Limestone, Sandstone	3915m (-3900mss)
Fahliyan		Limestone	

Note: the formation depths change a lot due to formation tends to ascend from north to south. The above depth figures of formation bottom are predicted according to drilled well information and subject to future adjustment during WORK. Major reservoirs: Sarvak, Khazdumi, Gadvan and Fahliyan.

1.6 Logging Program

The logging program has been tabulated in Table, which shall be performed by Contractor in open-hole and cased-hole, section by section.

Hole Size	Casing / Liner	Formation	Open Hole Wire Line Logging	Cased Hole Wire Line Logging
26"	20"	Aghajari	-	-
17 1/2"	13 3/8"	Aghajari		
		Gachsaran		
		Asmari		
		Jahrum		
		Pabdeh		
		Gurpi		
12 1/4"	9 5/8"	Gurpi	Full Suite Logs/CDR	SBT/VDL/CCL/GR
		Ilam		
		Laffan		
		Sarvak		
		Kazhdumi		
		Dariyan		
		Gadvan		
8 ½"	7" Liner	Fahliyan (Reservoir)TD	Full Suite Logs/CDR	SBT/VDL/CCL/GR

Note: After Perforation job, Perforation Intervals should be confirmed by CCL/GR.

1.6.1 Mud Logging Activities

Mud logging unit will be spudded at the start of the drilling project containing all standard sensors for controlling the activities. In geological operation, the required mud logging activites are as below:

- Depth controlling
- Lag depth calculation through mud Pump GPM and drilling depth
- Gas samples gathering, analysis and giving the related reports

- Oil show samples gathering and including the results in daily reports.
- Colaboration with directional drilling section in order for performing related calculations
- Catching the cutting samples, describe the samples and providing graphic logs
- Providing daily, weekly and monthly reports according to the company formats
- Preparing the Well Geological Final Report according to the standard formats.
- Kill sheet must be submitted during drilling.

1.6.2 Sampling plan

For identification, description and interpretation of lithology, sampling intervals and preservation of cuttings are required which listed in Table.
Well-site geologist is fully responsible to determine the frequency of cutting sampling, and mud logging crew shall take sample based on well site geologist advice. In this well, the sampling should not be limited to 10 m, and the well-site geologist should take care of sampling and collecting sample of any cuttings with different characteristics.

Hole Size	Formations	Sampling Interval	Distribution List
26"	Aghajari	Every 5 m MD	MSA
17 ½"	Aghajari		
	Gachsaran		
	Asmari		
	Jahrum		
	Pabdeh		
	Gurpi		
12 ¼"	Gurpi		
	Ilam		
	Laffan		
	Sarvak	Every 2 m MD	
	Kazhdumi	Every 5 m MD	
	Dariyan	Every 2 m MD	
	Gadvan		
8 ½"	Fahliyan		
Note: Only Washed Samples Shall Be Stored			

Note: Well-site geologist is fully responsible to determine the frequency of cutting sampling, and mud logging crew shall take sample based on well site geologist's advice.

1.6.3 Calcimetry Analysis

Calcimetry analysis should be performed from starting second hole size to TD off the hole continuously. All the calcimetry analysis result should be used on Graphic well log.

1.6.4 Gas Logging

The gas logging shall be in action continuously since the spud-in.

- Total Gas
- Gas Chromatograph (HC types: C1; C2; C3; iC4/nC4; iC5/nC5)
- Non-Hydrocarbon Gas（H2S and CO2）
- Accuracy: within 10 ppm.

1.7 Off-set Information (Nearby Wells)

1.7.1 History of near by Wells: AZNS049

1.7.1.1 General

Well Name/Pad No.	AZNS-049 (F12) / Pad 69
Well Classification	Development Well
Well Objective	Primary Oil Producer : Fahliyan
Well Type	Vertical Well
Country	Iran
Province	Khuzestan
Field	South Azadegan
Grid Geodetic system	UTM39N,WGS84
Wellhead Coordinates	E: 193,836.00 N: 3,442,671
Down hole Target Coordinates	E: 193,836 N: 3,442,671
Reference Wells	AZN#05 , AZN#01 & AZN#18
Rig Contractor	National Iranian Drilling Company
Rig Name	32/27 Fath
Rig Type	Land Rig , 2000 HP
RTE	12.95/12.95m
Ground Elevation	3.95 m
Mud Type	Bentonite, KCL Polymer
Designed TD	4300m MD
Actual TD	4170m MD
Well Completion	Completed
Spud in / Spud in[1]	00:00 15-Sep-2014 (24 Shahrivar 1393)/12:00 01-Aug-2016 (11 Mordad 1395)
Rig Release / Rig Release[2]	24:00 20-Mar-2015 (29 Esfand 1393) /06:00 02-Feb-2016 (13 Bahman 1395)
Actual Rig Days	393.5 Days
Delay Days	273.5 Days
Benchmark	All the depths refer to RTE. Unless otherwise specified

1.7.1.2 Casing Depth

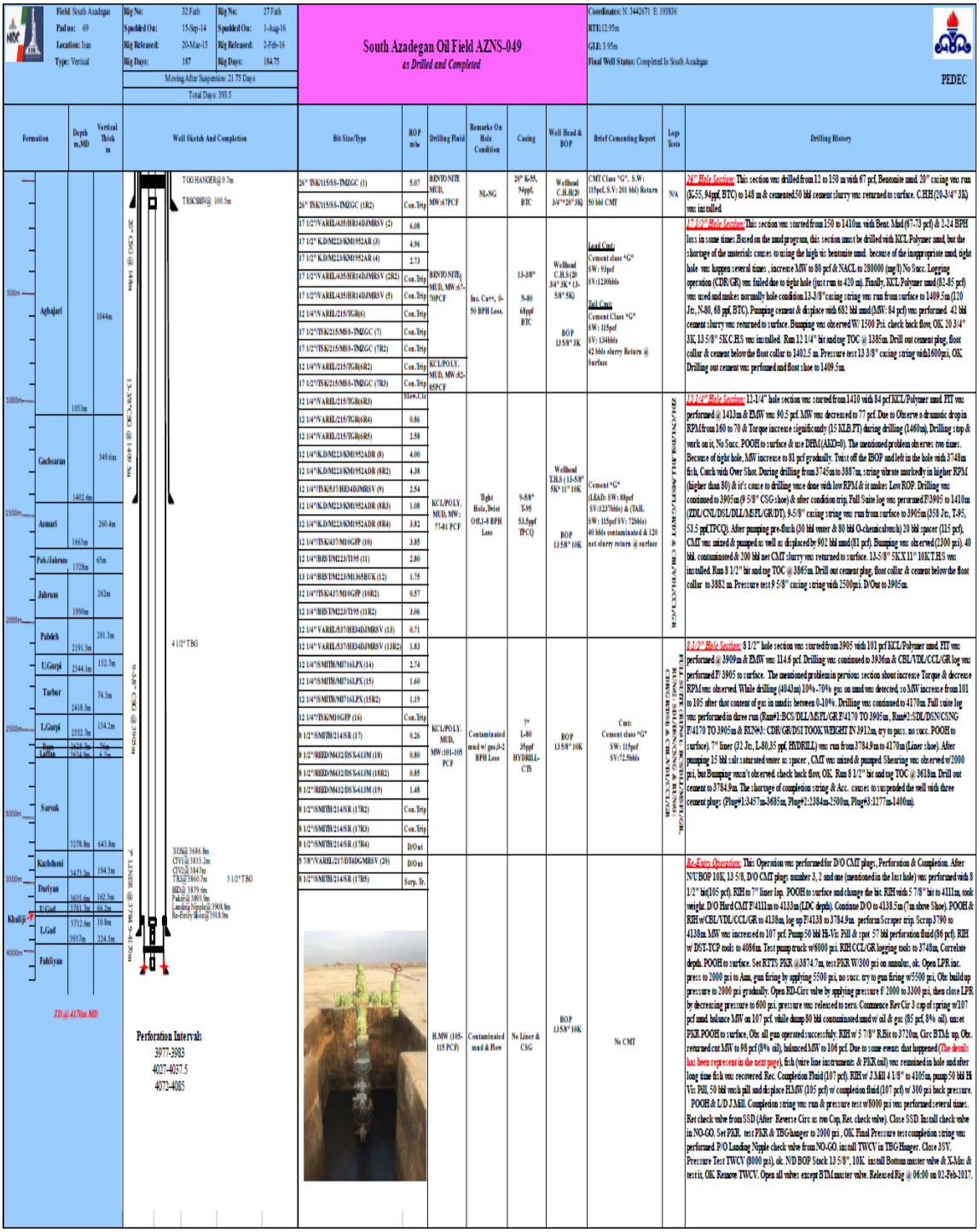
Hole		Casing/Liner Specifications					Mud Weight (pcf)	FIT/LOT EMW (pcf)	Cement Weight		Actual Consumed Time (Hrs.)
Size (In)	MD (m, RTE)	Size (In)	MD (m, RTE)	Grade	Weight (ppf)	Connection			Lead (pcf)	Tail (pcf)	
26	150	20	148	K-55	94	BTC	67	Not Performed	115	115	158
17-1/2	1410	13-3/8	1409.5	N-80	68	BTC	67-85	Not Performed	93	115	1246.5
12-1/4	3905	9-5/8	3905	T-95	53.5	TPCQ	77-81	90.5	88	115	2106
8-1/2"	4170	7"	3784.9-4170	L-80	35	HYDRILL -CTS	101-105	114.6	115	115	5933.5

1.7.1.3 Formation Tops

Group	Formation	Member	Formation Tops (Forecast)			Formation Tops (Actual)		
			MD (m MD)	TVDss (m MD)	Vertical Thickness (m MD)	MD (m MD)	TVDss ((m MD)	Vertical Thickness (m MD)
Fars	Aghajari		Surface	Surface	1043	9.0	3.95	1044.0
	Gachsaran		1043	-1030	514	1053.0*	-1040.0	349.6
Undefined	Asmari		1557	-1544	160	1402.6*	-1389.6	260.4
	Pabdeh/Jahrum		-	-	-	1663.0*	-1650.0	65.0
	Jahrum		-	-	-	1728.0*	-1715.0	262.0
	Pabdeh		1717	-1704	440	1990.0*	-1977.0	201.3
	Gurpi	U.Gurpi	2157	-2144	195	2191.3*	-2178.3	152.7
		Tarbur	2352	-2339	79	2344.1*	-2331.0	74.5
		L.Gurpi	2431	-2418	125	2418.5*	-2405.5	134.2
Bangestan	Ilam		2556	-2543	81	2552.7*	-2539.7	76.0
	Laffan		2637	-2624	7	2628.7*	-2615.7	6.2
	Sarvak		2644	-2631	646	2634.9*	-2621.9	643.8
	Kazhdumi		3290	-3277	204	3278.8*	-3265.7	194.5
Khami	Dariyan		3494	-3481	154	3473.2*	-3460.2	162.3
	Gadvan	U. Gadvan	3648	-3635	65	3635.6*	-3622.5	66.2
		Khalij	3713	-3700	11	3701.7*	-3688.7	10.8
		L. Gadvan	3724	-3711	221	3712.6*	-3699.5	224.5
	Fahliyan	Fahliyan-Top	3945	-3932	355	3937.0*	-3924.0	39.1
		Fah-1a	-	-	-	3976.2*	-3963.1	16.9
		Fah-1b	-	-	-	3993.0*	-3980.0	10.3
		Fah-1c	-	-	-	4003.4*	-3990.3	37.3
		Fah-2a	-	-	-	4040.6*	-4027.6	18.0
		Fah-2b	-	-	-	4058.6*	-4045.6	12.3
		Fah-3a	-	-	-	4071.0*	-4057.9	8.7
		Fah-3b	-	-	-	4079.7*	-4066.6	12.3
		Fah-4a	-	-	-	4091.9*	-4078.9	10.1
		Fah-4b	-	-	-	4102.0*	-4089.0	25.4
		Fah-5a	-	-	-	4127.4*	-4114.4	21.0
		Fah-5b	-	-	-	4148.5*	-4135.4	21.6**
	TD	-	4300	-	-	4170.0*	-4157.0	-

* Based on Log ** Drilled Thickness

1.7.1.4 Well Montage



1.7.1.5 Section Drilling History

26” Hole Section

General Data

FIELD	South Azadegan	WELL NO.	049	LOCATION	3442671	N	
RIG	32				193836	E	
HOLE SIZE	26"	HOLE DEPTH	150	m MD	MUD WT. Range	68	pcf
FORMATION	Aghajari				MUD TYPE	WBM	

Mud Weight Change

MW	Depth (m)	Reason of MW change
		MW wasn't changed.

Drilling Problems

Problem	Depth (m)	Corrective Action
		This section was drilled without any problem

Cementing

	VOL (bbl)	WT.(pcf)	CMT (SXS)	Addetive
Spacer	70	62.4		No Add.
Lead CMT	201	115	773	CACL2 (3211 GAL), CFR3 (159 LBS) & Antifoam (15 GAL)

COMMENTS: PER JOB SAFETY MEETING, PRESS TEST W/3000 PSI FOR 10 MIN, OK, PUMP 70 BBLSPACER (FRESH WATER) MIX & PUMP 773 SXS CMT "G" (S.V= 201 BBL, SW= 115 PCF), RETURN 50 BBL CMT, DISPLACE W/ 6 BBL FRESH WATER (2 BBL SHORT DISPLACE), CHECK BACK FLOW, OK.

Wellhead & Casing

Equipment	Property
20" Casing	K-55, 94 PPF, 148m (Surface to 148m)
C.H.H	20 3/4" 3K (BTM WELDING TYPE)

COMMENTS:

Bit Record

Bit Run.	From (m)	To (m)	Type/IADC/Nozz.	ROP (MPH)	Bit Run.	From(m)	To(m)	Type/IADC/Nozz.	ROP (MPH)
1	13	150	TSK/115/3*18+1*16	5.07	7				
1R2	Cond. Trip		TSK/115/3*18+1*16		8				
					9				
					10				
					11				
					12				

COMMENTS:

LOGGING

Logging wasn't performed.

BHA

26" R.Bit+ 9 3/4" Bit Sub+ 9 1/2" DC+9 1/2" DC+26" S.Stabliser+ 9 1/2" DC 4 Jts+ X.O.S

17 ½" Hole Section

General Data

FIELD	South Azadegan	WELL NO.	AZNS 049		LOCATION	3442671	N
PAD NO.	69					193836	E
RIG NO.	32	HOLE DEPTH	1410	m MD	MUD WT. Range	67-85	PCF
HOLE SIZE	17-1/2"				MUD TYPE	W.B.M.	
FORMATION	Aghajari & Gachsaran						

Mud Weight Change

MW (PCF) From	MW (PCF) To	Depth(m)	Reason of MW change
67	68	351	As per Program
68	72	1296	Inc. Ca++
72	73	1410	Tight Hole
73	79	1410	Treat Mud property for curing hole condition
79	82	1410	Treat Mud property for curing hole condition
82	85	1410	Treat Mud property for curing hole condition
85	82	1410	Treat Mud property for curing hole condition
82	84	1410	Treat Mud property for curing hole condition

COMMENTS:

Drilling Problems

Problem	Depth (m)	Corrective Action
Inc. Ca++	1296	Inc. MW F/68 to 72 PCF.
Loss	1328	Pumping 50 bbl Hi Vis Pill & Decreased to 4 BPH
Bad Hole Condition	1410	KCL Mud was used & condition of hole was cured.

COMMENTS: In this section, due to shortage of mud material (KCL Mud), Ben. Mud was used from 150 to TD section (1410m) & its causes to bad hole condition. KCL Mud was used in TD section & the condition of hole after several day was cured.

Cementing

	VOL (bbl)	WT.(PCF)	CMT (SXS)	Additive
Spacer	80	75		No Add.
Lead CMT	1230	93	2718	BENTONITE (6908 LBS), O-R5 (352 LBS) & ANTIFOAM (50 GAL)
Tail CMT	134	115	520	OCFR3 (167 LBS), O-R5 (67 LBS) & ANTIFOAM (12 GAL)

COMMENTS: SAFETY MEETING, R/UP CMT EQUIPMENT & CMT SURFACE LINE, TEST IT WITH 500 PSI 5 MIN & 3000 PSI 10 MIN, OK. PUMPING 80 BBL SALT SATURATED WATER AS THE SPACER MIXED AND PUMP 2718 SXS CMT " G "DELEJAN CLASS (LEAD SL.VOL = 1230 BBL, SL.WT= 93 PCF) & MIX AND PUMP 520 SXS CMT "G" DELEJAN CLASS (TAIL SL.VOL= 134 BBL, SL.WT 115 PCF) DROP TOP CMT PLUG , DISPLACE WITH 682 BBL MUD 84 PCF OBS. BUMPING W/ 1500 PSI CHECK BACK FLOW ,OK. RETURN 42 BBL CONTAMINATE CMT. WHILE CMT PUMPING 100 BBLS LOSS.

Wellhead & Casing

Equipment	Property
13.375" Casing	N-80, 68 PPF, BTC & Length:1409.5m
C.H.S	20-3/4"3K" 13-5/8"5K C.H.S
BOP Stack	21 1/4" BOP Stack 3K (HYDRILL, DSA, SPACER SPOOL, SHEAR, TOP & BTM RAMS)

COMMENTS:

Bit Record

Bit No.	From (m)	To (m)	Manufacture/IADC/Nozz	ROP (MPH)	Bit No.	From(m)	To(m)	Type	ROP (MPH)
2	150	792	VAREL/435/3*18	6.08	7			CONDITION TRIP FOR LOGGING	
3	792	1328	K.D/M223/7*12	4.96	7R2			CONDITION TRIP	
4	1328	1410	K.D/M223/7*13	2.73	6R2			CONDITION TRIP	
2R2			CONDITION TRIP		7R3			CONDITION TRIP FOR 13 3/8" CSG	
5			CONDITION TRIP		6R3			SLOW CIRCULATION @ 1370 M	
6			CONDITION TRIP						

COMMENTS: x*Ry that means the bit No. *x* was run for "*y*th time.

LOGGING

BHA

17 1/2" Bit + 9 1/2" Bit Sub +2 Jts 9 1/2" DC +17 1/2" S.Stabiliser + 7 Jts 9 1/2" DC +9 1/2" Jar+1 Jts 9 1/2" DC+X.O.S+2 Jts 8 1/2"DC+X.O.S+9 Jts 5" HWDP

17 1/2" Bit + 9 1/2" Bit Sub +2 Jts 9 1/2" DC +17 1/2" S.Stabiliser + 7 Jts 9 1/2" DC +9 1/2" Jar+1 Jts 9 1/2" DC+X.O.S+2 Jts 8 1/2"DC+X.O.S+9 Jts 5" HWDP,

12 1/4" Bit" + 9 1/2" Bit Sub" +2 Jts 9 1/2" DC" +12 1/4" S.Stabiliser" + 7 Jts 9 1/2" DC +9 1/2" Jar+1 Jts 9 1/2" DC+X.O.S+2 Jts 8 1/2"DC+X.O.S+9 Jts 5" HWDP

17 1/2" Bit + 9 1/2" Bit sub +2 jnts 9 1/2" DC +17 1/4" S.Stabiliser + 7 jnt9 1/2" DC +9 1/2" jar+1 jnts 9 1/2" DC+X.O.S+2jnt 8 1/2"DC+X.O.S+9 jnt 5" HWDP

12 1/4" Bit + 8 1/2" Bit sub+2 jnts 8 1/2" DC +8 1/2" S.Stabiliser + 10 jnt 8 1/2" DC +9 1/2" jar+3 jnts 8 1/2" DC+ X.O.S+15 jnt 5" HWDP

12 ¼" Hole Sectoin

General Data							
FIELD	South Azadegan	WELL NO.	AZNS 049	LOCATION	3442671	N	
PAD NO.	69				193836	E	
RIG NO.	32	HOLE DEPTH	3905	m MD	MUD WT. Range	77-84	PCF
HOLE SIZE	12-1/4"				MUD TYPE	W.B.M.	
FORMATION	Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Dariyan & Gadvan						

Mud Weight Change

MW (PCF) From	To	Depth(m)	Reason of MW change
84	77	1413	As per Program.
77	78	2737	Tight Hole
78	79	3183	Tight Hole
79	81	3636	Tight Hole

COMMENTS:

Drilling Problems

Problem	Depth (m)	Corrective Action
Inc. TORQ	1460	POOH & use DHM
Tight Hole	2738	Inc. MW to 78
Tight Hole	3183	Inc. MW to 79
Tight Hole	3636	Inc. MW to 781
Twist off IBOP	3748	Catch with Over Shot.
Inc. TORQ	3745-3887	Drilling with Low RPM.

COMMENTS: In this section, increasing TQ is the important problem that using DHM wasn't solved this problem. For decreasing TQ, drilling was performed w/ low RPM that cause to low ROP.

Cementing

	VOL. (bbl)	WT.(PCF)	CMT (SXS)	Additive
Pre Flush	40	62.4		O-CHEMICAL WASH
Spacer	20	82		O-SPACER (480 LBS), O-SURFACTANT (120 GAL), LST POWDER (9240 LBS) & ANTIFOAM (20 GAL)
Lead CMT	1237	88	1887	O-MICA SILICA (65604 LBS), O-R12 (584 LBS), O-MICRO BLOCK (1486 GAL) & ANTIFOAM (43 GAL)
Tail CMT	72	115	260	O-R12 (25 LBS), SILICA FLOUR (8239 LBS), CFL-350 (684 LBS) & ANTIFOAM (9 GAL)

COMMENTS: SAFETY MEETING R/UP CMT EQUIPMENT ,TEST IT W/3000 PSI "OK", PUMP 40 BBL WATER MIX WITH 80 GAL O-CHEMICALWASH AS PRE FLUSH & PUMP 20 BBLSPACER MIX & PUMP 1887SX CMT "G" DELUAN BLEND W/O-MICRO SILICA (S.V 1237 BBL ,SL.WT= 88 PCF) AS LEAD & 260 SX CMT "G" DELUAN BLEND W/SILICA FLOUR (S.V = 72 BBL , SW= 115 PCF) AS TAIL DROP TOP PLUG .DISPLACE W/ 902 BBL MUD 81 PCF OBS. BUMPING W/2300 PSI, RETURN 40 BBL CONTAMINATE CMT & 200 BBL PURE CMT NO LOSS RECORD BY MUD ENGINEER .

Wellhead & Casing

Equipment	Property
9.625" Casing	T-95, 53.5 PPF, TPCQ & Length:3905m
T.H.S	T.H.S (13-5/8" 5K*11" 10K)
BOP Stack	BOP Stack 13 3/8", 10K

COMMENTS:

Bit Record

Bit No.	From (m)	To (m)	Manufacture/IADC/Nozz	ROP (MPH)	Bit No.	From(m)	To(m)	Manufacture/IADC/Nozz	ROP (MPH)
6R4	1410	1413	VAREL/215/3*18	0.85	12	2323	2413	BEST/2*14+4*15	1.75
6R5	1413	1453	VAREL/215/3*18	2.58	10R2	2413	2415	TSK/437/3*18	0.57
8	1453	1465	K.D/M223/3*14+1*15	4.00	11R2	2415	2701.5	BEST/M223/7*14	3.06
8R2	1465	1866	K.D/M223/3*14+1*15	4.38	13	2701.5	2704	VAREL/537/3*18	0.71
9	1865	1979	TSK/537/3*18	2.54	13R2	2704	2871	VAREL/537/3*18	1.83
8R3	1979	1993	K.D/M223/3*14+1*15	1.08	14	2871	3570	SMITH/7*16	2.74
8R4	1993	2035	K.D/M223/3*14+1*15	3.82	15	3570	3745	SMITH/7*16	1.60
10	2035	2162	TSK/437/3*18	3.85	15R2	3745	3905	SMITH/7*16	1.19
11	2162	2323	BEST/M223/7*14	2.80	16	3905	3905	TSK/OPEN	Cond. Trip

COMMENTS: x*Ry that means the bit No. *x* was run for "*y*"th time.

LOGGING

FULL SUITE LOG (CDR/GR/MSF/DLL/DIPOLE/ORT/ZDT/CNT/DST) WAS PERFORMED FROM 3905 TO 1410M.

BHA

12 1/4" Bit +junk sub+ 8 1/2" Bit sub + 12jnt 8 1/2" DC +8 1/2" jar+3 jnts 8 1/2" DC+8 3/16" xos+15 jnt 6 5/8" HWDP
12 1/4" Bit + 8 1/2" Bit.sub+2jnts 8 1/2" DC +12 1/4" S.Stablizer + 10 jnt 8 1/2" DC +8 1/2" jar+3 jnts 8 1/2" DC+XOS+15 jnt 5" HWDP
12 1/4" Bit + 8 " motor +7 1/2" F.sub+1jnts 8 1/2" DC +12 1/4" S.Stablizer + 11 jnt 8 1/2" DC +8 1/2" jar+3 jnts 8 1/2" DC+XOS+15 jnt 5" HWDP
12 1/4" Bit + 8 1/2" Bit.sub+2jnts 8 1/2" DC +12 1/4" S.Stablizer + 10 jnt 8 1/2" DC +8 1/2" jar+3 jnts 8 1/2" DC+XOS+15 jnt 5" HWDP
12 1/4" Bit + 8 " motor +7 1/2" F.sub+1jnts 8 1/2" DC +12 1/4" S.Stablizer + 11 jnt 8 1/2" DC +8 1/2" jar+3 jnts 8 1/2" DC+XOS+15 jnt 5" HWDP
12 1/4" Bit + 8 1/2" Bit.sub+2jnts 8 1/2" DC" +12 1/4" S.Stablizer + 10 jnt 8 1/2" DC +8 1/2" jar+3 jnts 8 1/2" DC+XOS+15 jnt 5" HWDP
12 1/4" Bit + 8 " motor +7 1/2" F.sub+1jnts 8 1/2" DC +12 1/4" S.Stabliszr + 11 jnt 8 1/2" DC +8" jar+3 jnts 8 1/2" DC+XOS+15 jnt 5" HWDP
12 1/4" Bit + 8 1/2" Bit sub +2 jnts 8 1/2" DC +12 1/4" S.Stablizer + 10 jnt 8 1/2" DC +8" jar+3 jnts 8 1/2" DC+XOS+15 jnt 5" HWDP
12 1/4" Bit + j.sub+8 1/2" Bit sub +2 jnts 8 1/2" DC +12 1/4" S.Stabliser + 10 jnt 8 1/2" DC +8" jar+3 jnts 8 1/2" DC+ XOS+15 jnt 5" HWDP
12 1/4" Bit +8 1/2" Bit sub +2 jnts 8 1/2" DC +12 1/4" S.Stablizer + 10 jnt 8 1/2" DC +8" Jar+3 jnts 8 1/2" DC+ XOS+15 jnt 5" HWDP

8 ½" Hole Section

General Data							
FIELD	South Azadegan	WELL NO.	AZNS 049	LOCATION	3442671	N	
PAD NO.	69				193836	E	
RIG NO.	32	HOLE DEPTH	4170	m MD	MUD WT. Range	101-105	PCF
HOLE SIZE	8-1/2"				MUD TYPE	W.B.M.	
FORMATION	Fahliyan						

Mud Weight Change

MW (PCF) From	To	Depth(m)	Reason of MW change
101	105	4043	10-70% Gas was detected.

COMMENTS:

Drilling Problems

Problem	Depth (m)	Corrective Action
Gas Detected	4043	Inc. MW F/101 to 105 PCF

COMMENTS:

Cementing

	VOL (bbl)	WT.(PCF)	CMT (SXS)	Additive
Pre Flush	15	75		O-CHEMICAL WASH
Spacer				
Lead CMT				
Tail CMT	72	115	215	SILICA FLOUR (8278 LBS), CFL-109 (246 LBS), CFL-350 (99 LBS), D-065 (99 LBS), O-R12 (81 LBS), D400 (22.3 GAL), ANTIFOAM (4.5 GAL)

COMMENTS: R/U CMT LINE & TEST IT W/ 4000 PSI, OK, PERFORM PER JOB SAFETY MEETING, PUMP 15 BBL S.SALT AS AHEAD SPACER , MIX & PUMP 215 SX CMT "G" DELIGAN BELEND W/ SILICA FLOUR (S.W: 115 PCF & S.V 72.5BBL), RELEASE PUMP-DOWN PLUG DISPLACE W/ 260 BBL MUD WHILE OBS SHEARING W/ 2000 PSI AFTER PUMP 220 BBL MUD & NO OBS BUMPING. CHECK THE BACK FLOW, OK W/ NL-NG.

Wellhead & Casing

Equipment	Property
7" Liner	L-80, 35 PPF, HYDRILL-CTS & Length:385.1m
BOP Stack	10K BOP STACK

COMMENTS:

Bit Record

Bit No.	From (m)	To (m)	Manufacture/IADC/Nozz	ROP (MPH)	Bit No.	From(m)	To(m)	Manufacture/IADC/Nozz	ROP (MPH)
17	3905	3909.5	SMITH/214/3*18	0.26	17R4			SMITH/214/3*18	Cond. Trip
18	3909.5	3936	REED/M432/5*14+1*15	0.80	20			VAREL/217/OPEN	Cond. Trip
18R2	3936	3996	REED/M432/5*14+1*15	0.85	17R5			SMITH/214/3*18	Cond. Trip
19	3996	4170	REED/M432/5*14+1*15	1.48					
17R2	4170	4170	SMITH/214/3*18	Cond. Trip					
17R3	4170	4170	SMITH/214/3*18	Cond. Trip					

COMMENTS: **x*Ry** that means the bit No. *x* was run for "*y*"*th* time.

LOGGING

RUN CBL/VDL/CCL LOG F/3936 TO 1350M. RUN FULL SUITE LOG (Run#1:BCS/DLL/MSFL/GR F/4170 TO 3905M , RUN#2:SDL/DSN/CSNG F/4170 TO 3905M & RUN#3: CDR/GR/DSI TOOK WEIGHT IN 3912M, TRY TO PASS, NO SUCC. POOH TO SURF.).

BHA

8 1/2" Bit + 6 1/2"Junk sub+6 1/2" Bit sub +18 jnts 6 1/2" DC + 6 1/2" jar+3 jnts 6 1/2" DC+15 jnt 5" HWDP

8 1/2" Bit +6 1/2" Bit sub +2 jnts 6 1/2" DC + 8 1/2"stabilizer+16 jnts 6 1/2" DC+ 6 1/2" jar+3 jnts 6 9/16" DC+15 jnt 5" HWDP

8 1/2" Bit +6 1/2" Bit sub +18 jnts 6 1/2" DC + 6 1/2" jar+3 jnts 6 1/2" DC+15 jnt 5" HWDP

8 1/2" Rock. Bit*0.25m +6 1/2" Bit sub +18 jnts 6 9/16" DC + 6 1/2" jar+3 jnts 6 9/16" DC+15 jnt 5" HWDP

5 7/8" Bit +4 7/8"Junk sub+ 4 3/4" Bit sub +17 jnts 4 3/4" DC + 4 3/4" jar+3 jnts 6 1/2" DC

1.7.1.6 Re- Entry Operation Summary

1- Spud Rig @ 01 Aug 2016.
2- N/U BOP, and drill out suspension cement plugs.
3- CBL/VDL/CCL/GR was performed F/4134m (top of cement) to 3784.9m.
4- Under balanced perforation has been done in 3977-3983m, 4027-4037m, 4027-4037m, and 4072-4085m.
5- Balancing MW to 106 pcf before running completion.
6- Displacing Brineseal (Completion Fluid) with 101 pcf.
7- Due to pumping 100 bbl Hi Vis pill (90 pcf) during displacing Completion Fluid, Well flow was observed. Closing the well, well head pressure was increased to 650 psi (28 hrs).
8- Well control was done with displacing Brineseal (101 pcf) with mud (105 pcf).
9- Due to shortage of completion fluid, 500m Brineseal was placed in bottom hole.
10- POOH of drilling string to surface.
11- Running completion string (10K) to 3923m while fill up completion string w/ Brineseal (99 pcf). It should be noted plug & prong was placed in SSD & L.Nipple.
12- Trying pull out plug and prong from SSD, it didn't work. It should be mentioned that pulling out plug & prong from SSD was performed with several methods (Slick line operation, Coil Tubing Operation, and...), but they didn't work.
13- Due to observing metal filing, acidizing was performed, so finally plug and prong was pulled out.
14- When plug and prong had pulled out, well flow was observed (due to settling solids of mud), so well was secured with mud (111 pcf).
15- Closing SSD, annulus was tested with 2000 psi against mud (111 pcf).
16- Testing completion string with 8000 psi, OK. Trying to test 3SV, no success.
17- According to failure of 3SV test, Coil Tubing was run to 3903m, took weight.
18- Displacing mud inside of tubing with non-solids mud (86 pcf).
19- RIH to 3913m with pumping mud (86 pcf).
20- POOH and L/D Coil Tubing, trying to close 3SV, no success.

21- For testing 3SV, SSD was supposed to be open, so after trying several times, SSD was opened.

22- Displacing mud in annulus with 86 pcf mud, trying to test 3SV, no success.

23- Due to failure of 3SV, replacing 3SV was supposed to be considered.

24- Closing well, and wellhead pressure was increased to 500 psi., so mud was displaced with 111 pcf.

25- The completion string was pulled out from anchor seal. POOH to surface.

26- 3SV was changed. RIH to bottom (2.5m above packer).

27- Took weight. R/U Coil tubing and RIH to 2.5m below packer. Trying to wash, no success. POOH to surface and L/D it.

28- MW was increased from 111 to 115 pcf. One way plug was installed in SSD. Anchor Seal was placed inside of packer. Trying to pulling out, no success.

29- Due to trying to pulling out one way plug from SSD, Slick line got stuck. Trying to pulling out wire line, no success.

30- Slick Line was free from Rope Socket.

31- RIH with Coil Tubing to 3878m with pumping. Circulation was performed as completion string cap. Spot hi vis pill on fish.

32- RIH w/ impression block to 3866m, RIH w/ fishing tools to 3866m. Took weight, POOH and L/D slick line. Packer was released w/50 Klbs.

33- POOH completion fluid to surface. The Slick line instrument was left in hole.

34- Impression block with Slick Line fishing tools to 7" liner lap.

35- Coil Tubing was run to 7m above No-Go. Trying to wash, no success. Spot hi vis pill. POOH to surface.

36- Running one time gauge cutter and two times fishing tools to 3893m (6m above Packer).

37- Due to shortage of O.Shot tools, pulling tools was run two times with drilling string, no success.

38- Pulling tools with wire line was run to 3887m, no success.

39- Running 4.5" wash over shoe to 3886.5m, took weight. Wash to 3896.5m.

40- Running wire line w/ pulling tools in six times, no success (It should be noted that in the last time with pulling tools, the pulling tools engage with fish, but it wasn't released, so the pulling tools was left in hole).

41- RIH with 5 5/8" O.Shot to 3889.25m, engaging with fish, POOH to surface. Some part of Wire Line BHA was retrieved.

42- Trying to recover fish with 2.9" and 5.5" Skirt in two times.

43- Due to shortage of O.Shot tools, 3.125" Die Collar was run two times,no success

44- Three times O.shot with drilling string was run to top of fish, wire line fish was recovered (this wire line was left in hole at the first time).

45- 6 7/8" Mill was run to 3896.9m. Milling was performed from 3896.9 to 3898.1m. Packer was fallen to 3926.8m. Milling was done from 3926.8 to 3928.2m.]

46- Packer Picker was run to 3928m. Trying to engage with fish, no success. the fish was fallen down to 3929m.

47- 6 7/8" Mill was run to 3929.4. Milling was performed from 3929.4 to 3931.7.

48- 4 Junk baskets were run to clean the hole. 5 ¾" Impression Block was run to 3925.25m (weight on 15 Klbs without trace).

49- 4 ¾" Taper Tap was run to 3929.5m, took weight, wash & ream to 3930.2m. POOH to surface and wash out (3-3.5mm) was observed.

50- 5 7/8" mill with 3 4 Junk baskets were run to 3929.5m. Milling was performed to 3930m. POOH and 3 Ibs iron debris was recovered. Due to fill all baskets, 4 junk baskets were run again.

51- Two fishing operation was performed with 5 5/8" O.shot and 5 ¾" Taper Tap, no success.

52- Running 5 7/8" mill and push down to 4043m.Milling was performed from 4043 to 4049.5m. POOH to surface.

53- RIH with 5 5/8" O.Shot to 4043m. Fish was pushed down to 4049m. Trying to engage, no success.

54- - RIH 5 7/8" mm mill and 4No. J-subs to 4046m, wash down to 4049.5m, circ. & work on J-subs, start pushing and milling to 4052m, cut Mw F/ 110 to 106.5pcf.

55- - Cont. circ. & cut Mw to 103pcf w/ Nl-Ng.- POOH to surface recover 19 LB junk.

56- - RIH 5 7/8" mill & 4No. J-subs to 4052m, milling & pushing to 4052.4m, due to No. progress, Circulation & work on J-sb & weight up Mw f/ 103 to 105pcf Due to obs cut Mw.

57- Cont. cut Mw F/ 103 to 105pcf Due to obs cut Mw, POOH to surf recover 19lb junk. - RIH w/ 5 1/2" washover shoe & 2 Jts wash pipe to 4049.5m, W&R to 4052m. Cont. W&R to 4052.5m, try to wash down, No success. -POOH to surf, L/d washover shoe.

58- P/U 5 7/8" mill & 7 No. J-subs, RIH to 4051.5m, took weight, Circ. mud & pump hi-vis pill clean hole, wash down to 4052.4m.

59- Work on J-subs, POOH to surface, recover 38Lb junk. Again P/U 5 7/8" mm mill & 7 No. J-subs 4052.4m, work on J-subs, pooh to surf(recover 39 Lb junk), Rih 5 7/8" mm mill & 7 No. J-subs to 2052.4m, took weight, pump 20bbls clean the well. Work on J-subs several times. POOH to surf recover 22Lb junk. RIH w/ 5 1/2" washover shoe, 1 jt 5 1/2" washpipe to 4051m, took weight, Circ clean the well.

60- Wash down f/ 4051 m to 4053.11m, No progress, POOH to surf.- RIH w/ 5 1/2" washover shoe, 1 jt 5 1/2" washpipe to 4052m, took weight, Circ clean the well, wash around the fish to 4053.3m, REV. Circ recover 6.6 LB junk F/ magnet. W& R to 4052.4m, OBS plug strg, POOH to surface.- P/U 5 5/8" O. Shot RIH to 4052.7m, try to engage w/ fish, POOH to surf w/ obs no fish. - P/U 5 7/8" mill & 4 No j-subs to 4052.4m, work on J-subs, POOH to surf, L/D J-subs (recover 6LB junk). - P/U 5 1/2" washover shoe & 1jt washpipe & RIH to 4052.4, wash over to 4053.4m, due to No progress, CBU, clean the well, POOH to surface.

61- P/U 5 5/8" O.Shot & 3 5/8" B. Grapple to 4052.4m, try to engage by fish several times w/ No succeed. POOH to surface, RIH w/ 5 3/4" IB to 4052.4m, apply 10 Klb on IB, POOH to surface, L/D IB, OBS mark of Junk.

62- P/U 5 7/8" mill & J-sub to 4052.4, W & R to 4052.7m, milling to 4052.82m, due to No progress CBU clean the well. POOH to surface L/D 5 7/8" mill & J-sub w/ no recover junck. - P/U 5 3/4" Die-collar(3 1/2" x 4 3/8") to 4052.8m, took weight, CBU clean hole, engage by fish, work on stuck by 100KLB, string got free, POOH to surface, recover mule shoe, 3 1/2" TBG, LN, plug, P. Jts totally: 11.75m.

1.7.2 History of Near-by Well: AZNS-070

1.7.2.1 General Data

Well Name/Pad No.	AZNS-070 (Pad 75)	
Well Classification	Vertical Well	
Well Objective	To Produce Oil from FAHLIYAN Lime St. FM To Increase the Production Rate of the Field	
Well Type	IIIF	
Country	Iran	
Province	Khuzestan	
Field	South Azadegan	
Wellhead Coordinates	X: 189,683 Mt	Y: 3,439,014 Mt
Down hole Target Coordinates	X: 189,683 Mt	Y: 3,439,014 Mt
Offset Well(S)	AZNS-06 , AZNS-12	
Rig Contractor	PEDEX	
Rig Name	203PEDEX	
Rig Type	Land Rig	
Rotary Table Elevation From Ground	14.50 m	
GL Elevation	3.95 m	
Mud Type	H.Vis.Bentonite, S. KCL Polymer, KCL Polymer	
Designed TD	4095 M MD / 4095 M TVDRT	
Actual TD	4095 M MD / 4095 M TVDRT	
Well Completion	Fahliyan Completion is run & Tested is well.	
Start Rig Mobilization Date	94.01.29, April 18 2015	
Mobilization & Commissioning Days	15 Days	
Date Spudded	94.02.14 , @00:00AM May 04 2015	
Released	94.10.29 @06:00AM	
Actual Rig Days	260.25 DAYS	

Re-entry	
Rig Name	PEDEX 203
Rig Type	Land Rig
Rotary Table Elevation From Ground	14.50 m
GL Elevation	3.95 m
Mud Type	WBM, Poly Mud
Designed TD	4095 M MD / 4095 M TVDRT
Actual TD	4095 M MD / 4095 M TVDRT
Well Completion	3 ½" * 4 ½" TBG
Start Rig Mobilization Date	
Mobilization & Commissioning Days	
Date Spudded	95.10.21 00:00
Released	95.11.14 24:00
Actual Rig Days	24 Days

1.7.2.2 Casing Depth

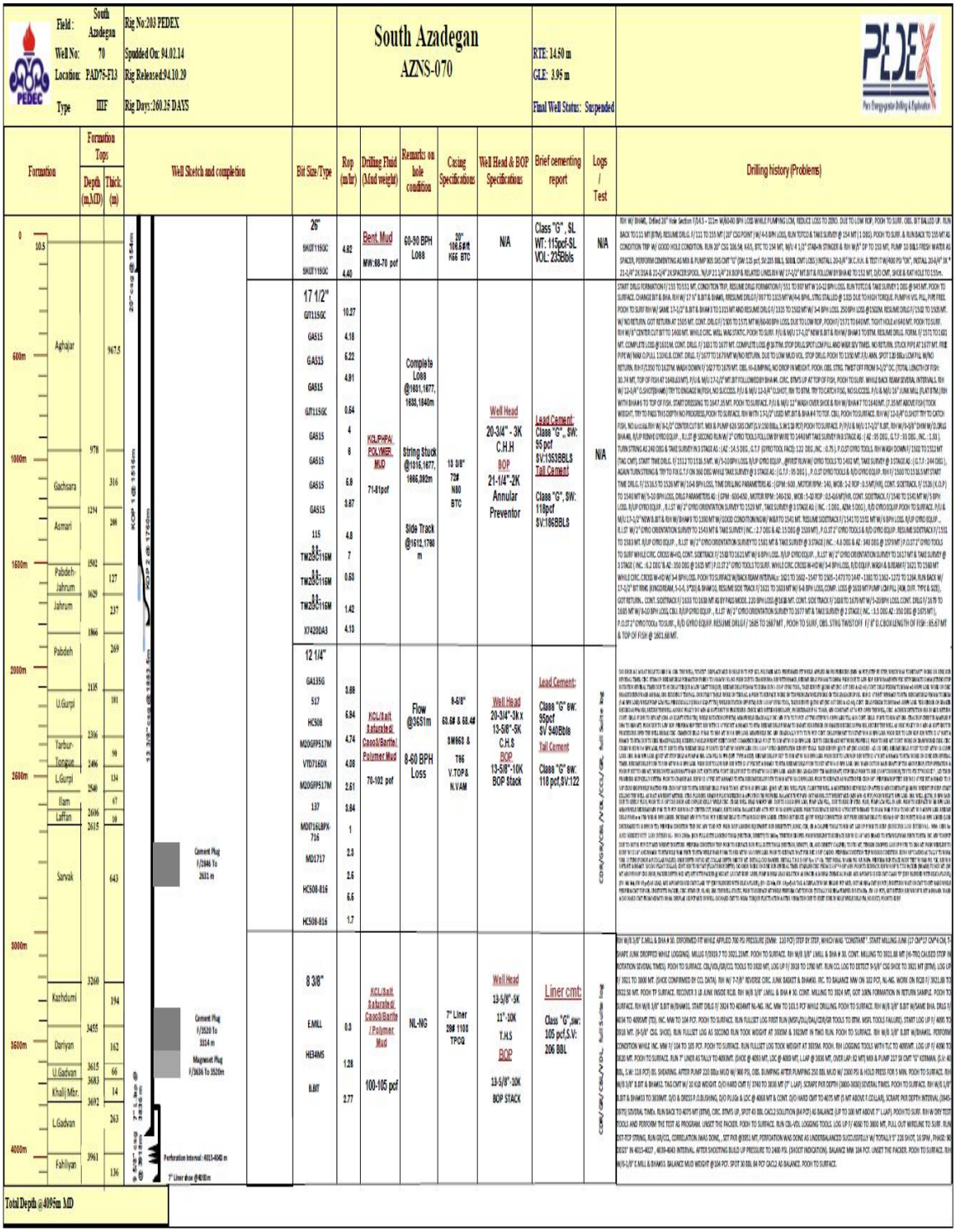
Hole		Casing/Liner Specifications					Mud weight pcf	FIT/LOT pcf	Cement Weight	
Size (In)	Depth (m, RKB)	Size (In)	Depth (m, RKB)	Grade	Weight	Connection			Lead	Tail
26	155	20	0-155	K55	106.5	BTC	68-70		115	-
17.5	1884.5	13.375	0-1883.5	N80	72	BTC	71-81		95	118
12.25	3920	9.625	0-1861	SM95S	53.5	VAMTOP	70-102	84	95	118
12.25	3920	9.625	1861-3566	T95	53.5	VAMTOP	70-102	84	95	118
12.25	3920	9.625	3566-3918	T95	58.4	N.VAM	70-102	84	95	118
8.375	4095	7	3836-4093	110S	29	TPCQ	100-105	110	118	-

1.7.2.3 Formation Tops

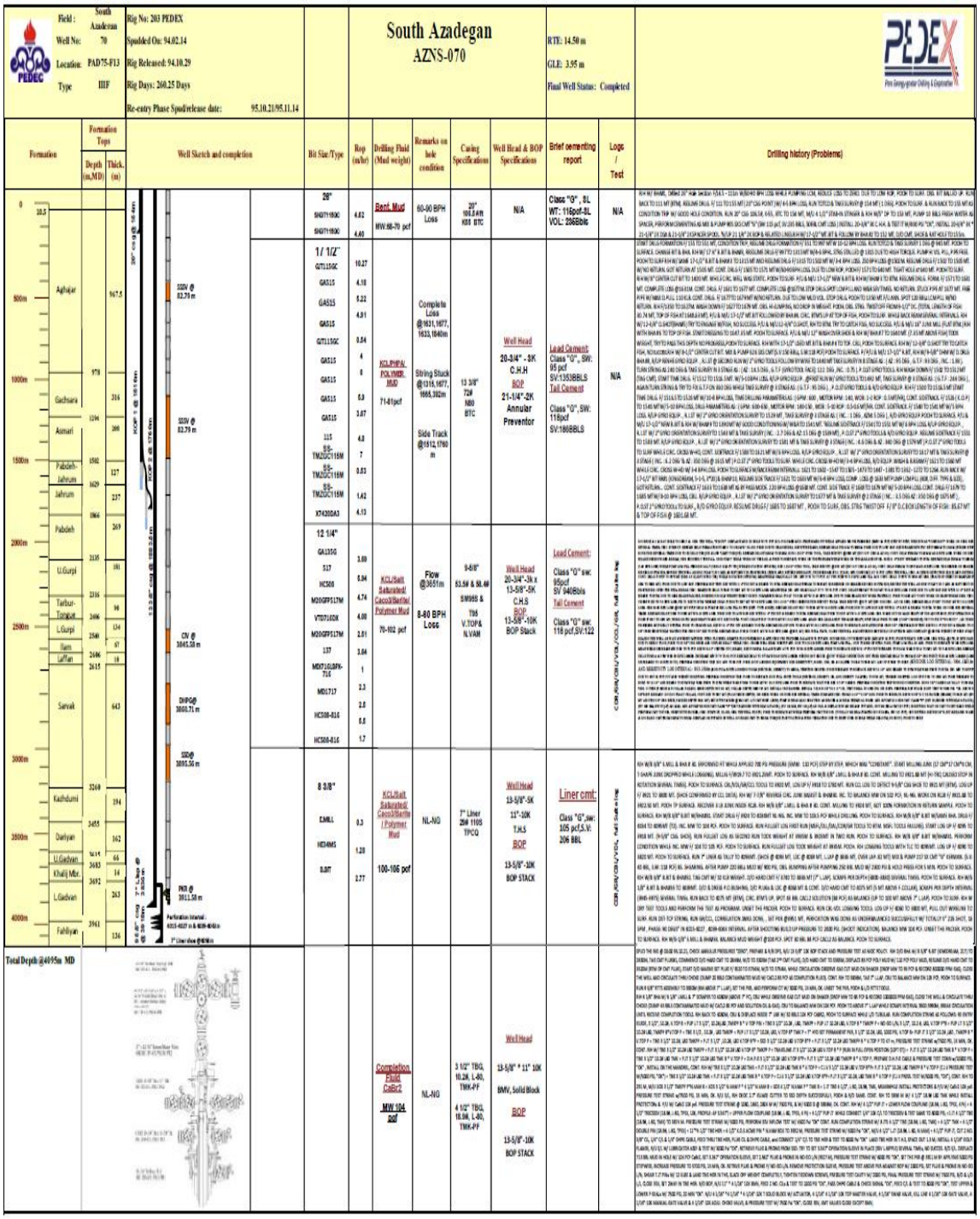
General		Location/Target		Parties/Reference	
Well Name	F 13	Surface Location		Client	NIOC/PEDEC
Wel Type	Vertical Well	Easting	189683m	Contractor	PEDEX
Field	South Azadegan			Rig Name	PEDEX-203
Location	Khozestan, Abadan Plain	Northing	3,439,014	RTE	14.5m
Country	Iran			TD	4095mTVD
Spud Date	940307; May, 28, 2015	Target	Gadvan		
Rig Release	941028;Jan, 18, 2015				

Age	Formation	Formation Top			A.Thick	Lithology	Lithology description	CSG Points/Mud properties	Logging
		Forecast	TVD	Log(MD)	(cutting)				
Quaternary to Pliocene	Aghajari	10.5	10.5	N.A	967.5		Sst: lse, v.fn. - fn. gr. Mrl: rd. clac.; With sm. sft, occ. Frm., stky. Clst.	Mud Type: Bentonite Mud MW: 70pcf 20" @ 154m	NA
Miocene	Gachsaran	965	978	N.A	316		Anhy: off wht.,-wht., md.hd. Mrl: gy. Calc. clst., sft, occ. Frm., stky. Mrl: dk.gr.-gr., sft., hi. Cal. Salt		N.A
Oligocene	Asmari	1275	1294	N.A	208		Lst: off wht., lt.gr., lt. br., crm-br., pacnstone, mudstone, type I/II, arg. Sst: Loose sand		
Paleocene-Eocene	Pabed-Jahrum	1455	1502	N.A	127		Lst: offwht, off wht., wht.-lt.gr., gr., type I/II		
	Jahrum	1593	1629	N.A	237		Lst: wht., wht.-lt.gr., gr., type I/II		
	Pabdeh	1800	1866	N.A	269			Mud Type: KCL polymer Mud MW: 70-77pcf 13 3/8"@1883.5m	
Upper Cretaceous	U. Gurpi	2030	2135	2135	181		Lst: wht., gr., lt.gr., dk.gr., mudstone, wackstone type I/II, II, arg., pyr., rexln.		CDR/GR; CBL/VDL/CCL/GR; Fullsuite
	Tarbur Tongue	2290	2316	2316	90				
	L. Gurpi	2385	2406	2406	134				
	Ilam	2530	2540	2540	67		Mrl: gr., stf., stk.,calc		
	Laffan	2603	2607	2606	10		Mrl & shale		
Lower Cretaceous	Sarvak	2610	2617	2615	643		Lst: off wht., lt.crm.-xln., Lt..gr-gr-dkgr, type I/II, II, II/I		
	Kazhdumi	3240	3260	3260	194		Lst: lt.gr.-br.gr. Lst., chky., foss., part bituminous		
	Dariyan	3450	3454	3455	162		Lst.		
	U. Gadvan	3615	3616	3615	66				
	Khalij Mbr.	3676	3682	3683	14				
	L. Gadvan	3693	3696	3692	263		Mrl., Lst., Sst.	7" linerlap@3836m Mud Type: KCL polymer MW: 73-82pcf 9 5/8"@3918m	
	Fahliyan	3915	3959	3961	36			Mud Type: KCl polymer MW:103-104 pcf 7"@4002mMD TD: 4095m	CBL/GR; BLAS Lithol ogite

1.7.2.4 Well Montage After Suspension



1.7.2.5 Well Montage After Completion



1.7.2.6 Brief Summary of Operation

Well AZNS-070(Pad-75) was drilled by PDX203 which mobilization was initiated on 94.01.29, @ 24:00. PDX203 was spudded on 94.02.14 @ 00:00 & Reached to TD at 4095 m 94.08.22 @ 23:00, Rig was released on 94.10.28 @ 24:00PM

1 Spud the Rig on 94.02.14 @ 00:00.

2 26" hole was spud 94.02.14 @ 00:00.& drilled to TD at 155m @ 10:00 on 94.02.15.

3 20" CSG was run to 154m and cement.

4 17-1/2" hole was commenced @ 06:00AM on 94.02.20 & drilled to 1884.5m @ 03:00 94.05.01.

5 13-3/8" CSG was run & cemented @ 1883.5.

6 12-1/4" hole was spud @11:00 on 94.05.07 & reach section TD at 3920m @ 20:00 on 94.07.07.

7 Full Suite Log is run from 3920m to 1883.5m.

8 9-5/8" CSG was run & cemented via RTTS @ 3918m.

9 Run 9-5/8" CBL/VDL/CCL/GR f/1780 to 3916 m.

10 8-3/8" hole was spudded @ 17:00 94.08.03. It was drilled to 4095m @ 23:00 on 94.08.22.

11 Full Suite Log is run from 4095m to 3918m.

12 7" liner was run & cemented to 4093m.

13 Run 7" CBL/VDL/CCL/GR f/3840 to 4056 m.

14 The Completion was run unsuccessfully, so pooh completion string, @ 18:00. on 94.10.18

15 Rig was released on 94.10.29 @ 06:00AM.

16 The well suspended by 2 cement plug and displaced by non-solid fluid @ 18:00. on 94.10.19

1.7.2.7 Brief Summary of Re-Entry Operation

- Spud the Rig @ 00:00 95.10.21
- D/O 1st CMT plug (2630-2846 m), 2nd plug f/ 3305-3363 m, displace 85 pcf poly mud w/ 110 pcf poly mud, resume D/O hard CMT to 3520 m, (BTM of CMT plug), start D/O Magne-set plug f/ 3520 to 3754m, RIH & tag 7" L.Lap
- Displace & balance MW on 105 pcf poly mud
- Run 9 5/8" RTTS assembly to 3830m (6m above 7" L.Lap), Set the PKR, and perform CIT w/ 3000 psi, 15 min, OK
- Displace inside 7" LNR w/ 30 bbls 104 pcf cabr2
- Run completion string to PKR depth, displace the well thru string w/ 104 pcf CaBr2
- Set the PKR @ 3911m & Perform related pressure tests
- N/U XMT & pressure test
- Rig release @ 95.11.14 24:00

1.8 General Assumption

1.8.1 Well Objective, Target and Total Depth

1.8.1.1 Purpose of the Well

This well will be completed as a fahliyan oil producer at vertical hole.

1.8.1.2 Wellbore Quality Evaluation Criteria:

1) Maximum Deviation and Horizontal Displacement for Vertical Wells

Depth(m)	Deviation (°)	Bottom Hole Shift (m)	DLS (°/30m)
≤2000	≤3°	≤50	0≤2.25°
2000~3000	≤5°	≤80	≤2.50°
3000~4000	≤7°	≤120	≤2.75°
4000~5000	≤9°	≤160	≤3°

2) Other requirements for Wellbore Quality

- Average borehole diameter enlargement rate less than 15%
- All casing/liner are run to designed depth (normally 1-2m above TD/section TD)
- No fills at bottom of hole
- No junk lost in hole
- No abnormal casing wear and casing deformation caused by CONTRACTOR's malpractice

3) Well trajectory control key points

- Drill upper formation before KOP using Pendulum BHA, low WOB and high RPM, to ensure vertical hole drilled.
- Control the distance between two holes in one well pad is not less than 2 meters. Single-shot and multi-shot will be used to monitor the well trajectory. Take single shot survey at interval of 100m; run multi-shot survey each 300m from surface to kick off point. If necessary, survey shall be run more frequently.
- Run build up BHA (Motor + MWD) whenever collision risk increases to build up a little inclination (2~4°) and bottom hole departure to avoid collision at vertical section if necessary.
- If magnetic interference exists, gyroscopic inclinometer shall be used in order to avoid collision.
- Amend the planned profile according to the data obtained by electronic multi-shot or single-shot survey in vertical hole, and locate the kick off point.
- MWD/GR will be used in directional drilling to ensure that the actual well path will coincide with the planned well trajectory to the extent that it shall not pass through the boundaries of the corridor that surrounds the planned hole trajectory in case a corridor has been specified.
- LWD/GR will be utilized in horizontal drilling in order to controlling well trajectory, especially in Sarvak-3.
- The downhole motor and MWD shall be used in vertical hole section if necessary.
- Monitors and analyses the build-up ability of the BHA used in directional/horizontal section. Due to the lagged survey data measured by MWD, inclination, azimuth and the buildup rate should be predicted based on the instant survey at all times. If necessary, more surveys should be taken so as to adjust drilling parameters and BHA, control the well trajectory within the specified target tolerance strictly.
- In buildup section, to avoid pipe-stuck, BHA should be designing such way to minimize sliding mode time especially in clay formations. To reduce torque in rotating mode and consequently reduce sliding mode time, using stabilizer with short gauge and profile.
- Using circulating sub above MWD is recommended due to sever loss in this area.
- Adjust the drilling method according to the well survey data, and control the change amplitude of inclination and azimuth to ensure the smoothness of the well trajectory.
- In this section, the torque and drag are relatively higher, torque and drag analysis for each BHA run should be performed, and torque and drag reduction measures (for example simplifying the BHA or running cuttings bed impeller) are recommended.

1.8.1.3 Expected H2S and CO2

Reservoir	H2S from PVT Reservoir Oil (mol%)	CO2 from PVT Reservoir Oil (mol%)	H2S from PVT Separator Gas (mol%)	CO2 from PVT Separator Gas (mol%)
Sarvak	0.06	1.45	0.13	6.11
Kazhdumi	0.00	2.00	0.00	2.65
Gadvan	0.11	1.87	0.16	2.91
Fahliyan	0.28	2.68	0.37	3.60

1.8.1.4 Depth Reference

All depths are based on RTE=14.45m

1.8.1.5 Unit of Measure

Angle: deg	Gas / oil ratio: scf/stb	Pump / Flow rate: USgal/min	Thermal conductivity: Btu/hr*ft*F
Area: ft2	Gas rate: ft3/min	Rotation velocity: rpm	Thermal conductivity: Btu/hr*ft2*F2
Compressibility: psi-1	Length: m	Shear rate: rpm	Time: min
Density: lbm/ft3	Nozzle diameter: 1/32 in	Shear stress: lbf/100ft2	Torque: lbf-ft
Density gradient: lbm/ft3*F	Percent: %	Specific heat capacity: Btu/lbm*F	Tripping velocity: m/min
Dogleg severity: deg/100 ft	Permeability: md	Specific heat capacity: Btu/lbm*F2	Viscosity: cp
Diameter: in	Power: hp	Temperature: Fahrenheit	Viscosity gradient units: cp/F
Fluid velocity: ft/s	Pressure: psi	Temperature (relative): Celsius	Volume: bbl
Force: lbf	Pressure gradient: psi/ft	Temperature gradient: F/ft	Weight/length: lbm/ft

2. Well Plan

2.1 Well Montage

General Information:
Well Name: F-20
Pad No.: 93
Well Type: IIIF
Well Progile: Vertical
Block: III
Field: South Azadegan
Country / Area: Iran / Khuzestan
Reservoir: Fahlian

Wellhead Coordinate:
Grid Geodetic System: (UTM39N, WGS84)
E: 192,581
N: 3,441,012

Nearby Wells:
AZN-070-ST2
AZNS-049
AZNS-173

Parties/Refrence:
Client: PPL/PEDEC
Contractor: MSA/OEOC
Rig Cont: OEOC
Rig: OEOC 208

Well Montage

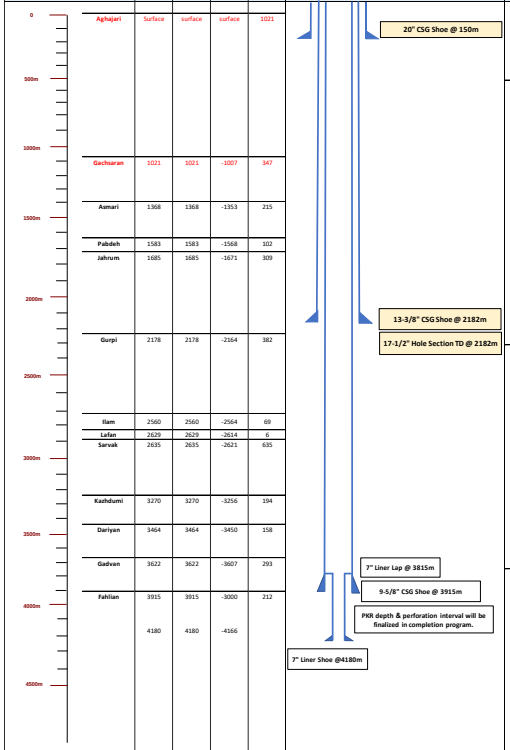
F-20 Development Well

(PAD-93), IIIF

Drilling Depth:
TD (MD): 4180m
TD (TVD): 4180m
GLE: 3.95m
RTE: 14.45m

Timing:
Moving Days: 10 Days
Spud Date: TBA
Total Rig Days: 145 Days (With out rigless job)
Rig Released Date: TBA

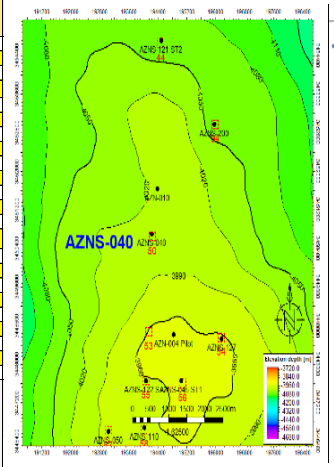
Field Location/Objective:
***Field Location:** The AZNS oilfield is located in Khuzestan province, approximately 80 Km to the west of Ahwaz city.
***Well Objective:** Development well, Fahliyan producer as a vertical well.

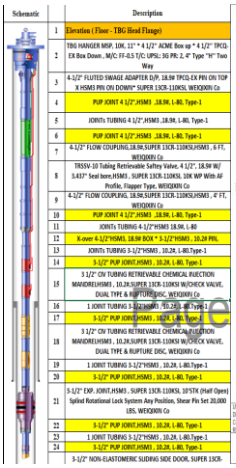


Formation	Formation TOP MD (M)	Formation TOP TVD (M)	Formation TOP TVD ss (M)	Formation Thickness (M)
Aghajari	Surface	surface	surface	1021
Gachsaran	1021	1021	1007	347
Asmari	1368	1368	-1353	215
Pabdeh	1583	1583	-1568	102
Jahrum	1685	1685	-1671	309
Gurpi	2178	2178	-2164	382
Ilam	2560	2560	-2564	69
Lafan	2629	2629	-2614	6
Sarvak	2635	2635	-2621	635
Kashdumi	3270	3270	-3256	194
Dariyan	3464	3464	-3450	158
Gadvan	3622	3622	-3607	293
Fahlian	3915 4180	3915 4180	-3000 -4166	212

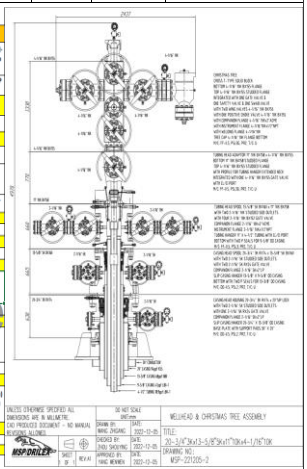
Hole Size Bit Type Nozzle Sizes	BHA	Drilling Parameters	Drilling Fluid	CSG/Liner Design	Cementing	Logging Testing Coring	BOP and Well Head	Remarks (Drilling Issues/Lesson Learned)
26" 26" MT Bit 1-1-1 (1-1-5)	26" Bit + 9 3/4" Bit Sub + 4 Jts 9 3/4" DC + 26" S. Stab. + 2 Jts 9 3/4" DC + X.O.S + 6 Jts 8 1/2" DC + X.O.S	**Flow Rate:** 900-1000 GPM **SPP:** 100-1000 Psi **WOB:** 10-30 Klbs **RPM:** Max. Available **Av. ROP:** 5 m/hr	Pre Hydrated Bentonite Mud 65-67 PCF	**0-150m:** 20", 94 ppf, K-55, BTC Burst: 2110 psi Colapse: 520 psi Body Yield: 1,480 Klbs	**Lead:** Class "G" CMT S.V: bbl, SW: 117.5 pcf	NIL	20-3/4", 3K CHH (Weld On 20" CSG)	1. Bit Balled-up with 65 pcf Bentonite Mud in AZNS-045. 2. Cement Losses in AZNS-017 with 115 pcf Sw. 3. Bit balled-up with bentonite mud in AZNS-127.
17-1/2" 17-1/2" Tri Cone Bit 1-1-5 or 1-3-5 or 5-3-7 17-1/2" PDC Bit S223/M223	17 1/2" Tri Cone Bit + 9 3/4" B. Sub + 2 Jts 9 3/4" DC + 17 1/2" S. Stab + 4 Jts 9 3/4" DC + X.O.S + 6 Jts 8 1/2" DC + 8" Drlg Jar + 3 Jts 8 1/2" DC + X.O.S + 6 Jts 5" HWDP 17 1/2" PDC Bit + 9 3/4" B. Sub + 2 Jts 9 3/4" DC + 17 1/2" S. Stab + 4 Jts 9 3/4" DC + X.O.S + 6 Jts 8 1/2" DC + 8" Drlg Jar + 3 Jts 8 1/2" DC + X.O.S + 6 Jts 5" HWDP	**Flow Rate:** 800-900 GPM **SPP:** 850-1000 Psi **WOB:** Max Available Klbs **RPM:** 80-140 **ROP:** 3-6 m/hr **Flow Rate:** 800-900 GPM **SPP:** 1500-2500 Psi **WOB:** Max Available Klbs **RPM:** Max. Available **ROP:** 5-6 m/hr	Pre Hydrated Bentonite Mud 65-67 PCF Salt Saturated Kcl Polymer WBM 67-88 PCF	**0-2182m:** 13-3/8", 68 ppf, N-80, BTC Burst: 5020 psi Colapse: 2260 psi Body Yield: 1,556 Klbs	**Lead:** Class "G" CMT S.V: bbl, S.W: 90.5 pcf **Tail:** Class "G" CMT S.V: bbl, S.W: 115.5 pcf		20-3/4", 3K BOPs 20-3/4", 3K x 13-5/8", 5K CHS	1. Bit and Stabilizer balling with 67-86 pcf Kcl/Polymer mud in GS formation in AZNS-045. 2. Calcium ion contamination in GS formation. 3. Frequent BHA/Bit balling and tight hole in AZNS-017 due to low Mw and improper mud properties. 4. Wash down 13-3/8" CSG and set CSG off bottom due to bad hole condition in AZNS-017 (mainly low Mw in compare to hole condition). 5. Obs back flow after Cementing 13-3/8" CSG, apply pressure and close well in several steps in AZNS-045. 6. String stuck while POOH & B.Ream in GS formation with 84 pcf Mw in AZNS-045 well. 7. Frequent tight hole and hole stability problem in GS formation in well AZNS-017 (12-1/4" Hole). 8. WL logging tool held up in GS Fm., in well AZNS-017 (12-1/4" Hole). 9. Bit balled-up in base of GS formation in AZNS-127. 10. Frequent tight hole and string washout in GS in AZN-010 with 75-90 pcf Mw.
12-1/4" 12-1/4" Tri Cone Bit 2-1-4 or 5-1-7 12-1/4" PDC Bit S223,M223	12 1/4" Tri cone Bit + 9 3/4" B.Sub + 2 Jts 8-1/2" DC + 12 1/4" S.Stab + 13 Jts 8 1/2" DC + 8" D. Jar +3 Jts 8-1/2" DC + XOS +6 Jts 5" HWDP 12 1/4" PDC Bit + 9 3/4" B.Sub + 2 Jts 8-1/2" DC + 12 1/4" S.Stab + 13 Jts 8 1/2" DC + 8" D. Jar +3 Jts 8-1/2" DC + XOS +6 Jts 5" HWDP	**Flow Rate:** 500-650 GPM **SPP:** max 2850 Psi **WOB:** 5-20 Klbs **RPM:** 80-110 **Flow Rate:** 600-750 GPM **SPP:** max 2850 Psi **WOB:** 20-30 Klbs **RPM:** 160-180	Salt Saturated KCL/Glycol/Polymer/CaCo3 WBM 72-85PCF	**0-3915m (3915m):** 9-5/8", 53.5 ppf (SD), 110SS, Premium Gas Tight Con. **ID:** 8.535" (SD) **Burst:** 10900 psi **Colapse:** 7950 psi **Body Yield:** 1,710 Klbs **B.SF:** 1.17 **C.SF:** 1.99 **A.SF:** 2.33 **TA.SF:** 1.42	**Lead:** Class "G" CMT S.V: bbl, S.W:81-85 pcf **Tail:** Class "G" CMT S.V: bbl, S.W: 118pcf	**Cased Hole Logs:** GR/CCL/VDL /SBT, from 9-5/8" Casing **Open Hole Logs:** Full Suite/CDR	13-5/8", 10K BOPs 13-5/8", 5K x 11", 10K THS	**Main Drilling Challenges:** Losses in AGH/GAS/KAZ, Chert formation in base of Pabdeh, Tight hole and string stuck, Washout string, Oil/Gas Flow, CMT Loses, Complete Loss, Hole Deviation 1. Frequent tight hole while drilling/tripping through GU (L) in AZNS-045 well. 2. Seepage losses 2-4 BPH with 75-87 pcf Mw in ASD?DGU formation in AZNS-045. Drilling F/ SV to GU with 0-2 BPH losses wi 84-87.5 pcf Mw. 3. Differential Sting stuck in KZ of AZNS-045 wi 87.5 pcf Mw, free same with HCL. Acid and cut MW to 81 pcf. 4. Perform top cement job in 9-5/8" Ann due to failure while displacing cement and remained cement inside casing in AZNS-045. 5. Obs2 BPH losses in KZ Fm., in AZNS-017 and frequent string stuck while drilling horizontal hole in KZ with 76-78 pcf Mw. 6. Obs 2-12 BPH losses while drilling through SV in AZNS-017 with 75-77 pcf Mw. 7. Obs 3-5 BPH losses with 80-81 pcf Mw in AZNS-127, losses inc to 12 BPH in KZ & 15 BPH in GD formation. 8. Frequent string washout and twist off string in AZNS-127 while performance drilling with 80-81 pcf Mw. 9. Noticeable CMT losses while cementing 9-5/8" CSG w/ 88 pcf Lead CMT Sw in AZNS-127. 10. Low ROP in Cherty Zone (PD) from 1920-1989m in AZNS-127. 11. Obs 45-50 BPH losses while drilling with 80 pcf Mw in top of SU in AZN-010.
8-1/2" 8-1/2" MT Bit 214 1No. 8-1/2" PDC Bit S223,M223 S422,M422 1No. 5-7/8" B. Bit S27	D.Out BHA:Bit + 1.Sub + B.Sub + 9Jts 6-1/2" DC + 6-1/2" D.Jar + 3Jts 6-1/2" DC + XOS + 6Jts 5" HWDP Bit + B.sub + 1Jt 6-3/4" NMDC + MWD sub + 1Jt 6-3/4" NMDC + 8-1/2" S.STB + 9 Jt's 6-1/2" DC + 6-1/2" D.Jar + 3 Jts 6-1/2" DC> 9 Jts 5" HWDP	**Flow Rate:** 450-500 GPM **SPP:** max 2850 psi **WOB:** 5-15 Klbs **RPM:** 70-80 **Flow Rate:** 400-500 GPM **SPP:** max 2850 Psi **WOB:** 10-20 Klbs **RPM:** 150-170	Salt Saturated KCL/Polymer/CaCo3 WBM: 100-106 PCF	**3815-4180m (365m):** 7", 32ppf, 110SS, Premium Gas Tight Con. **ID:** 6.094" **Burst:** 12460 psi **Colapse:** 10780 psi **Body Yield:** 1,025 Klbs **B.SF:** 1.24 **C.SF:** 1.06 **A.SF:** 1.51 **TA.SF:** 1.49	**Lead:** Class "G" CMT S.V: bbl, SW: 118pcf	**Open Hole Logs:** Full Suit/CDR Cased Hole Logs: SBT/VDL/CCL/GR	13-5/8", 10K BOPs 4-1/16", 10K X-Tree	**Main drilling Challenges:** Tight hole and string stuck, Oil/Gas Flow, Complete Loss, Hole Deviation 1. String stuck in FH formation with 105-106 pcf MW, free same with Jar/pulling/Pipe-Lax in AZNS-045. 2. String stuck while coring in FH., LIH and side track well in AZNS-045. 3. 7" Liap leakage as Dry Test result in AZNS-127 even with good amount of hard CMT on TOL. Run T.Back Liner and cement same. 4. Seepage losses 2-8 BPH in AZNS-045 with 100-106 pcf Mw, 1-3 BPH w/ 104-105 pcf Mw in AZNS-127 & AZN-010 w/ 100-110 pcf Mw.

Time Breakdown Hole Size	26"	17 1/2"	12 1/4"	8 1/2"	Completion	Contingency plan		Sum (Hrs)	Total %	Total (Days)
Drilling	31	386	786	100		115		1418	43.2%	59.1
Round Trip/BHA	5	67	67	85				224	6.8%	9.3
BOP Test	6	12	18	13	6			55	1.7%	2.3
Hole Condition, Reaming, Circ	3	17	20	52				92	2.8%	3.8
Deviation Survey		15	14					29	0.9%	1.2
Directional Drilling								0	0.0%	0.0
Casing/Liner	13	47	45	40				145	4.4%	6.0
Cementing	13	11	17	10				51	1.6%	2.1
WOC								0	0.0%	0.0
N/UBD BOP	20	24	18					62	1.9%	2.6
N/U WH	9	24	24					57	1.7%	2.4
Drill Out	4	22	9	90				125	3.8%	5.2
Logging		13	45	75				133	4.1%	5.5
Testing (FIT, LOT, CIT, Dry Test, etc.)								0	0.0%	0.0
DST								0	0.0%	0.0
Coring								0	0.0%	0.0
Plug Back								0	0.0%	0.0
Completion					421.5			421.5	12.9%	17.6
Well Testing						429		429	13.1%	17.9
Rig Repair								0	0.0%	0.0
Slip & Cut		3	3	3	3			12	0.4%	0.5
De-suspend	2	6	7	6	5			26	0.8%	1.1
Workover								0	0.0%	0.0
Total Hrs per Hole	106.0	647.0	1073.0	474.0	435.5	544.0	3279.5			
Total Days per Hole	4.4	27	44.71	19.750	18.146	22.667	136.6			
Total Days per Hole with 10% Contingency	4.9	29.7	49.2	21.7	20.0	24.9	150.3			
Cummulative Time (Days)	4.4	31.4	76.1	95.8	114.0	136.6				
Cummulative Time (Days) with 10% Contingency	4.9	34.5	83.7	105.4	125.4	150.3				

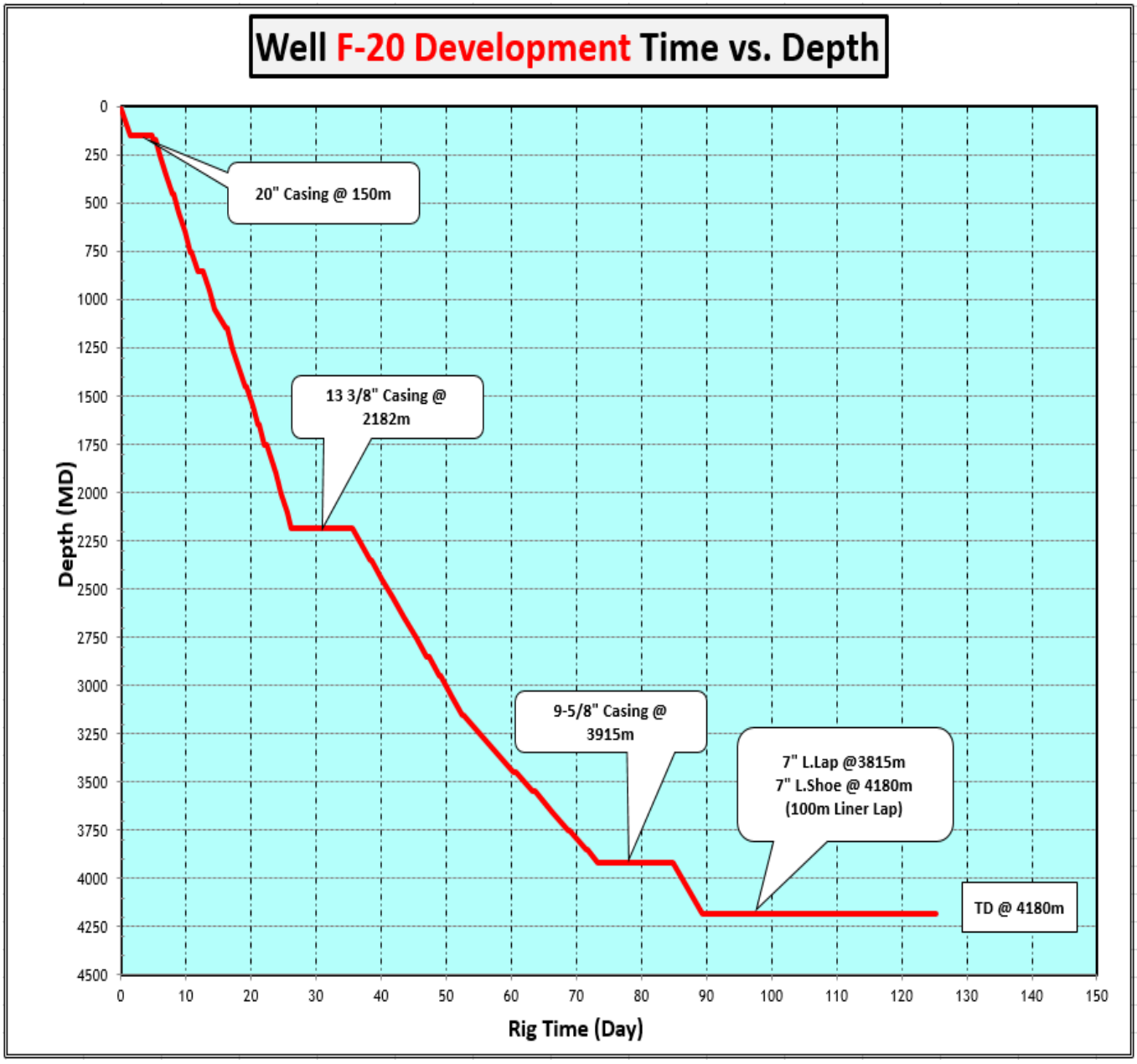




	Description
1	Elevation (Floor - TBG Head Flange)
2	TBG HANGER MSP, 10K, 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ, EX Box Down , MJC: FF-63.5 T/C: UPS: 35 PR: 2, 4" Type "H" Two Way
3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ, EX PIN ON TOP X HSM5 PIN ON DOWN" SUPER 13CR-110KSI, WEIQIXIN Co
4	PUP JOINT 4 1/2" ,HSM5 , 18.9#, L-80, Type-1
5	JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1
6	PUP JOINT 4 1/2",HSM5 ,18.9#, L-80, Type-1
7	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT., WEIQIXIN Co
8	TRSSV-10 Tubing Retrievable Safety Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type, WEIQIXIN Co
9	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4" FT., WEIQIXIN Co
10	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1
11	JOINTs TUBING 4 1/2",HSM3 18.9#, L-80
12	X-over 4 1/2"HSM3, 18.9# BOX * 5 1/2"HSM3 , 10.2# PIN,
13	JOINTs TUBING 5-1/2"HSM3 , 10.2#, L-80,Type-1
14	5-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1
15	5 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDREL,HSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co
16	1 JOINT TUBING 5-1/2"HSM3 ,10.2#, L-80,Type-1
17	5-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1
18	5 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDREL,HSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co
19	1 JOINT TUBING 5-1/2"HSM3 , 10.2#, L-80,Type-1
20	5-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1
21	5-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10 STK (Half Open) Spind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS, WEIQIXIN Co
22	5-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1
23	1 JOINT TUBING 5-1/2"HSM3 , 10.2#, L-80,Type-1
24	5-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1
25	5-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-



2.2 Time vs. Depth



2.3 Time Breakdown

Item No.	Drilling and Completion Operation TBD of Well: F-20	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP Average (MPH)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
	26" Hole Section / 20" Casing												
1	P/U & M/U 26" bit #1 (R.Bit) and 9-3/4" drill collars, RIH to tag formation @ ±14m.	Drilling	5	Agh.	14	0.0	0.0	0.00	0.0	0.0	0.0	0.00	0.0
2	Drilling 26" hole from 14 to 150m (20" casing point in Aghajari formation) using Hi-vis bentonite pill (Run Totco each 50m).	Drilling	136	Agh.	150	5.0	31.0	1.29	1.3	1.4	34.1	1.42	1.4
3	Circulate and clean the hole. Spot high viscosity pill at bottom, drop Inclinometer (Totco).	Trip		Agh.	150		3.0	0.13	1.4	0.1	3.3	0.14	1.6
4	Perform hole condition prior to run casing. If observe any indication of drag, run back to bottom, POOH to surface.	Trip		Agh.	150		5.0	0.21	1.6	0.2	5.5	0.23	1.8
5	R/U casing running equipment & run 150 m approximately 13 Jts. 20" casing to 150 m.Fill casing by non-LCM fresh mud.	Csg. Running		Agh.	150		13.0	0.54	2.2	0.6	14.3	0.60	2.4
6	Circ mud, RIH W/ Stringer on 5" DP, Circulate mud.Cement Casing as per program.POOH to Surf., L/D stinger, Wait on cement to get hard meanwhile perform top job(s) to fill annulus	CMT		Agh.	150		13.0	0.54	2.7	0.6	14.3	0.60	3.0
7	Cut 30" conductor pipe, Cut & bevel 20" casing. Weld and Installed 20-3/4" x 3K C.H.H & pressure test same with 80% Casing Collapse.	well Head		Agh.	150		9.0	0.38	3.1	0.4	9.9	0.41	3.4
8	N/U BOP and releated lines and valves, test BOP & choke manifold as per NIOC policy.	N/U BOP & BOP Test		Agh.	150		26.0	1.08	4.2	1.2	28.6	1.19	4.6
9	Drill out cement and shoe with 17-1/2" bit (R.Bit), work on junk sub.	D/Out		Agh.	150		4.0	0.17	4.3	0.2	4.4	0.18	4.8
	Total Duration of 26" Hole Section:							***4.3***	***4.3***			***4.8***	***4.8***
	17-1/2" Hole Section / 13-3/8" Casing												
10	Commence drilling 17-1/2" hole to 174m (bit#2).	Drilling	24	Agh.	174	5.0	5.0	0.21	4.5	0.2	5.5	0.23	5.0
11	Circulate and clean hole, POOH to surf.	Trip		Agh.	174		4.0	0.17	4.7	0.2	4.4	0.18	5.2
12	RIH w/ same bit (R. bit #2RR1) and S.Stab @ 20m to bottom.Survey shall be taken by running single shot (Each 100 m) & multishot (Each 300 m) in this hole section.	Trip		Agh.	174		4.0	0.17	4.9	0.2	4.4	0.18	5.4
13	Continue drilling 17-1/2" hole to ±250m while pumping hi-vis pill periodically	Drilling	76	Agh.	250	6.0	15.0	0.63	5.5	0.7	16.5	0.69	6.1
14	Continue drilling 17-1/2" hole to ±350m while pumping hi-vis pill periodically	Drilling	100	Agh.	350	6.0	19.0	0.79	6.3	0.9	20.9	0.87	6.9
15	Take survey	Survey		Agh.	350		2.0	0.08	6.4	0.1	2.2	0.09	7.0
16	Continue drilling 17-1/2" hole to ±450m while pumping hi-vis pill periodically	Drilling	100	Agh.	450	6.0	19.0	0.79	7.2	0.9	20.9	0.87	7.9
17	Condition trip to 150m Run back, Take survey as multi-shot	Trip		Agh.	450		5.0	0.21	7.4	0.2	5.5	0.23	8.1
18	Continue drilling 17-1/2" hole to ±550m while pumping hi-vis pill periodically	Drilling	100	Agh.	550	6.0	19.0	0.79	8.2	0.9	20.9	0.87	9.0
19	Continue drilling 17-1/2" hole to ±650m while pumping hi-vis pill periodically	Survey	100	Agh.	650	6.0	19.0	0.79	9.0	0.9	20.9	0.87	9.9
20	Continue drilling 17-1/2" hole to ±750m while pumping hi-vis pill periodically	Drilling	100	Agh.	750	6.0	19.0	0.79	9.8	0.9	20.9	0.87	10.7
21	Condition trip to 450m and Run back , displace mud in hole by 74pcf KCL/Polymer mud @ ±750m,	Survey		Agh.	750		4.0	0.17	9.9	0.2	4.4	0.18	10.9
22	Continue drilling 17-1/2" hole to 850m (bit hrs), while pump hi-vis Pill periodically.	Drilling	100	Agh.	850	6.0	19.0	0.79	10.7	0.9	20.9	0.87	11.8
23	Circulate clean hole, Take survey as multi-shot	Trip		Agh.	850		13.0	0.54	11.3	0.6	14.3	0.60	12.4
24	POOH to surface, change bit to PDC (bit #3), RIH to bottom while pick up drill pipe.	BOP Test		Agh.	850		6.0	0.25	11.5	0.3	6.6	0.28	12.7
25	Continue drilling 17-1/2" hole to 950m while pump hi-vis Pill periodically.	Drilling	100	Agh.	950	6	19.0	0.79	12.3	0.9	20.9	0.87	13.5
26	Continue drilling 17-1/2" hole to 1050m while pump hi-vis pill periodically.	Drilling	100	Agh.-Gas.	1050	6.0	19.0	0.79	13.1	0.9	20.9	0.87	14.4
27	Take survey	Survey		Agh.-Gas.	1050		2.0	0.08	13.2	0.1	2.2	0.09	14.5
28	Continue drilling 17-1/2" hole to 1150m while pump hi-vis pill periodically.	Drilling	100	Gas.	1150	3.3	35.0	1.46	14.6	1.6	38.5	1.60	16.1
29	Perform Wiper Trip as POOh to 850 m & Run back to BTM. Take survey as multi-shot	Trip		Gas.	1150		6.0	0.25	14.9	0.3	6.6	0.28	16.4
30	Continue drilling 17-1/2" hole to 1250m while pump hi-vis pill periodically.	Drilling	100	Gas.	1250	6.0	19.0	0.79	15.7	0.9	20.9	0.87	17.2
31	Continue drilling 17-1/2" hole to 1350m while pump hi-vis pill periodically.	Drilling	100	Gas.	1350	6.0	19.0	0.79	16.5	0.9	20.9	0.87	18.1
32	Take survey	Survey		Gas.	1350		2.0	0.08	16.5	0.1	2.2	0.09	18.2
33	Continue drilling 17-1/2" hole to 1450m while pump hi-vis pill periodically.	Drilling	100	Gas.-As.	1450	6.0	19.0	0.79	17.3	0.9	20.9	0.87	19.1
34	Perform Wiper Trip as POOH to 1150 m & Run back to BTM. Take survey as multi-shot	Trip		As.	1450		7.0	0.29	17.6	0.3	7.7	0.32	19.4
35	Continue drilling 17-1/2" hole to 1550m while pump hi-vis pill periodically.	Drilling	100	As.	1550	6	19.0	0.79	18.4	0.9	20.9	0.87	20.3

36	Continue drilling 17-1/2" hole to 1650m while pump hi-vis pill periodically.	Drilling	100	AS/Pb.-Ja.	1650	6.0	19.0	0.79	19.2	0.9	20.9	0.87	21.1
37	Take survey	Survey		Pb.-Ja.	1650		2.0	0.08	19.3	0.1	2.2	0.09	21.2
38	Continue drilling 17-1/2" hole to 1750m while pump hi-vis pill periodically.if observe Chert particle while formation sampling Follow Contingency Plan#01	Drilling	100	Ja.	1750	6.0	19.0	0.79	20.1	0.9	20.9	0.87	22.1
39	Perform Wiper Trip as POOH to 1450 m & Run back to BTM. Take survey as multi-shot	Trip		Ja.	1750		7.0	0.29	20.4	0.3	7.7	0.32	22.4
40	Continue drilling 17-1/2" hole to 1900m while pump hi-vis pill periodically.	Drilling	150	Ja.	1900	6.0	29.0	1.21	21.6	1.3	31.9	1.33	23.7
41	Take survey,	Survey		Ja.	1900		2.0	0.08	21.7	0.1	2.2	0.09	23.8
42	Continue drilling 17-1/2" hole to 2000m while pump hi-vis pill periodically.	Drilling	100	Ja.-Pb.	2000	6.0	18.0	0.75	22.4	0.8	19.8	0.83	24.7
43	Continue drilling 17-1/2" hole to 2100m while pump hi-vis pill periodically.	Drilling	100	Pb.	2100	6.0	18.0	0.75	23.2	0.8	19.8	0.83	25.5
44	Continue drilling 17-1/2" hole to 2182m (13-3/8" Casing point) while pump hi-vis pill periodically.	Drilling	82	U-Gu.	2182	6.0	15.0	0.63	23.8	0.7	16.5	0.69	26.2
45	Take survey	Survey		U-Gu.	2182		2.0	0.08	23.9	0.1	2.2	0.09	26.3
46	Circulate and clean hole, Condition hole prior to run Log, POOH to surface	Hole Cond/Circ		U-Gu.	2182		16.0	0.67	24.5	0.7	17.6	0.73	27.0
47	Run CDR/GR/Caliper log, POOH and L/D log	Log		U-Gu.	2182		13.0	0.54	25.1	0.6	14.3	0.60	27.6
48	R/U casing running equipment, Run 13-3/8" casing as per program to bottom, Rig pump efficiency should be check in trip-tank before cement job.	Csg. Running		U-Gu.	2182		47.0	1.96	27.0	2.2	51.7	2.15	29.7
49	Circulate Mud one casing vol., Perform Cementing, Bump plug w/1500psi for 15min for CIT. (In case, not observing bumping during CMT job, perform CIT with 1500psi /15min after D/O 7m CMT above F.shoe)	CMT		U-Gu.	2182		11.0	0.46	27.5	0.5	12.1	0.50	30.3
50	Set 13-3/8" casing slip and test to 800-1000psi, N/D BOP, Cut and L/D landing Joint, N/U 20-3/4" 3000 x 13-5/8" 5000psi C.H.S., inject and pressure test p-seals and cavity with maximum 80% of collapse pressure rating.(1800 psi)	well Head		U-Gu.	2182		24.0	1.00	28.5	1.1	26.4	1.10	31.4
51	N/U 13-5/8" 10000psi BOP's stack meanwhile fill 13-3/8" casing annulus (Top Job) if required, perform BOPs test as NIOC policy. (Set 4-1/2" * 7" Rams in top Rams)	N/U BOP & BOP TEST		U-Gu.	2182		36.0	1.50	30.0	1.7	39.6	1.65	33.0
52	P/UP 12-1/4" bit#5 & RIH joint by joint , Drill out f.collar and cement to 7m above shoe (In case not bumping plugs during cementing, perform casing integrity test with 1500psi/15min) drill out cement and shoe track, work on junk sub.	D/Out		U-Gu.	2182		22.0	0.92	30.9	1.0	24.2	1.01	34.0
	Total Duration of 17-1/2" Hole Section:							***26.6***	***30.9***			***29.2***	***34.0***
	12-1/4" Hole Section / 9-5/8" Casing												
53	Drilling new formation about 2-3m below shoe.	Drilling	3	U-Gu.	2186	2.0	2.0	0.08	31.0	0.1	2.1	0.09	34.1
54	Work on junk sub, Circulate and clean hole, POOH to Surf.	Cir.		U-Gu.	2186		15.0	0.63	31.6	0.7	16.5	0.69	34.8
55	P/U & M/U New 12 1/4" PDC bit (bit#6) and S.Stab @ 20m, RIH to bottom.	Trip		U-Gu.	2186		13.0	0.54	32.2	0.6	14.3	0.60	35.4
56	Circulate and displace hole with new KCL polymer mud (76 pcf)	Cir.		U-Gu.	2186		5.0	0.21	31.8	0.2	5.5	0.23	35.6
56	Perform FIT (2-5m below shoe: EMW should be equal to max. planned mud weight for next hole section plus 2 pcf, 550 PSI)	Testing		U-Gu.	2186		5.0	0.21	32.4	0.2	5.5	0.23	35.8
57	Drilling 12-1/4" hole to 2350m while adjust MW based on hole condition .Use single-shot every 100m multi-shot survey every 300m to control well deviation in this hole section.	Drilling	164	Tar.	2350	3.0	60.0	2.50	34.9	2.8	66.0	2.75	38.6
58	Take survey	Survey		Tar.	2350		2.0	0.08	35.0	0.1	2.2	0.09	38.7
59	Cont. drlg 12 1/4" hole section to 2450m,while pump hi-vis Pill periodically.	Drilling	100	L-Gurpi	2450	3.0	37.0	1.54	36.5	1.7	40.7	1.70	40.4
60	Cont drlg 12 1/4" hole section to 2550m while pump hi-vis Pill periodically.	Drilling	100	L-Gurpi	2550	3.0	37.0	1.54	38.0	1.7	40.7	1.70	42.1
61	Cont drlg 12 1/4" hole section to 2650m while pump hi-vis Pill periodically.	Drilling	100	Il-La-Sv	2650	3.0	37.0	1.54	39.6	1.7	40.7	1.70	43.8
62	Take survey	Survey		Il-La-Sv	2650		2.0	0.08	39.7	0.1	2.2	0.09	43.9
63	cont drlg 12 1/4" hole section to 2750m while pump hi-vis Pill periodically.	Drilling	100	SV	2750	3.0	37.0	1.54	41.2	1.7	40.7	1.70	45.6
64	cont drlg 12 1/4" hole section to 2850m while pump hi-vis Pill periodically.	Drilling	100	SV	2850	3.0	37.0	1.54	42.7	1.7	40.7	1.70	47.3
65	BOP Test.	BOP Test		SV	2850		6.0	0.25	43.0	0.3	6.6	0.28	47.5
66	cont drlg 12 1/4" hole section to 2950m while pump hi-vis Pill periodically.	Drilling	100	SV	2950	3.0	37.0	1.54	44.5	1.7	40.7	1.70	49.2
67	Take survey	Survey		SV	2950		2.0	0.08	44.6	0.1	2.2	0.09	49.3
68	cont drlg 12 1/4" hole section to 3050m while pump hi-vis Pill periodically.	Drilling	100	SV	3050	3.0	37.0	1.54	46.2	1.7	40.7	1.70	51.0
69	cont drlg 12 1/4" hole section to 3150m while pump hi-vis Pill periodically.	Drilling	100	SV	3150	3.0	37.0	1.54	47.7	1.7	40.7	1.70	52.7
70	CBU, POOH to surf (due to rotating HRS), change BHA & Bit, RIH W/BIT#7 to btm.	Trip		SV	3150		2.0	0.08	47.8	0.1	2.2	0.09	52.8
71	cont drlg 12 1/4" hole section to 3250m while pump hi-vis Pill periodically.	Drilling	100	SV	3250	2.0	56.0	2.33	50.1	2.6	61.6	2.57	55.4

72	Take survey	Survey		SV	3250		2.0	0.08	50.2	0.1	2.2	0.09	55.5
73	cont drlg 12 1/4" hole section to 3350m while pump hi-vis Pill periodically.	Drilling	100	Kz.	3350	2.0	56.0	2.33	52.5	2.6	61.6	2.57	58.0
74	cont drlg 12 1/4" hole section to 3450m while pump hi-vis Pill periodically.	Drilling	#REF!	Kz.	3450	2.0	56.0	2.33	54.9	2.6	61.6	2.57	60.6
75	Perform Wiper Trip as POOH till 3150m & Run back to BTM. Take survey as multi-shot	Trip		Kz.	3450		6.0	0.25	55.1	0.3	6.6	0.28	60.9
76	cont drlg 12 1/4" hole section to 3550m while pump hi-vis Pill periodically.	Drilling	100	Da.	3550	2.0	56.0	2.33	57.5	2.6	61.6	2.57	63.4
77	Take survey	Survey		Da.	3550		2.0	0.08	57.5	0.1	2.2	0.09	63.5
78	BOP Test.	BOP Test		Da.	3550		6.0	0.25	57.8	0.3	6.6	0.28	63.8
79	cont drlg 12 1/4" hole section to 3650m while pump hi-vis Pill periodically.	Drilling	100	Da/Ga.	3650	2.0	56.0	2.33	60.1	2.6	61.6	2.57	66.4
80	cont drlg 12 1/4" hole section to 3750m while pump hi-vis Pill periodically.	Drilling	#REF!	Ga.	3750	2.0	56.0	2.33	62.5	2.6	61.6	2.57	68.9
81	Perform Wiper Trip as POOH till 3450m & Run back to BTM. Take survey as multi-shot	Trip		Ga.	3750		6.0	0.25	62.7	0.3	6.6	0.28	69.2
82	cont drlg 12 1/4" hole section to 3850m while pump hi-vis Pill periodically.	Drilling	100	Ga.	3850	2.0	56.0	2.33	65.0	2.6	61.6	2.57	71.8
83	Take survey	Survey		Ga.	3850		2.0	0.08	65.1	0.1	2.2	0.09	71.9
84	cont drlg 12 1/4" hole section to 3915 m(section TD) while pump hi-vis Pill periodically.	Drilling	65	Ga-FH.	3915	2.0	36.0	1.50	66.6	1.7	39.6	1.65	73.5
85	Take survey	Survey		FH.	3915		2.0	0.08	66.7	0.1	2.2	0.09	73.6
86	Circulate and clean hole, Condition hole prior to logging. POOH to surface.	Trip		FH.	3915		40.0	1.67	68.4	1.8	44.0	1.83	75.4
87	R/U logging equipment, run Full Suite logs & Deviation tools as per program. POOH, L/D logging equipment.	Log		FH.	3915		45.0	1.88	70.2	2.1	49.5	2.06	77.5
88	R/U casing running equipment.Run 9-5/8" casing to bottom, rig pump efficiency should be check in trip-tank before cement job.	Csg. Running		FH.	3915		45.0	1.88	72.1	2.1	49.5	2.06	79.6
89	Circulate one casing vol. mud, L/U CMT line & pressure test, Perform Cementing casing and displacing. in case of observed bumping; perform CIT With 3000psi/15 Min. (Max 80% Casing Burst) Record and Report the final pressure.(In case of not observing bumping during CMT job, perform CIT with 3000psi after D/O 7m CMT above F.shoe)	CMT		FH.	3915		17.0	0.71	72.8	0.8	18.7	0.78	80.3
90	Hang up BOPs. Set the 9-5/8" casing slip & Pressure test same w/ 800-1000psi. (open side outlet valves while testing), Cut and L/D landing Joint, N/U 13 5/8"-5000 x 11"-10000 T.H.S Pack & test p-seals & cavity w/ max 80% of CSG collapse pressure.(3800 psi)	well Head		FH.	3915		24.0	1.00	73.8	1.1	26.4	1.10	81.4
91	N/U 13-5/8" x 10000 BOPs stack meanwhile fill the 9-5/8" casing annulus if required, Perform BOP's test as per NIOC Policy	N/U BOP & BOP Test		FH.	3915		24.0	1.00	74.8	1.1	26.4	1.10	82.5
92	Drill out shoe track with 8-1/2" bit to 7m above shoe, in case not bumping plugs during cementing; perform casing integrity test with 3000psi, cont drill out shoe track and rat hole,	D/Out		FH.	3915		9.0	0.38	75.2	0.4	9.9	0.41	83.0
	Total Duration of 12-1/4" Hole Section:							***44.5***	***75.2***		***48.9***	***83.0***	
	8-1/2" Hole Section / 7" Liner												
93	Drilling new formation about 3-5m below shoe	Drilling	3	FH	3918	2	2.0	0.08	75.3	0.1	2.1	0.09	83.0
94	Work on junk sub, Circulate and clean hole, POOH to Surf.	Trip		FH	3918		20.0	0.83	76.1	0.9	22.0	0.92	84.0
95	P/U & M/U New 8 1/2" PDC bit (bit#9) and S. Stab @ 20m, RIH to bottom.	Trip		FH	3918		18.0	0.75	76.9	0.8	19.8	0.83	84.8
96	Displace hole with KCL Polymer mud (100 pcf),	Hole Cond/Circ		FH	3918		5.0	0.21	77.1	0.2	5.5	0.23	85.0
97	Perform FIT. (EMW (720 psi) should be equal to max. planned mud weight for 8-1/2" hole section plus 2 PCF),	Testing		FH	3918		5.0	0.21	77.3	0.2	5.5	0.23	85.2
98	Drilling 8-1/2" hole to 4180m (TD) While use single-shot every 100m, multi-shot survey every 300m to control well deviation in this hole section.	Drilling	262	FH-5b	4180	3	98.0	4.08	81.2	4.5	107.8	4.49	89.5
99	Circualte clean hole, perform condition trip f/4180 to 3915m & run back to BTM, C.B.U and POOH to Surface.	Trip		FH-5b	4180		27.0	1.13	82.3	1.2	29.7	1.24	90.7
100	BOP Test.	BOP Test		FH-5b	4180		6.0	0.25	82.5	0.3	6.6	0.28	91.0
101	Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (Full Suite/CDR/GR Logs,), R/D Logging Eq., change Ram.	Log		FH-5b	4180		54.0	2.25	84.8	2.5	59.4	2.48	93.5

102	RIH w/ 8-1/2" PDC Bit BHA (Without MWD), condition hole prior to 7" Liner running (RIH to Btm, CBU, Spot pill, POOH to surface)	Hole Cond/Circ		FH-5b	4180		47.0	1.96	86.7	2.2	51.7	2.15	95.6
103	R/U 7" Liner running Eq., Run 7" liner (+/- 365m LNR) to Btm @ 4180m, R/U CMT lines, Set L.Hanger +/- 3815m (100m overlap), release L.Hanger setting tool.	CSG		FH-5b	4180		40.0	1.67	88.4	1.8	44.0	1.83	97.5
104	Cement 7" Liner as per program.. perform liner integrity test w/ 3000 psi for 15 min while cementing.	CMT		FH-5b	4180		10.0	0.42	88.8	0.5	11.0	0.46	97.9
105	POOH w/ L.Hanger setting tool to 3500m, Flush string & CBU, POOH to 3300m.	Trip		FH-5b	4180		7.0	0.29	89.1	0.3	7.7	0.32	98.3
106	Open the Well. Cont' POOH to surface & L/D L.Hanger setting tool.	Trip		FH-5b	4180		13.0	0.54	89.7	0.6	14.3	0.60	98.9
107	P/U & M/U 8-1/2" MT Bit (8RR1) BHA, RIH to L.Lap @ +/- 3815m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH.	D/Out		FH-5b	4180		45.0	1.88	91.5	2.1	49.5	2.06	100.9
108	P/U & M/U 5-7/8" MT Bit(bit#10) BHA, RIH to tag POB, D/Out POB, RIH to LDC depth while D/Out any cement, work on junk sub and CBU. Perform 7" Liner CIT. (80% of Burst press) with 3000 psi,for 10 min, (If not obs bumping). POOH to top of Liner Lap @ +/- 3815m, CBU, perform Liner lap positive test w/ 2000 psi, POOH to Surface. based on client approval for dry test 7" Liner lap foloow contingency plan#2	D/Out		FH-5b	4180		45.0	1.88	93.4	2.1	49.5	2.06	103.0
109	BOP Test.	BOP Test		FH-5b	4180		7.0	0.29	93.7	0.3	7.7	0.32	103.3
110	Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform CMT quality Logging of 7" Liner & 9-5/8" Casing from 7" liner LDC depth to Surface. (SBT/VDL/CCL/GR), R/D Logging Eq, change Ram.	Log		FH-5b	4180		21.0	0.88	94.6	1.0	23.1	0.96	104.3
111	Hold pre-job safety meeting.Rig Up WL & Pressure Test,RIH CCL & Dummy Gun,POOH & Rig Down.	Completion		FH-5b	4180		13.5	0.56	95.1	0.6	14.9	0.62	104.9
112	Rig Up for Perforation,RIH String (Firing head & shooting ,CCL+Guns),Correlate Depth,Shooting the gun,POOH String & RIH again to Perforate all the Depths then Rig Down	Completion		FH-5b	4180		96.0	4.00	99.1	4.4	105.6	4.40	109.3
113	Rig Up WL & Pressure Test,RIH CCL & Dummy Gun to cross checking perforated intervals,POOH & Rig Down.	Completion		FH-5b	4180		13.0	0.54	99.7	0.6	14.3	0.60	109.9
114	Change Ram, Install shooting nipple, R/U WL Eq. RIH WL with CCL/GR Log to confirm Perforated intervals, POOH and R/D Logging Eq, change	Completion		FH-5b	4180		10.0	0.42	100.1	0.5	11.0	0.46	110.3
115	P/U and M/U 5-7/8" Mill and 7" Scraper. RIH to Perforation depth while Scrap 7" PKR setting depth interval, CBU. Displace well volume with CaBr2 Completion fluid.	Completion		FH-5b	4180		36.0	1.50	101.6	1.7	39.6	1.65	112.0
116	POOH to surface while L/D all Tubulars.	Completion		FH-5b	4180		63.0	2.63	104.2	2.9	69.3	2.89	114.9
117	BOP Test.	BOP TEST		FH-5b	4180		6.0	0.25	104.5	0.3	6.6	0.28	115.2
118	R/U Completion running Eq, Run 4-1/2" * 3-1/2", 10K Completion string as per program, R/D completion running Eq.	Completion		FH-5b	4180		140.0	5.83	110.3	6.4	154.0	6.42	121.6
119	Install BPV in TBG Hanger, N/D BOPs.N/U X-mass Tree, perform required test as per program,	Completion		FH-5b	4180		40.0	1.67	112.0	1.8	44.0	1.83	123.4
120	Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion		FH-5b	4180		10.0	0.42	112.4	0.5	11.0	0.46	123.9
121	R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
122	R/U CTU, Stimulation and Well Testing Eq. RIH w/ CT to Btm, displace the well with Diesel, POOH CT to surface.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
123	R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
124	Flow the well as Pre-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 72 hrs.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
125	R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
126	RIH w/ CT and perform Acid Wash as per program, POOH CT to surface.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
127	R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
128	Flow the well as Post-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 96 hrs.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
129	R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
130	RIH w/ CT to Btm, displace well with Diesel, POOH CT to surface, R/D CTU and well Testing Eq.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
131	R/U Slick Line, RIH w/ SL and Install Plug and Prong into NO-GO L.NPL, POOH.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
132	RIH w/ SL and Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
133	Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion		FH-5b	4180		Rig Less	0.00	112.4	0.0	0.0	0.00	123.9
	Total Duration of 8-1/2" Hole Section:							***37.4***	***112.4***			***41.1***	***123.9***
	NOTE:												
134	Total Duration for Cut & Slip DL in all hole section.	Cut & Slip		FH	4180		12.0	0.50	112.9	0.6	13.2	0.55	124.4
135	Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service		FH	4180		26.0	1.08	114.0	1.2	28.6	1.19	125.6
	Total Duration of F-20 Well:			**FH**	**4180**				**114.0**				**125.6**
	Contingency Plans TBD of Well: F-20	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency	Time Duration w/ contingency	Total Duration w/ contingency (Days)
	Based on operation situation in each step, the following contingency plans might be followed												
136	Contingency plan #1 POOH to Surafce to change PDC Bit, P/U & M/U TCI Bit & RIH to	Drilling		FH-5b	4180		115.0	4.79	4.8	5.3	126.5	5.27	5.3
137	Contingency plan #2 Perform scraper trip, RIH with 9-5/8" Dry test BHA, RIH to L.Lap @ +/- 3815m, Perform 7" Liner lap Dry Test (based on fahliyan Reservoir Datum Pressure), POOH.	Testing		FH-5b	4180		429.0	17.88	17.9	19.7	471.9	19.66	24.9
	Total Duration of Contingency Plan F-20 Well:								**22.7**				**24.9**

3. Sequence of Drilling Operation

3.1 Rig Move

- Hold pre-move meeting on location. Discuss contingency plans for:
  - Safe arrangements for rig up and rig down.
  - Safe standby areas for rig
  - In case move have to be suspended.
  - Loss of trucks in case of repair.
  - Moving at night.
- Hold pre-move safety meeting. Ensure everyone understands individual responsibilities:
- Commence rigging down. The ordering of trucks will be done by rig manager.
- Commence rig move to F20, PAD NO. 93
  - Position rig for optimal operations
- Rig up on cellar
  - Rig up and commence operational procedure for (F20)
  - Before leaving previous location, all equipment should be checked.
- Test all manifold and surface safety valves
  - RKB – GLE and Elevation
  - Spud date and time.
- Mud tanks should be empty during move. After moving Fill water tanks and keep as full as possible.

3.2 Pre Spud

3.2.1 Certificate

- All rig equipment is on location, properly rigged up and function-tested. Check availability and content of hoisting equipment inspection & maintenance records according to API.
- All rig equipment including all Tubulars, Subs, Tools & … should be inspected & according to API.

3.2.2 Safety Device

- Test draw-works crown block saver and record results on daily drilling report; test derrick man escape device (Geronimo line) and record time required for test weight on daily drilling report; test rig intercom system, especially for Driller-Derrick Man and Driller-Mud Logger communication link.
- Check that all rig safety equipment, e.g., fire extinguishers, eye-wash stations etc., is installed at the proper locations as per rig layout diagrams.
- Verify that substructure is level and foundation adequate to support drilling unit during whole well period and that mast is centered above rotary table.
- Ensure rig alignment with Top Drive System. Make sure rig tongs and rig floor hoisting equipment are adequate for handling 8-1/2″ and 9-1/2″ drill collars. Make sure that mechanical tong torque gauges for drill pipe/drill collar are properly functioning and have been calibrated; re-calibrated as required.

3.2.3 Rig Equipment

- Verify that the 26″ tri-cone bit is ready to be picked up, all nozzles installed, and 26″ bit breakers are on the rig floor. Make sure single-shot or multi-shot survey tools available for vertical section survey.
- Make sure all solids control equipment are correctly installed and have been function-tested with charging pressure gauges in place where applicable. Make sure adequate screen inventory for the shale shakers is available in the sizes required for the well. Make sure water tanks are completely filled and transfer lines from water tanks to mud system are hooked up. Make sure all mud materials and chemicals required to drill 26″ section is on location, in sound condition and suitably stored. Make sure that bentonite is pre-hydrated and sufficient volume has been built for top section and for well-kill.
- Make sure that all mud pumps have been fitted with new consumable parts: liners, pistons, valves, seats etc. Make sure that pulsation dampeners are charged with Nitrogen to the proper pre-charge pressure, 30% of expected standpipe pressure during drilling of the 26″ hole section; check if sufficient Nitrogen bottles are available on location to adjust pulsation dampener pre-charge pressure as required. Make sure that mud pump pop-off valves are set at the proper pressure for the liner sizes installed and that the mud discharge lines from the pop-off valves to the mud tanks are properly secured. Check that all high-pressure hoses are properly secured with safety chains. P/T should be done to high-pressure surface line of circulating system.

3.2.4 **Wellhead & BOP**

- Have 2″ line pipe nipple with 2″ ball valve welded to 30″ conductor pipe 0.2m above cellar bottom to drain cement from conductor/riser pipe after cementing 20″ surface casing. Verify that pre-installed 30″ conductor pipe is vertical and that no angle/bend exists between conductor and 30″ riser pipe.
- Make sure that adequate supply of 20″ casing is on location, inspected and ready to be run, float shoe should already be made up, centralizers should be pre-installed on the pipe-rack prior to spud in the well.
- Ensure mud logging unit is functioning and equipment are calibrated and tested properly.

3.2.5 **Others**

- Verify that the rig's fuel tanks are full to meet minimum of three days rig consumption available on site.
- Make sure all waste treatment equipment available and commissioning test is ok.
- Hold pre-spud meeting with relevant service Company personnel on location. Discuss procedures involved throughout 26″ hole section drilling, reporting system and critical safety operation measures.
- OEOC prepared a pre-spud checklist and distributed it in rigs. The checklist will be performed based on the aforementioned checklist. In addition to the checklist, following matters should be checked:

- Check & observe the alignment of the rig. This check will be performed in four stages; rig levelling, derrick verticality, derrick alignment with rotary table, rotary table alignment with conductor pipe. To do this pick up 3 joints of 9 3/4" DC and check the verticality of the derrick.
- Mechanical facilities for mud treatment such as:

  1. Desander, Desilter, (Degasser, mud centrifuge), mud cleaner, proper size of shaker–screens, hoppers, mud guns & mixers, should be all available on rig and functioning properly.
  2. All mud tanks, mud transfer lines & connection should be checked by filling them with water & cross wellhead circulation.
  3. Mud engineer laboratory should be equipped with a complete set of testing equipment. The calibration should be done under supervision of well site drilling supervisor before starting the operation.
  4. Weight indicator, rotary speed gauge, any smart drilling instrumentation, Top Drive, Stroke meter, pressure gauges on manifold and in the drilling console, loss / gain gauge and any other gauges should be checked and calibrated.

5. (Grand manifold valves & related lines), Stand pipe manifold (lines) valves & connection must be checked & pressure tested.
6. All motors, (SCR & VFD) should be checked under load prior to start drilling.
7. Check & control mud stock & cement stock every day & if there is any necessity to request mud material, cement or any drilling tool, it should be prepared & requested at proper time.

Note: Fill up following forms and send to drilling office:

- Pre-spud check list
- Location Handed Over

3.3 Drilling

3.3.1 26" Hole Section, 20" Casing

After commencing spud-in, 26" hole drilling will be started. The drilling would be anticipated to go on fast.

Casing Point	20" Casing @ 150 mMD	
Formation	Aghajari	
Risks & Hole problems	Bit balled up	
Deviation control	INCLINOMETER (TOTCO)	
Logging	---	
Bit Type	R. Bit (1-1-5)	
Drilling Fluid	Pre-Hydrated Bentonite mud	
	MW	65-67
	PV	5-18 CP
	Vis.	>50
Drilling Parameters	WOB	10-30
	GPM	900-1000
	RPM	Max available

3.3.1.1 Sequence of Operation

- Make up 26" BHA as per BHA design. Tag formation at ±14m.
- Drill ahead formation down to 150 m (20" casing point in Aghajari formation) using Hi-vis bentonite pill. (pump Hi-Vis Pill Periodically) Pay enough attention to drill straight hole.

Note:

In Case of complete loss while drilling 26" hole Cure by pump LCM pill or cement as cement plug.

- TD should be adjusted to have 1m of rat hole below casing shoe.
- The exact TD will be determined based on casing tally.
- Run Inclinometer (Totco) each 50m to evaluate the verticality of the hole.
- Ensure that all drilling tools and equipment are on well site and are in serviceable condition.
- Ensure that adequate fishing tools for this section of hole are available.
- Enough Mud Material, LCM and cement should be available on rig site to use in case of needing.

- At TD, Circulate and clean the hole. Spot high viscosity pill at bottom, drop Inclinometer (Totco) POOH to surface. Check Bit.
- Perform hole condition prior to run casing. If observe any indication of drag, run back to bottom. Wash and ream to make sure that the hole condition is good.
- Perform safety meeting.
- R/U casing running equipment.
- Run ±150m (Approx. 13 Jts) 20", 94 ppf, K-55, BTC casing with stab-in duplex shoe to bottom as per casing program. (function test of F.shoe should be checked on surface by WSS). Casing shoe should be adjusted to have 1m of rat hole and enough stick up (1 m on rig floor). Fill casing by non-LCM fresh mud.

Note:

- Take care when running casing not to set down excessive weight if tight hole or ledges are encountered.
- Run stinger on DP's using the "C" Plate & 5" x 20" centralizer. Sting in. M/U & test cementing lines. Circulate for one casing annulus volume. While circulating, check the DPs annulus for any return. Use all DPs weight on the string if observing any minor leakage through DPs annulus.
- Cement casing as per program. Any minor adjustment regarding the corrected depth will be done in due time. Displace with mud (1-2 bbls Short Displace). Monitor for pressures. Record and report the final pressure, Sting out. Check the float. POOH one stand, Circulate to flush/clean drill pipe.
- After cement returns to cellar floor, open the 2" nipple on conductor & flush it.
- POOH to surface. L/D stinger.
- Keep the casing on the elevator until cement surface samples get hard. Check for casing to be center inside conductor pipe.

Note:

- Based on the results obtained from cementing, decision will be made to perform top jobs.
- If there is no cement return to surface, a top job should be performed to fill the annulus. Use same slurry weight as the primary cement slurry. Ensure the availability of adequate X-over for connecting line pipe to cement line. The cement should be accelerated with max. (2%) Calcium Chloride BWOC and no other additives. Wait for the top job cement to get hard.
- Check casing by slack off weight.
- Cut conductor pipe. Rough cut landing joint and L/D C.P. and L.J. Dress and bevel 20" suitable to install and to weld casing head housing.
- Install and weld Casing Head Housing 20 ¾" – 3000 psi on casing and pressure test the sealing of welded casing. (Limit pressure to a maximum of 80% of the collapse strength of the casing, recommended testing pressure: 400 psi). Outlets must be aligning with cellar stairs.
- Before installing the casing head housing, open all the test ports and clean inside it with water and air.
- Check the 20 ¾" wear bushing, 20 ¾" Plug Tester and 13-3/8" Casing Slip and seal assembly in CHH before installing CHH (on Surface).
- N/U 21 ¼"-3000 BOP stack and related lines and valves and test all to 2400psi.
- N/U Bell Nipple & flow line
- P/U & M/U Next BHA and stand drill pipes to 2182m.
- Fill up the following forms and send to drilling office:

  Casing Tally
  BHA reports
  Bit Reports
  Casing Report
  Cement Report

Wellhead diagram
Hole Section Summary for EOWR

3.3.2 17 ½" Hole Section, 13 3/8" Casing

3.3.2.1 General

Casing Point	13-3/8" Casing @ ±2182m (5m TVD below top of Gurpi FM)	
Formation	Aghajari, Gachsaran, Asmari, Pabdeh or Pd./Ja., Jahrum, Pabdeh, Upper Gurpi	
Risks & Hole problems	Tight holes, wash & ream (especially in pipe connections), minor stucks and bit balling, Hole Deviation, Wahsout & chert existence inside Pd. Fm.	
Deviation control	Single shot each 100m, multi shot in 300m	
Logging	N/A	
Bit Type	Tri Cone Bit (115,537,517), PDC Bit	
Drilling Fluid	Pre-hydrated Bentonite Mud and KCL-Polymer Mud (3-5 % KCL)	
	MW	67-88pcf
	PV	5-27
	Vis.	40-50
Drilling Parameters	WOB	Maximum Available
	RPM	120-180
	GPM	800-1000

3.3.2.2 Formation Sampling

Every 10 meters a sample would be to section TD.
NOTE: In addition to this general rule, geologist may decide to take more samples.

3.3.2.3 Sequence of Operation

- Make up 17-1/2" BHA as per BHA Design. RIH, tag top of cement, Drill out cement & shoe and rat hole cement. Work on junk sub.
- Watch carefully for any indication of abnormal torque, since there might be a possibility of shoe being backed off. Report problem immediately.
- Continue drilling 3m below shoe. Clean out the rat hole, circulate clean.
- Drill ahead formation down to 25m below 20" casing shoe with hi-vis pill. (Pump Hi-Vis pill Periodically)

Note:
If use stabilizer to drill out shoe track, make sure the RPM & WOB are in low as possible until the stabilizer is out from shoe Then Cont. Drilling 17-1/2" hole section with same BHA.

- Circulate and clean hole, POOH to surface and P/U the same bit & S. Stab @ 20m according to BHA program, RIH to bottom.

- Drill ahead 17-1/2" hole section down to 750 m. CBU.
- Condition trip to 150m, run back and displace mud in hole by 74 pcf KCL/Polymer.
- Continue drilling to 850m or End of Bit Life, circ to clean hole, POOH to surface.
- P/U new PDC bit and Drill ahead 17-1/2" hole section down to 1750 m, 5m meanwhile perform condition trip each 300 m. If observe Chert particle while formation sampling, Follow Contingency Plan#01
- Cont. Drill ahead 17-1/2" hole section down to casing point @ ±2182m, 5m TVD below top of Gurpi formation meanwhile perform condition trip each 300 m.

Note: In case of observe Cherty zone (Pabdeh formation), Stop Drilling, circulate to clean the hole, POOH to surface and change PDC bit by Tri Cone Bit.

Note:

- For this well Single-shot every 100m and multi-shot survey every 300m to Control Well Deviation or take gyro shot (5-6 shots each 500-600m)
- Adjust mud weight according to program and as hole condition dictate.
- Tight holes, minor stucks and bit balling are hole problems which happen in offset wells.
- TD should be adjusted in such a way to have 1 meters of rat hole for probable fillings, before running casing.
- Choke manifold and choke line must be lined up for SOFT SHUT-IN through pit or burning line.
- Ensure that adequate fishing tools for this section are available.
- Ensure that enough amount and type of LCM are available and keep ready to mix and pump LCM Pills to cure losses.

- At section TD, circulate, clean hole. Perform wiper condition trip to shoe and observe well @ shoe. RIH (back to BTM, check for any drag, CBU), Check for any fill. CBU on bottom. Spot 100 bbl. hi-vis pill @ bottom. POOH to surface
- If observed any indication of tight hole, run back to TD. W&R to make sure that the hole condition is good prior to run Log.
- Rig up Wireline logging equipment and run CDR/GR /CAL log as per program, POOH and L/D log
- Perform pre-job safety meeting. R/U casing running equipment. Run casing 68#, N-80, BTC with F. shoe below first joint & F. collar as per casing program (function test of F. shoe & F. Collar should be checked on surface) while Circ. BTM's up at shoe with low GPM.
- Rig pump efficiency should be check in trip-tank before cement job.
- Casing shoe depth should be adjusted in such a way to have 1 meters of rat hole and enough stick up (1.5 m) on rig floor.
- When casing is at casing setting depth circulate while preparing for cementing M/U & test cementing lines. Circulate for 120% of casing annulus volume.
- Cement casing as per program. Any minor adjustment regarding the corrected depth will be done in due time. Displace with mud. Monitor for pressures.
- Bump plug with 1500 psi (Max 80% Casing Burst) / 15 min as CIT (In case of not observing bumping during CMT job, perform CIT W/1500psi (Max 80% of 13-3/8" Casing Burst)/15min with consider mud weight after D/O CMT 7m above F. Shoe also consider MW while perform CIT), Record and report the final pressure, Bleed off pressure, check for back flow, if any back flow observed, hold the pressure inside casing until the cement get hard.
- H/U 20-3/*4" BOP Hold on casing string weight. Drop casing hanger. Slack off 80% of casing string weight. Test hanger to 800-1000 psi pressure. (Open base flange side outlet valves while testing).
- , Rough cut L.J. & L/D same, N/D 20-3/4" BOP, Dress and bevel casing neck.
- Install Casing head spool 20 ¾"-3000*13 ⅝"-5000, Pack p-seals and pressure test p-seals and cavity with maximum 80% of rated casing collapse pressure. (1800 psi)
- Ensure that the casing is drifted; cleaned and tallied priors to RIH.
- Fill casing by non-LCM fresh mud.

- Use Bow Type centralizer each 4 jts in open-hole and at least 2 no. Rigid type centralizer in cased-hole. (In shale/sticky Marl layers use bow types each 10 jts)

Note: Final centralizer arrangement will be issued with final cementing program.

- Based on the results obtained from cementing, decision will be made to perform top-up jobs.
- Before installing the casing head spool, open all the test ports and clean inside it with water and air.
- Check the 13-5/8" wear bushing, 13-5/8" Plug Tester and 9-5/8" Casing Slip in CHS and adopter flange before installing CHS (on Surface).
- N/U 13 5/8" x 10000 psi BOP stack. N/U bell nipple. Test BOP stack and related lines to 8000psi pressure, using plug tester.
- L/D 9 3/4" drill collars, P/U & M/U Next BHA and stand drill pipes.
- Fill up the following forms and send them to drilling office:

BHA Report
Bit Report
13-3/8" Casing Tally,
13-3/8" Casing Report
13-3/8" Cement Report
Hole Section Summary
Wellhead diagram along with wellhead installation report
BOP stack Schematics with all related dimensions
BOP test Report
CDAS Data
CIT Graph

3.3.2.4 Logging Program

N/A

3.3.3 12 ¼" Hole Section, 9 -5/8" Casing

3.3.3.1 General

Casing Point	9-5/8" Casing @±3915m(top of Fahliyan)	
Formation	Upper Gurpi, Tarbur, L. Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan, Top of Fahlian	
Risks & Hole problems	Tight Hole & Stuck, Wash & Ream, Hole Deviation,	
Deviation control	Single shot every 100m, and multi shot every 300m	
Bit Type & Nozzle	217,537, PDC	
Logging	Full Suite logs/ CDR (Open hole)	
	SBT/VDL/GR/CCL (Cased Hole)	
Drilling Fluid	KCL Polymer Mud	
	MW	71-85
	PV	10-25
	Vis.	45-60
Drilling Parameters	WOB	15-40
	RPM	80-160
	GPM	500-750

3.3.3.2 Formation Sampling

Sample would be taken F/Gurpi to Laffan Every 5 meters & from Sarvak to Gadvan Every 2 Mmd
NOTE: In addition to this general rule, geologist may decide to take more samples.

3.3.3.3 Sequence of Operations

- Make up 12-1/4" BHA as per BHA design include Junk Sub. RIH, Tag top of cement. Drill out plugs, collar, drill out F. collar and cement to 7m above shoe (In case of observe no bumping during cementing perform casing integrity test with 1500 psi/15min) drill out cement and shoe track (while RIH pick up and rack back 500m 5"DP.)

Note: if shoe and float collar are PDC drillable, use PDC bit with stabilizer at first trip without junk sub and take care while drill shoe track to avoid bit damage.

- Drill rat hole to 3-5m below shoe. Clean out the rat hole, circulate clean. Work on junk sub. POOH to surface.
- R/U Logging Equipment, Run GR/CCL/CBL/VDL Log, POOH To Surface. L/D Logging Equipment.
- RIH W/12-1/4" PDC Bit and BHA to BTM
- Displace hole with new mud ±76 pcf. Perform FIT (3-5m below shoe: EMW (550 psi) should be equal to max. planned mud weight for next hole section plus 2 pcf),
- Resume DRLG 12-1/4" Hole Section to 3915m (9-5/8" CSG point top of Fahliyan) while perform condition trip each 300 m.

Note: use single shot each 100m and multi shot each 300m to Control Well Deviation or take gyro shot (5-6 shots each 500-600m)

Note:

- Pre-plan for different loss control conditions including LCM materials in different sizes & cement plug.
- BOP's test every week. (every 2 weeks)
- Choke line and choke manifold must be lined up for **Soft SHUT-IN**.
- Adjust mud weight according to nearby wells and as hole condition dictation.
- The Section TD maybe changed based on formation tops detected by the wellsite geologist.
- The mud engineer shall check the mud properties as pre-planned time schedule.
- Ensure that all drilling tools and equipment are on well site in serviceable condition.
- Ensure that adequate fishing tools are available for this section.
- Ensure that enough amounts of LCM are available at location.

- At TD, circulate and clean the hole prior to condition trip. Observe the hole for one hour @ casing shoe. RIH to bottom and Pump 50 bbl hi-vis pill, circulate to clean hole (check bottoms up for any influx), Spot 70 bbl hi-vis Pill prior to POOH.
- It's necessary to control drilling fluid rheology and especially solid content. It's practically proven that excessive solid percent can reduce ROP significantly and cause major drilling difficulties. So, it's necessary to check solid control equipment's and shaker screen inventory of the rig prior to drilling this hole section.
- POOH to surface. If observed any indication of drag, run back to bottom. Wash and ream to make sure that the hole condition is good prior run log.
- R/U wireline logging equipment. Run Wire line Loggin (Full Suite logs/ Deviation tool/MDT) as program. R/D wireline.
- RIH and condition hole prior to run casing and spot hi-vis pill.
- Change top rams to 9-5/8".
- R/U casing running equipment. Perform pre-job safety meeting.
- Use Torque Monitoring System (TMS).
- Run casing as per casing program. Run 9-5/8" casing 53.5#, 110SS, Permium Gas tight Connection with F. shoe below first joint & F. collar, as per casing program (function of float shoe and float collor should be ckecked on surface)while Circ. BTM's up at shoe with low GPM.

- Rig pump efficiency should be check in trip-tank before cement job.
- Casing shoe depth should be adjusted for having short rat hole and enough stick up (1.5 m) on rig floor. Fill casing by non-LCM fresh mud each 10 joints.
- When casing is at casing setting depth circulate while preparing for cementing M/U & test cementing lines. Circulate for 120% of casing annulus volume.
- M/U cementing head. Load the plug. WSS shall witness the loading. Test the lines to 4000psi for 10 minutes. Make sure that the cement head is capable of holding pressure while displacing and testing cement (Visually).
- Perform cementing as per program. Any minor adjustment of cement volume in regards to correction of depth will be done in due time. Displace with mud. Monitor for pressures. Record and report bumping pressure in case of observed bumping; perform CIT With 3000psi/15 Min. (in case of bumping with max 80% casing burst), Record and Report the final pressure. (In case of plug was not bumped during CMT job, D/O cement to 7m above F. shoe & perform CIT with 3000psi (Max 80% of 9-5/8" Casing Burst)/ 15min and also consider MW while perform CIT). Bleed off pressure. Check for back flow. If any back flow observed, hold the pressure inside casing until the cement get hard.
- "Any minor adjustment of cement volume in regards to correction of depth will be done in due time." Monitor for pressures. Record and report bumping pressure. Bleed off pressure. Check for back flow. If any back flow observed, hold the pressure inside casing until the cement is hard.
- Hold on casing string weight. Drop casing hanger. Slack off 80% of casing string weight. Test hanger to 800-1000 psi pressure. (Open side outlet valves while testing).
- H/U BOP. Rough cut landing joint. L/D same. Dress and bevel casing neck.
- Install T.H.S. 13 5/8"-5000 PSI x 11"–10000 PSI. Pack p-seals and pressure test p-seals and cavity with maximum 80% of rated collapse pressure. (3800 psi)

Note:

- Before installing the casing head spool, open all the test ports and clean inside it with water and air.
- Use Bow Type centralizer each 3 jts in open-hole and at least 4 no. Rigid type centralizer in cased-hole. (In shale/sticky Marl layers use bow types each 10 jts)
- During Casing running TMS (Torque Monitoring System) shall be used.
- N/U bell nipple. Test BOP stack and related lines to 8000 psi pressure, using plug tester.
- L/D 8-1/2" drill collars, P/U & M/U Next BHA and stand drill pipes.
- Fill up the following forms and send them to drilling office:

BHA Report
Bit Report
9-5/8" Casing Tally
9-5/8" Casing Report
9-5/8" Cement Report
12-1/4" Hole Section Summary
Wellhead diagram along with wellhead installation report
BOP stack Schematics with all related dimensions
BOP test Report
TMS graph
CDAS data & Graph
FIT chart
CIT Chart

3.3.3.4 Logging Program

Open Hole: Full Suite logs/ CDR
Cased hole: SBT/VDL/GR/CCL

3.3.4 8 ½” Hole Section, 7” Liner

3.3.4.1 General

Liner Point	7” Liner Piont @ 4180 mMD	
Formation	Fahliyan	
Risks & Hole problems	Tight hole and stuck pipe	
Logging	Full Suite logs/ CDR (Open hole)	
	SBT/VDL/CCL/GR (Cased Hole)	
Bit Type & Nozzle	Tri cone 2-1-4, 5-2-7, PDC (M1365)	
Drilling Fluid	KCL/NaCl-Polymer mud	
	MW	100-106
	PV	26-46
	Vis.	40-50
Drilling Parameters	WOB	10-30
	RPM	100-130
	GPM	400-600

3.3.4.2 Formation Sampling

NOTE: In addition to this general rule, geologist may decide to take more samples.
Well site geologist is fully responsible to determine the frequency of cutting sampling, and mud logging crew shall take sample based on well site geologist’s advice. In deviated/Horizontal interval the sampling shouldn’t be limited to 20 m, and the well site geologist should take care of sampling and collecting sample of any cuttings with different characteristics.

3.3.4.3 Sequence of Operations

- Pick up 8-1/2” bit and bit sub and junk sub, RIH to Tag top of cement. Drill out plugs, collar, drill out F. collar and cement to 7m above shoe (In case of observe no bumping during cementing perform casing integrity test with 1500 psi/15min) drill out cement and shoe track and shoe.
- Drilling 8 1/2" hole (3-5m) to 3918m.
- Work on junk sub, Circulate and clean hole, POOH to Surf. L/D 8-1/2” bit & junk sub.
- R/U logging equipment, Run GR/CCL/CBL/VDL/SBT, POOH & R/D logging Equip.
- P/U & M/U New 8 1/2" PDC bit (Optional performance) and S. Stab @ 20m, RIH to bottom.
- Circulate and clean hole, displace hole with new mud (100 pcf KCL Polymer Mud), perform FIT. (EMW (720 psi) should be equal to max. planned mud weight for 8-1/2” hole section plus 2 PCF),
- Drilling 8-1/2” vertical Hole Section to 4180m (7” Liner Point) while use single-shot every 100m and multi-shot survey every 300m)
- Perform Condition trip as POOH to 3915m, Observe the well, & run back to BTM, C.B.U, POOH to surface.
- R/U logging equipment, Run Full Suite logs/ Deviation tool/MDT, POOH & R/D logging Equip.
- R/U casing running equipment. Run and hang 7" liner from ±3815 mMD to ±4180mMD, (Hang 100 m above 9 5/8" CSG shoe, procedure of setting LNR will be issued in due time).

- M/U and torque up the shoe-to-shoe joint (32ppf, 110SS, gas tight premium connection. F/U first joint then P/U & M/U second joint of 7" liner, run & F/U same. Break circulation and function test shoe.
- P/U float Collar, clean threads, M/U and use baker lock for its box. M/U 1 joint 7" liner (32ppf, 110SS, gas tight premium connection) above float Collar & run & F/U same, break circulation and check float valve. P/U LDC, clean threads, M/U and use baker lock for its box, add one more 7" liner (32ppf, 110SS, gas tight premium connection), Break circulation and check float valve.
- Run 7" liner as per Tally, (Centralizer should be installed based on 7" LNR tally) F/U all 5 first joints, and then F/U every 5 joint. M/U liner hanger assembly as per job specialist instruction.
- Record pick-up and slack-off weight of liner.
- RIH 1 stand DP. Break & continue circulation as 1.2 cap of liner string. Check liner hanger for leaks.
- R/D liner running equipment
- Continue RIH on DP's while drifting DP's. F/U each 10 stands. With the liner assembly shoe at 9 5/8" casing shoe, break circulation and check free circulation through the liner assembly. Circulate at least 1.2 volume of Liner + Drill Pipe capacity.
- Continue RIH to liner setting depth, wash down last stand to bottom. Tag bottom. Continue circulation for at least 120% of the liner and DP's volume. Space out the string to have cement manifold at proper position.
- Circulation test at following flow rates: 20, 30, 40, 50 SPM, checking for losses, record results in IADC and advice DSV.
- Record pick-up and slack-off weight of liner at bottom.
- Drop ball and set liner hanger as per job specialist instruction. Test hanger setting in place. Shear ball catcher and establish circulation. Release setting tool from liner and make sure that it's free, as per job specialist procedure.

Note: in case if it is not possible to set liner hanger, set liner shoe in TD then release liner hanger setting tools.

- Perform Pre-Job safety meeting with all personnel involved on the operation as well as the client representative. Read and follow the risk assessment for this job, Ensure communication between cement compant and rig floor
- Cement liner as per program. Perform liner integrity test w/ 3000 psi /15 min while cementing (max 80% of liner burst with consider mud weight)
- Pump Chemical wash.
- Pump spacer.
- Pump cement slurry.
- Release pump down plug.
- Pump 3 bbl cement slurry (as a part of displacement).
- Displace cement slurry with mud (decrease flow rate to 4 BPM for last 10 bbl displacement volume).
- POOH to 3350 m (depth will be corrected after finalize CMT program), flush string, Cont. POOH to Surface. L/D setting tools,

- **Note:** LNR running procedure will be issue in due time.

Note:

- Check the rig pump efficiency in order to prevent any kind of failure in the cementing job.
- Surge tank, cement unit, batch mixer and dry cement lines should be checked before the cementing job commencement in presence of well site supervisor, Tool pusher and cement supervisor to prevent any blockage during the cement job. (Get a confirmation from well site supervisor)
- Drill water, spacer, slurry solution and dry cement samples from each silo should be taken during the job and sent to lab for analyzing.
- Check the pit for spacer mixing to be clean, to prevent any contamination.
- Amount of cement and additives to be calculated based on the final cementing program.
- Double check the calculation with the well site supervisor (or well site engineer) before the job.

- Perform risk assessment and discuss contingency plan with the well site supervisor.
- Any deviation to the design (volume, rate, fluid sequence) MUST be notified to the cementing engineer for risk assessment.

➢ P/U & M/U 8-1/2" Bit & new BHA & RIH to top of cement, D/O cement to L. Liner 7", circulation and clean the hole, POOH to surface, L/D 8-1/2" bit.
➢ P/U & M/U 5-7/8" Bit & new BHA (3-1/2" H.W. Dp & 3-1/2" DP) & RIH, drill POB and clean out inside 7" Liner to 7m above LDC, Circ. Clean the hole. (In Case of Observe No Bumping, Perform CIT W/ 3000PSI (Max 80% of liner Burst)/ 15 MIN and consider MW while perform CIT
➢ POOH to 10m Above L. Lap, perform 7" L. Lap positive test (w/ 2000psi), POOH to surface.
➢ Fror dry test, Follow Contingency plan #2
➢ Pressure test BOPs as per NIOC Policy.(change rams before BOP test)
➢ R/U logging equipment, Run GR/CCL/CBL/VDL/SBT, POOH & R/D logging Equip.
➢ R/U Wire line RIH as dummy run
➢ Run Perforation Gun Assembly and perforate 7" Liner.
➢ Run CCL & dummy gun to cross checking perforated intervals,, R/D Logging Equipment.
➢ Pick up 5-7/8" E. Mill and 7 " Scraper & RIH W/ Clean out BHA, Clean inside 7" Liner to TD
➢ Circ. Displace hole w/ completion fluid f/ 10m below Completion PKR depth POOH to surface while L/D All DPs.
➢ Perform BOP test.
➢ Run Completion String as per Program.

Note: DETAILED OF COMPLETION PROGRAM WILL BE ISSUED IN DUE TIME.
Note: DETAILED OF STIMULATION & PERFORATION PROGRAM WILL BE ISSUED IN DUE TIME.

NOTE:

- Rabbit/ drift all liner and strap. Ensure that sufficient number of liners has been delivered to allow for 10% excess. Caliper the ID of 10% of randomly chosen joints and use the average for displacement purposes. Ensure to caliper each joint in both NS and EW directions.
- Check all liner threads. All liner connections cleaned and all liner joints are drifted.
- Gauge the OD of pump down plug and ensure it is compatible with drill pipe tools joint ID.
- Ensure no debris is inside liner joints.
- Ensure that backups of liner hanger, shoe, latch down collar, cementing plugs, XOS, etc. are available on location.
- TMS (Torque Monitor system) shall be used during running 7" liner.
- Section TD maybe changed based on formation tops detected by wellsite geologist.
- Perform BOP Pressure test on every 2 week as per policy.
- Choke line and choke manifold must be lined up for HARD SHUT-IN.
- Ensure that all drilling tools and equipment are on well site and are in serviceable condition.
- Ensure that adequate fishing tools for this section are available.
- Ensure that enough amounts of LCM are available.
- Fill up the following forms and send them to drilling office:

BHA Report
Bit Report
FIT chart
7" Liner Tally -7" Liner Report - 7" Cement Report.
8-1/2" Hole Section Summary.
BOP's test report (each 7 Days)
BOP's Schematics with all related dimensions
Directional drilling reports
Well Schematic

TMS graph
CIT Chart
CDAS Data & Gragh
Final Perforation Report

3.3.4.4 Logging program:

Open hole: Full Suite logs/ Deviation tool/MDT
Cased hole: SBT, CAST or USIT

3.3.4.5 Well Flowing

- The main objectives of the 96 hrs clean up and well flowing is to displace completion fluid, possible mud losses, and flow back the reservoir fluid until achieving the clean up operation criteria. The clean up operation procedure by rig is categorized as follows:
- Rigging up all related-equipment and perform hydro-test
- Retrieving barriers (plug, TWV) and landing protection sleeve in TRSSSV nipple
- Displacing the wellbore via CT with suitable method and compatible fluid
- Flowing the well over 96hrs thru different choke sizes while testing integrity and surface sampling
- Retrieving protection sleeve, inject 5 bbl diesel Close TRSSSV, Set BPV in tubing hanger, secure well & release rig.

Note: Detail Clean Up and Well Flowing Program Will Be Issued in Due Time.

3.3.4.6 Contingency Plan

Contingency Plan #1

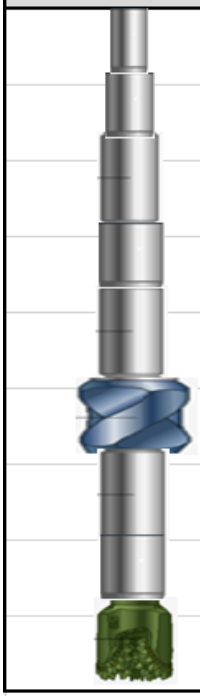
- POOH to Surface to change PDC Bit, P/U & M/U TCI Bit & RIH to Previous Depth, Perform Wiper Trip & Run back to BTM. Take survey as multi-shot, continue drilling to pass Cherty formation. Stand Back required & DPs, HWDPs and DC.

- **Contingency Plan #2**
- Perform Tandem scraper trip for dry test.
- Dry test with 2000psi differential press between water cushion and Fahliyan datum press. POOH to surface (inject Cement in L. Lap if required).

4. BHA Design

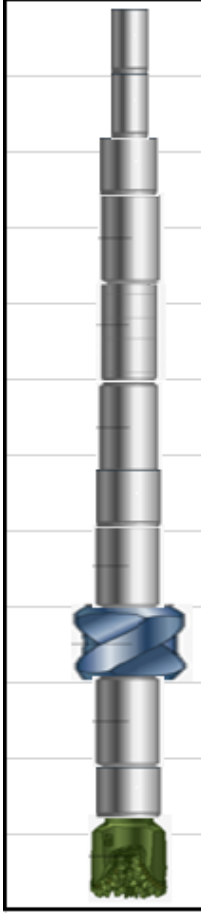
4.1 26” Hole Section

BIT SIZE (inch)	26"	BHA Configuration for Drilling 26" Hole Section	Collar	OD
Depth In/Out (m)	surface/150		DC-1	9- 3/4"
HOLE INCLINATION	0		DC-2	8- 1/2"
MUD WEIGHT (pcf)	67			
B.FACTOR	0.863			
Weight in Air (lb)	72,585			
Bouyed Weight (lb)	62,659			
Bouyed Weight Below Jar (lb)				

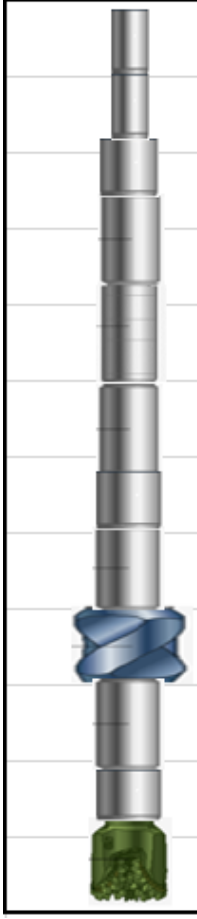
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	X.O.S	1	8"	3	6-5/8"Reg P x 4 -1/2" IF B	1	1
	8-1/2" Drill Collar	6	8-1/2"	3	6-5/8" Reg P x B	55.5	113.6
	XOS	1	9-3/4"	3	7-5/8" Reg. P x 6-5/8" Reg B	1	1
	9-3/4" Drill Collar	2	9-3/4"	3	7-5/8" Reg. P x B	18.5	58.1
	26" S.Stabilizer	1	26"	3	7-5/8" Reg. P x B	1	39.6
	9-3/4" Drill Collar	4	9-3/4"	3	7-5/8" Reg. P x B	37	38.6
	9-3/4" Bit Sub W/ Float Valve	1	9-3/4"	3	7-5/8" Reg. B x B	1	1.6
	26" MT Bit	1	26"	-	7-5/8" Reg. P	0.6	0.6

4.2 17 ½” Hole Section

BIT SIZE (inch)	17-1/2"	BHA Configuration for Drilling 17.5" Hole Section	Collar	OD
Depth In/Out (m)	150/850		DC-1	9- 3/4"
HOLE INCLINATION	0		DC-2	8- 1/2"
MUD WEIGHT (pcf)	88			
B.FACTOR	0.820			
Weight in Air (lb)	97,004			
Bouyed Weight (lb)	79,581			
Bouyed Weight Below Jar (lb)	59,557			

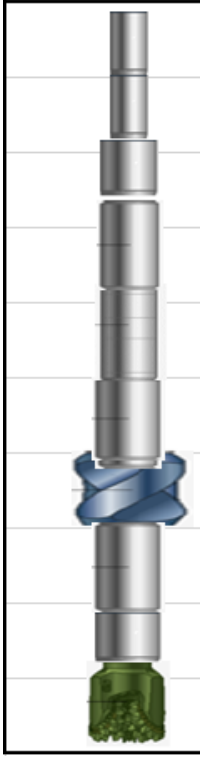
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" HWDP	6	5	3	4-1/2" IF Pin x Box	55.5	204.9
	X.O.S	1	8-1/2"	3	6-5/8"Reg P x 4 -1/2" IF B	1	149.4
	8-1/2" Drill Collar	3	8-1/2"	3	6-5/8" Reg P x B	27.75	148.4
	8" Drlg Jar	1	8"	3	6-5/8" Reg P x B	6	120.6
	8-1/2" Drill Collar	6	8-1/2"	3	6-5/8" Reg P x B	55.5	114.6
	XOS	1	9-3/4"	3	-5/8" Reg. P x 6-5/8" Reg	1	59.1
	9-3/4" Drill Collar	4	9-3/4"	3	7-5/8" Reg. P x B	37	58.1
	17-1/2" S.Stabilizer	1	17-1/2"	3	7-5/8" Reg. P x B	1	21.1
	9-3/4" Drill Collar	2	9-3/4"	3	7-5/8" Reg. P x B	18.5	20.1
	9-3/4" Bit Sub W/ Float Valve	1	9-3/4"	3	7-5/8" Reg. B x B	1	1.6
	17-1/2" R. Bit	1	17-1/2"	-	7-5/8" Reg. P	0.6	0.6

BIT SIZE (inch)	17-1/2"	BHA Configuration for Drilling 17.5" Hole Section	Collar	OD
Depth In/Out (m)	850/2182		DC-1	9- 3/4"
HOLE INCLINATION	0		DC-2	8- 1/2"
MUD WEIGHT (pcf)	88			
B.FACTOR	0.820			
Weight in Air (lb)	97,004			
Bouyed Weight (lb)	79,581			
Bouyed Weight Below Jar (lb)	59,557			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" HWDP	6	5	3	4-1/2" IF Pin x Box	55.5	204.9
	X.O.S	1	8	3	6-5/8"Reg P x 4 -1/2" IF B	1	149.4
	8-1/2" Drill Collar	3	8-1/2	3	6-5/8" Reg P x B	27.75	148.4
	8" Drlg Jar	1	8	3	6-5/8" Reg P x B	6	120.6
	8-1/2" Drill Collar	6	8-1/2	3	6-5/8" Reg P x B	55.5	114.6
	XOS	1	9-3/4	3	-5/8" Reg. P x 6-5/8" Reg	1	59.1
	9-3/4" Drill Collar	4	9-3/4	3	7-5/8" Reg. P x B	37	58.1
	17-1/2" S.Stabilizer	1	17-1/2	3	7-5/8" Reg. P x B	1	21.1
	9-3/4" Drill Collar	2	9-3/4	3	7-5/8" Reg. P x B	18.5	20.1
	9-3/4" Bit Sub W/ Float Valve	1	9-3/4	3	7-5/8" Reg. B x B	1	1.6
	17-1/2" PDC Bit	1	17-1/2	-	7-5/8" Reg. P	0.6	0.6

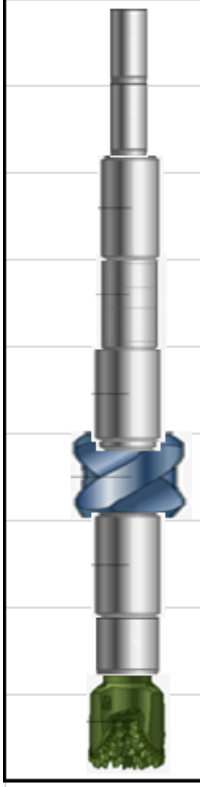
4.3 12 ¼” Hole Section

BIT SIZE (inch)	12-1/4"	BHA Configuration for Drilling 12.25" Hole Section	Collar	OD
Depth In/Out (m)	2182/3915		DC-1	8- 1/2"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	85			
B.FACTOR	0.827			
Weight in Air (lb)	101,150			
Bouyed Weight (lb)	83,602			
Bouyed Weight Below Jar (lb)	65,207			

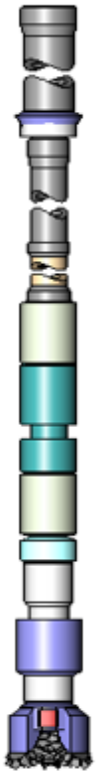
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" HWDP	6	5	3	4-1/2" IF Pin x Box	55.5	231.2
	X.O.S	1		-	6-5/8"Reg P x 4 -1/2" IF B	1	175.7
	8-1/2" Drill Collar	3	8-1/2	3	6-5/8" Reg P x B	27.75	174.7
	8" Drlg Jar	1	8	3	6-5/8" Reg P x B	6	146.95
	8-1/2" Drill Collar	13	8-1/2	3	6-5/8" Reg. P x B	120	140.95
	12-1/4" S.Stabilizer	1	12-1/4	-	6-5/8" Reg. P x B	1	20.95
	8-1/2" Drill Collar	2	8-1/2	3	6-5/8" Reg. P x B	18.5	19.95
	8-1/2" Bit Sub W/ Float Valve	1	8-1/2	3	6-5/8" Reg. B x B	1	1.45
	12-1/4" PDC Bit	1	12-1/4	-	6-5/8" Reg. P	0.45	0.45

4.4 8 ½” Hole Section

BIT SIZE (inch)	8-1/2"	BHA Configuration for Drilling 8.5" Hole Section	Collar	OD
Depth In/Out (m)	3915/4180		DC-1	6- 1/2"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	70,293			
Bouyed Weight (lb)	55,085			
Bouyed Weight Below Jar (lb)	42,411			

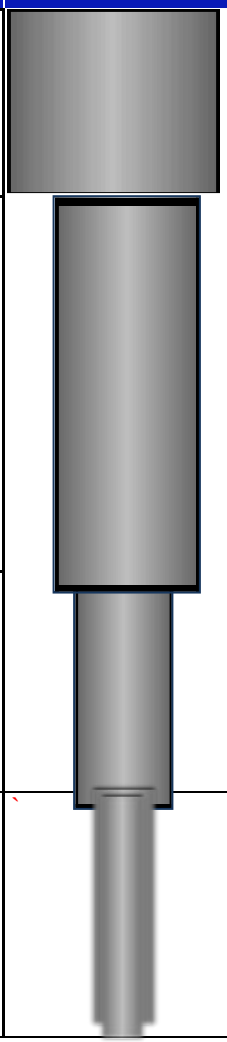
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF P x B	-	
	5" HWDP	6	5	3	4-1/2" IF P x B	55.5	258.2
	6-1/2" Drill Collar	3	6-1/2	2-1/2	4-1/2" IF P x B	27.75	202.7
	6-1/2" Drlg Jar	1	6-1/2	3	4-1/2" IF P x B	6	174.95
	6-1/2" Drill Collar	16	6-1/2	2-1/2	4-1/2" IF P x B	148	168.95
	8-1/2" S.Stabilizer	1	8-1/2	-	4-1/2" IF P x B	1	20.95
	6-1/2" Drill Collar	2	6-1/2	2-1/2	4-1/2" IF. P x B	18.5	19.95
	6-1/2" Bit Sub W/ Float Valve	1	6-1/2	3	4-1/2" Reg. B x 4-1/2" IF B	1	1.45
	8-1/2" PDC Bit	1	8-1/2	-	4-1/2" Reg. P	0.45	0.45

BIT SIZE (inch)	5-7/8"	BHA Configuration to C/Out Inside 7" Liner	Collar	OD
Depth In/Out (m)			DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,229			
Bouyed Weight Below Jar (lb)	16,462			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	5-7/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

5. Drilling Fluid Program

5.1 Proposed Drilling Fluid Program

Hole Size (in)	Casing Size (in)	Casing Program	Depth (m MD)	Drilling Fluid system	Mud Properties
26”	20”		150 m	Drill Water + PHG	MW: ALAP PV: ALAP (12-20) YP: 20 - 30 lb/100 ft² F.L: N.C PH:10.5 - 11 Vis: ±70 sec/1 qt
17 ½”	13 3/8”		2182 m	Gel Mud & KCl/Pol Mud	MW: 67-74 PV: ALAP (12-20) Vis:40 - 50 YP: 12 - 20 FL: N.C PH: 10 – 10.5 MW: 74 – 88 pcf PV: ALAP (15-25) YP: 12 - 22 lb/100 ft² FL: 12-5 cc/30 min PH: 9 - 10 Vis: 40 - 55 sec/qt
12 ¼”	9 5/8”		3915 m	KCl/$CaCo_3$/Polymer mud	MW: 72 - 85 pcf PV: ALAP (15-25) YP: 12 – 20 lb/100 ft² F.L: < 2.5 cc/30 min PH: 9 - 10 Vis: 40 - 50 sec/qt
8 1/2”	7” liner		4180 m	$CaCo_3$/Polymer mud	Mw:100 – 106 pcf PV: ALAP YP: 14 – 24 lb/100 ft² F.L: < 2.5 cc/30 min PH: 9 - 10 Vis: 45 - 60 sec/qt

5.2 Expected Challenges

5.2.1 26” Section drom surface to 150 m

Challenges

- Complete losses
- Hole cleaning issues due to large hole size
- Bit Blling

Recommendations

- Keep mud weight as low as possible to minimize formation losses.
- Optimize the procedure to fight the losses (see the guide at the end of this mud program) and be ready with LCM material to mix LCM pill as quick as possible, it is recommended to 8 ppb Mica M by 8 ppb Mica C to active system from start the Drilling up to 50 mt to prevent the probable Complete Losses.
- Keep Low GPM from Start drilling to 50 mt to prevent the Probable Complete Losses
- Adequate surface volume should be prepared in advance. A suitable quantitative of Drill water must be always available on location.
- Do not permit to use The Local Water (Hoor).
- Pump hi-vis pill at each connection to ensure good hole cleaning.
- Maintain adequate viscosity to clean the hole at anticipated penetration rates.
- All solids control equipment should be serviced and brought to optimum operating condition.
- Run all solids control equipment continuously while circulating.
- Utilize the finest mesh screens possible on flow line cleaners.
- Sand trap should be controlled and settled drilled solid should be dumped.
- A final high viscosity sweep should be pumped prior to pulling out of the hole to run casing.
- It is recommended that a high viscosity pill be spotted on bottom prior to running the casing.
- For Get sufficient pre-Hydrate gel its better after Mixing Gel left it on pit at least 6 hours before use.

- The Bentonite should be added slow enough to prevent balling and clogging. The initial yield depends in part on the quality of the surface equipment. Normally the slurry becomes thicker with time and agitation.

- The PH should be maintained around 10.5 - 11 with additions of Caustic Soda. This aids in hydrating the Bentonite, reducing corrosion rates, and minimizing the solubility of contaminants.

- In case of shortage in Drill Water XC-Polymer HV sweep pill should be pumped just as each single is drilled, the pill should then be pumped around, allowing at least one lag time before stopping pumping for connection.

5.2.2 17 ½” Section from 150 m to 2182 m

Challenges

- Hole cleaning issue.
- Possible severe losses.
- Shale
- Marl
- Tight Hole and Hole stability

Recommendations

- It is mandatory to monitor closely the rheological properties in order to ensure adequate hole cleaning, maximizing the flow rates to assist in cuttings removal.
  Poor hole cleaning is indicated by an increase in pump pressure, torque and drag, over-pull at connections and by a lack of an appropriate amount of cuttings on the shakers relative to the ROP and hole size.
- Periodic high viscosity sweeps will help assess the hole cleaning efficiency.
- When Faced with marl lithology must be pump Low Vis pill every single to prevent increase Rheology unusually. Process of the cutting removing in the Shaker Screens by Mud Engineer could be the best issue to pump Hi vis pill or Low Vis and Adjust The GPM amount due to have a good hole cleaning.
- Mud can be treated with additional shale inhibitor if deemed necessary to control shale reactivity and washouts.
- All the precautions have to be considered in order to take immediate reactions for any risk of circulation losses: Optimize the procedure to fight the losses (see the guide at the end of this mud program) and be ready with LCM materials to mix LCM pill as quick as possible.
- This section will be drill up to ±700 mt by Gel mud then Will be displaced by KCL Pol mud
- For Get sufficient pre-Hydrate gel its better after Mixing Gel left it on pit at least 6 hours before use.
- Test and report key mud properties three times per day. The MBT & Hardness should be tested and reported 3 times per day.
- Adequate surface volume should be prepared in advance. A suitable quantity of water must be always available on location.
- All solids control equipment should be serviced and brought to optimum operating condition.

- Also the mud density has to be carefully monitored at 68 pcf at the first then for prevent the tight hole and hole stability increase the mud weight gradually to 77 pcf .
- Require to maximum effitiency of solid control equipment to control the mud weight

- The shakers should be run with the finest screens possible in accordance with the flow rate and minimize fluid losses from the shakers by checking deck angle and running appropriate screens at all times. Also it should be monitored closely to ensure effective hole cleaning; if there is an inadequate or inconsistent supply of cuttings at the shaker, it may be necessary to increase the yield point further.

- The mud cleaner should be run with a 200 mesh screen, making sure that the cones are always clean and discharging heavy liquid. Check the working pressure in order to have the mud cleaner working at the maximum efficiency.
- Sand trap should be controlled and settled drilled solid should be dumped.

- Before tripping the shakers must be circulated clean at normal pump rate while reciprocating the drill pipes and moving the drill pipes up and down.
- Keep pH around the 9 - 9.5 while drilling the marl lithology to prevent the solving the marl in mud and cuases to dispersion and drop the PH.

- Measuring of the MBT is the direct indication of formation reactivity. MBT of range of < 12.5 lb/bbl is acceptable. Above this range, mud Rheology will start to shoot up to uncontrollable levels. Using fine screens & high speed centrifuges will help in maintaining lower values of MBT by discarding the fine solids and consequently minimize the dilution rate.
- The Aghajari formation in Azadegan Field from surface to 1274 m consists of a repetition of brown to grey calcareous sandstones, chert debris and gypsum-veined red marl. A fauna of brackish to freshwater was observed, together with more marine fauna.

5.2.3 12 ¼" Section from 2182 to 3915 m

Challenges

- Hole instability issues
- Wellbore losses.
- Shale
- Tight hole and well instability

Recommendations

- It is mandatory to monitor closely the rheological properties in order to ensure adequate hole cleaning, run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- Mud can be treated with additional shale inhibitor if deemed necessary to control shale reactivity and washouts.

- Optimize the procedure to fight the losses (see the guide at the end of this mud program) and be ready with adequate LCM material to mix LCM pill as quick as possible.
- The previouse mud in hole must be dump , due to have enoght surface volume , keep 300 bbl heavy mud in mud pits as kill mud.
- The concentration of KCl in the system is insufficient; shale cuttings will be soft and mushy. If the concentration of KCl is sufficient, they will retain their integrity. As drilling commences the potassium ion becomes depleted and must be replenished. The rate of depletion provides an indication of how reactive the shales are. While drilling gumbo, the appearance of soft mushy cuttings at surface is a good indication that the potassium concentration should be increased. Other indicators include a drilled solids increase, a CEC increase and the occurrence of tight hole or mud rings.

- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.

- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.

- Due to necessary of directional drilling, it is recommended to pump (at the first 20 bbl Low Vis pill follow by 20 bbl Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.
- Due to face with torque and drag , it is recommended to mix water base mud lubricant 1 – 3% vol reduce torque and drag while drilling horizontal or vertical
- In the directional drilling section it is recommended to use Glass beads 2 – 3 ppb as due to have torque and drag.
- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding CaCO3 F.

5.2.4 8 -1/2” Section from 3915 to 4180 m

Challenges

- Wellbore losses
- Possible HT environment
- Oil/Water Influx

Recommendations

- Hole cleaning is highly related to low shear rate parameters, so an adequate low shear rate yield point should have maintained to ensure hole cleaning. Run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- The previous mud will be transfer to this section.
- A suitable LCM material for reservoir should be available on location to combat down hole losses if occurred (see the guide at the end of this mud program).
- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.
- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.
- Due to necessity of directional drilling, it is recommended to pump (at the first 20 bbl Low Vis pill follow by 20 bbl Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.
- Due to face with torque and drag , it is recommended to mix WATER BASE MUD Lubricant 1 – 3% vol reduce torque and drag while drilling horizontal or vertical
- In the directional drilling section it is recommended to use Glass beads 2 – 3 ppb as due to have torque and drag.
- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding CaCO3 F.

5.3 Interval Description

5.3.1 26” Section from surface to 150 m

This section will be drilled as Gel mud system and the FV is between ±70 sec/qt so this mud system can be diluted with water after drill 65 meter below ground level for this thinks mentioned your plan for pumping Hi Vis sweep.

26" HOLE SIZE FROM : Surface	20" Casing Size TO : 150 MT
151 meter to drill in Aghajari Gel mud (Bentonite mud)	
DRILLING FLUID SYSTEM	Drill Water Sweep by Gel Mud Pill
Key products	Bentonite , Caustic , Soda ash

Water type	Drill Water
Solid Control	Shale shakers

Aghajari : Mainly Limestone inter bed upward with Marl and Gypsum

Interval Discussion

Gel Mud system will be simply formulated with Bentonite to obtain the required suspending properties and form a good filter cake. Makeup water should be checked and treated with SODA ASH to treat out hardness as needed. Adequate surface volume should be prepared in advance, allowing maximum pre-hydration time.

All solids control equipment should be serviced and brought to optimum operating condition.

Run all solids control equipment continuously while circulating.

The mud density has to be carefully monitored and adapted to the well conditions, it is suggested to maintain the mud weight at 65 - 67 pcf according to the well stability, this will be done by running the solids control equipment, control and dump settled drilled solids from sand trap and diluting the active system with new mixed mud as required.

Due to the large hole size, the hole should be swept frequently with HV pills made by Gel Mud (Viscosity: 70 - 80 sec/lt).

Losses may occur while drilling this top section specially between surface up to 50 mt which for prevent this matter it is recommended to add 4 ppb Mica M and 4 PPB Mica C together. Mud should be treated with SODIUM BICARBONATE before cement job to prevent cement contamination.

It is recommended that a high viscosity pill be spotted on bottom prior to running the casing.

Product Concentration and Function

Gel Mud		
Product N.	**lb/bbl**	**Product Function**
Bentonite	20 - 25	Viscosifier
Caustic Soda	0.25 – 0.5	Alkalinity Control
Soda Ash	0.25 – 0.5	Hardness Remover

Required Properties

Fluid Properties Summary		
MUD PARAMETERS		**Interval**
		I
Bit diameter	Inch	26"
Density	Pcf	ALAP
Marsh Viscosity	Sec	±70
PV	Cps	ALAP
Yield Point	lb/100ft²	20 - 30
API Filtrate	cc/30'	N.C

pH	-	10.5 - 11
Ca++	mg/l	< 300
Total Hardness	Mg/l	<400
LGS (drilled Solids)	% vol	<10
KCl	% by wt	-
NaCl	Mg/l	

Note: Recommended mud parameters that based on drill water which mentioned.

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud Volume

DRILLING FLUID TYPE			
Mud Density (pcf)	65 - 67	LGS Below (%vol)	10
Bit Size (inch)	26	Solid Control Equipment Eff. %	70
Section TD (m)	150	Hole Wash Out %	20
Casing OD/ID (inch)	0 / 0	Dilution Factor (bbl/m)	2.5
Water Depth (m)		Meters To Drilled (m)	150

Volume information (PHB)		
Surface System	500	bbl
Cased Hole Volume	0	bbl
New Hole Volume	323	bbl
Wash Out Volume	65	bbl
Sub Total	888	bbl
Dilution Volume	808	bbl
Sub Total	1,696	bbl
Volume of PHG	800	bbl
Hi vis pill Spot at TD	100	bbl
Sub Total	2,596	bbl
New Vol Required of PHG	2,596	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
PHG With Fresh water			
Bentonite	2,200	20-25	10
Caustic Soda	55	0.25 - 0.5	24
Soda Ash	110	0.25 - 0.5	12

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
MicaM , C	Loss control material	55 lb	each size 40sxs
Walnut shell M,c	Loss control material	55 lb	each size 40sxs
Lime	Alkalinity control	55 lb	20
Salt	non solid weighting agent	2200 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

5.3.2 17 ½" Section from 150 to 2182 m

The 17 ½" section will be drilled from 20" Casing Shoe @ ±150 m to ±850 mt with previous mud in hole (Gel mud) then at depth will be displace by KCL/Polymer mud and drill to an estimated section TD 2182 m.

At this system uses Salt/KCl as the source of density and Clay inhibition in the system. Mud weight will be Increase to 87 pcf by Barite as weighting agent; Rheology will be controlled by the use of Xanthan gum. Also Fluid loss will be keep under control by the addition of polymeric chemicals such as, Starch LV. Addition of Drilling Detergent (As contingency) and KCl (3 - 5 % wt.) to the mud system can prevent bit balling and consequently can help to optimize the drilling operation while drilling sticky Marl and Clay section. Fluid loss should be kept below 10 cc, meanwhile.

Also pH should be maintained within 9 - 9.5 while drilling Marl and keep within 9.5 – 10 while drilling Anhydrite.

At this section for the inhibition of system must be use PHPA (0.25 ppb) which adheres to the surface of the drilled cuttings, encapsulating them, and also coating the well bore reducing the hydration of clays in the formation.

17 1/2" HOLE SIZE FROM : 150 MT	13 3/8" Casing Size TO : 2182 MT
2032 meter to drill in Aghajari & Gachsaran, Asmari, Pabdeh by KCl/Polymer mud	
DRILLING FLUID SYSTEM	**Salt/KCl/PHPA/Polymer Mud**
Key products	Barite , Salt (NaCl) , KCL , Xanthan gum , Starch LV , Caustic Soda, Soda ash, PHPA , Biocide
Water type	Drill Water + Waste Water
Solid Control	Shale shakers , Desilters , Desander, Centrifuge

Interval Mud Discussion

After drilling shoe track and rate hole with carried Bentonite mud from previous section Salt/KCl/Polymer mud will be employed to drilling this hole section.

Saturated Salt Muds are primarily used to prevent excessive hole enlargement while drilling massive salt beds. They can also be used to reduce dispersion and hydration of shales and clays.

Potassium chloride is the primary source of the potassium ions in this system. In case of soft, sticky red Marl & Clay which encountered problems such as bit balling additions of drilling KCl (3 - 5 %) to inhibit clays in Aghajari is recommended.

Potassium is used in areas where inhibition is required to limit chemical alteration of shales. Potassium performance is based on cationic exchange of potassium for sodium or calcium ions on smectites and interlayered clays. Potassium-based muds perform best on shales containing large quantities of smectite or interlayered clays in the total clay fraction. Shallow shales, containing large amounts of montmorillonite.

Potassium interactions with clay surfaces can be traced to two effects: ionic size and hydration energy. Potassium ions are of the proper size to fit snugly into the spaces between the two silica tetrahedral layers which contact each other in the formation of a three-layer clay packet. The ionic diameter of potassium is 2.66 Å, very close to the available distance of 2.8 Å in the lattice space of the clay structure. A cation slightly smaller than 2.8 Å is desirable to allow for crystalline compaction.

In addition of using KCl, despite Aghajari is an un-permeable formation and seems that isn't necessary to control Fluid loss, Starch and PAC LV may be used to keep FL under control.

While drilling this section Xanthan gum will be used as base of rheology to extend hole cleaning and providing suitable YP and LSRV. An YP between 15-20 lb/100ft2 for sufficient carrying capacity and a LSRYP at range of 7-15 to preventing any Barite sag in system is required.

As it is seen in geological forecast, soft & sticky marls referred as Gumbo, contained large amount of Smectite and/or Illite could be associated with problems such as bit balling, mud rings, plugged flow lines and hole washout& cause low process of drilling and unnecessary trips and some more difficulties to drill the section as fast as possible. With all those problems reduction of ROP was inevitable, blinding of shaker screens, excessive over pull on trips, washout and stuck pipes has been experienced as well. For problem Solve can add 3-5 % KCl in system.

In case of necessary the inhibition of system can be increased by adding PHPA which adheres to the surface of the drilled cuttings, encapsulating them, and also coating the well bore reducing the hydration of clays in the formation.

Also while drilling this section encounter with Ca2+ based Gypsum is inventible. Gyp is calcium sulfate with water of crystallization (CaSO4.2H2O), Calcium sulfate contamination is similar to cement contamination because both liberate calcium ions, which in turn cause flocculation. Unlike cement, calcium sulfate does not because a PH increase since it supplies a sulfate ion in lieu of a hydroxyl ion. The sulfate ion contributes to flocculation of clay solids, although its effect is small compared to calcium ions. The severity of flocculation depends to a great degree on clay content. When CaSO4 causes the calcium concentration to increase and viscosity may fluctuate drastically. As calcium ion concentrations further increase, a base exchange occurs in which sodium montmorillonite becomes calcium montmorillonite. There are several methods for handling CaSO4 contamination. The drilling fluid can be maintained as a low-calcium fluid by chemically precipitating by adding soda ash.

Product Concentration and Function

Salt / KCL / Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.25 – 0.5	Alkalinity Control
Lime	0.5 - 1	Alkalinity Control
Soda Ash	1 – 2	Hardness Remover
Salt	91 - 100	non - solid weighting agent
KCL	10.7 - 18	Shale Inhibitor
Barite	Up to 110	weighting agent
Starch LV	8 - 12	Fluid loss controller
Xanthan gum	0.5 – 1	Viscosifier
PHPA	As Requirement	Encapsulate
Biocide	As Requirement	Anti-bacteria

MIXING PROCEDURE

1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer
2. Fill the mud tank with Drill water
3. add KCL as inhibitor
4. add Salt to reach mentioned mud weight
5. Remove Hardness by add Soda ash
6. adjust PH & alkalinity by Caustic soda and Lime
7. add Starch LV to control fluid loss
8. add Xanthan gum to reach good rheology
9. Add Barite powder to reach Mentioned Weight
10. Check the pH and keep in range by add caustic soda
11. Mix Biocide as need

Required properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		II
Bit diameter	inch	17 ½"
Density	pcf	67 - 88
Marsh Viscosity	sec	40 – 55
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 22
Gel 10 Sec	lb/100ft²	3 – 5
Gel 10 min	lb/100ft²	5 – 9
API Filtrate	cc/30'	5 - 12
pH	-	9 – 10
Ca++	mg/l	< 1200
Total Hardness	Mg/l	<1600
MBT	lb/bbl	<12.5
LGS (drilled Solids)	% vol	<10
KCl	%Wt	3 - 5
CL^-	Mg/l	170,000 – 188,000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud volume

Note: at ±800 mt Gel mud in hole will be displace with New KCL/PHPA/Pol Mud.

GEL MUD

DRILLING FLUID TYPE			
Mud Density (pcf)	67 - 74	LGS Below (%vol)	7
Bit Size (inch)	17.5	Solid Control Equipment Eff. %	70
Section TD (m)	600	Hole Wash Out %	20
Casing OD/ID (inch)	20 / 19.124	Dilution Factor (bbl/m)	3.2
Water Depth (m)		Meters To Drilled (m)	450

Volume information (Salt Pol Mud)		
Surface System	700	bbl
Cased Hole Volume	175	bbl
New Hole Volume	439	bbl
Wash Out Volume	88	bbl
Sub Total	1,402	bbl
Dilution Volume	1,406	bbl
Sub Total	2,808	bbl
Carried from last section	600	bbl
Volume Received	0	bbl
Sub Total	2,208	bbl
New Vol Required	2,208	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
PHG With Fresh water			
Bentonite	2,200	15-20	20
Caustic Soda	55	0.25 - 0.5	20
Soda Ash	110	0.25 - 0.5	10

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.
2-All the calculation based on max material Concentration which mentioned.

KCL / POL MUD

DRILLING FLUID TYPE					
Mud Density (pcf)	74 - 88		LGS Below (%vol)	8	
Bit Size (inch)	17.5		Solid Control Equipment Eff. %	80	
Section TD (m)	2182		Hole Wash Out %	15	
Casing OD/ID (inch)	20 / 19.124		Dilution Factor (bbl/m)	3	
Water Depth (m)			Meters To Drilled (m)	2032	
Volume information (KCL Poly Mud)					
Surface System				800	bbl
Cased Hole Volume				175	bbl
New Hole Volume				1,983	bbl
Wash Out Volume				298	bbl
Sub Total				3,256	bbl
Dilution Volume				5,950	bbl
Sub Total				9,206	bbl
Carried from last section				0	bbl
Volume Received				0	bbl
Sub Total				9,206	bbl
New Vol Required				9,206	bbl
Vol transfer out				0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1-2	167
Lime	55	0.5-1	167
Salt (NaCl)	2,200	91-100	418
KCL	2,200	10.7-18	75
Starch LV	55	8-12	2,009
Barite	3,300	Up to 110	279
Caustic soda	55	0.25 - 0.5	84
Xanthan Gum	55	0.5-1	167
PHPA	55	As Requirement	
Biocide	55	as Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
De foamer	De foamer	55 Gal	2
Mica F,M,C	Loss control material	55 lb	each size 80 sxs
Walnut shell M,c	Loss control material	55 lb	each size 80 sxs
Lignosulphonate	Deflocculants	55 lb	40
Sodiume Bicarbonate	Hardness remover/CMT Drilling	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

5.3.3 12 ¼" Section from 2182 to 3915 m

After cementing the 9 5/8" inch casing and confirmation of its proper cementing, The 8 1/2"section will be drilled from casing shoe to an estimated section TD KCl/Caco3 Polymer mud system necessitating a minimum requirement of 72 pcf to 85 pcf based on nearby well data.

In case of having problem with Marl and Shale layer 3 - 5wt. % of KCl, should be added to system to minimize associated problem.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

In the directional and horizontal section required amount of Lubricant such as Bit Lube and Glass beds will be added to active system to minimize torque and drag.

12 1/4" HOLE SIZE **FROM : 2182 MT**	**9 5/8" CSG** **TO : 3915 MT**
1733 meter to drill in Gurpi, Ilam, Lafan, Sarvak, Kazhdomi, Darian, Gadvan by KCl/Polymer mud	
DRILLING FLUID SYSTEM	**KCl/CaCo₃/Polymer Mud**
Key products	Limestone Powder , Salt (NaCl) , KCL , Xanthan gum , PAC LV , Starch LV , Caustic Soda, Soda ash, Biocide
Water type	Drill Water + Waste Water
Solid Control	Shale shakers , Desilters , Desander, Centrifuge

Asmari	:	Sandstone, sandy Limestone, Limestone, dolomite and contains salt aquifer, H2S, Salt water 1.15sg.
Pabdeh	:	Limestone, shale and silt interbed, interbeded horizons glauconitic, with Salt aquifer, H2S. Salt water 1.15sg.
Gurpi	:	Argillaceous limestone and marl with scattered glauconitic, asphalt, with salt aquifer. Salt water 1.15sg.
Ilam	:	Limestone, argillaceous limestone, shale interbed. Salt water 1.15sg.
Lafan	:	Shale and dolomite.
Sarvak	:	Limestone to argillaceous limestone, the lower part is shale and dolomite, with salt aquifer. Salt water 1.15sg.H2S
Kazhdomi	:	Shale interbedded with argillaceous limestone, sandy claystone, claystone, marl, sandy limestone, limestone beds
Daryan	:	shale interbedded with argillaceous limestone
Gadvan	:	limestone, claystone, sandy claystone and sandstone

Interval Mud Discussion

The water-based KCL Polymer system provides effective inhibition when drilling reactive shale's through below mechanism:

Potassium Chloride provides a source of Potassium ions, which are small enough to fit between the clay platelets without distorting the shale lattice. The Potassium ions are adsorbed on the exchangeable caution sites in the shale lattice and this

holds the platelets together, which helps to minimize clay and shale hydration, swelling and dispersion. A low solids, non-dispersed Salt/KCl/$CaCo_3$ Polymer drilling fluid is selected for drilling this section. This system is used primarily to drill moderately reactive shale's. It can be used effectively at temperatures up to 275˚ F – 300˚

If made a decision to run the Glycol System (KCL/Glycol/Polymer mud):

The primary shale inhibition mechanisms in the system are:

1. The cation exchange of potassium ions with the clay to reduce hydration and swelling in conjunction with;

2. A partially hydrolyzed polyacrylamide (PHPA) which adheres to the surface of the drilled cuttings, encapsulating them, and also coating the well bore reducing the hydration of clays in the formation.

3. Clouding out of Glycol: The above methods of shale inhibition from the KCl/PHPA system are enhanced by the clouding effect of the glycol which prevents pore pressure transmission. The stabilization mechanisms combine synergistically to provide inhibition close to that of non-aqueous fluids.

At a specific temperature (cloud point) a water soluble glycol starts to change from soluble to insoluble, forming a thermally activated emulsion (TAME). As the temperature increases the glycol starts to come out of solution and the liquid begins clouding. Eventually the single phase liquid separates into two separate phases.



A: Glycol and water together – at ambient temperature the glycol is 100% water soluble

B: Solution is heated until the glycol reaches 'cloud point' and becomes insoluble.

Cloud point is a function of the following variables:

- Salinity of the solution
- Molecular weight and type
- Type of electrolyte
- Concentration of the glycol

Research has concluded that the glycol is associated by hydrogen bonding to the reactive sites of clays. During this adsorption process, water is displaced and ordered structures of glycols are formed. There is little depletion of the glycol from the fluid when drilling, the glycol returning to solution as the cutting moves up the wellbore and the fluid temperature drops below the cloud point. This supports the theory that the bonding between the clay and the glycol is transitory, and there is some mechanism for association and disassociation due to changes in the environment.

It can be concluded that the shale inhibition and formation protection will be achieved by one or both of the following mechanisms:

- The glycol displaces water from adsorption sites on the clay minerals present in shales, especially in the presence of potassium Ions.
- Blocking the formation pores, preventing further ingress of invasive fluids by clouding out preventing pore pressure transmission.

During drilling operations, shale cuttings should be monitored continuously for inhibition. If the concentration of inhibitors in the system is insufficient, shale cuttings will be soft and mushy. If the concentration of inhibitors is sufficient, they will retain their integrity. In addition of this controlling API filter press is a key parameter to deal with.

Due to Shale layer in Pabdeh formation different additives such as PHPA and Glycol should be added to active system any prevent any hole problem.
The mud will be formulated and treated to in inhibit and encapsulate the reactive shale found in this area. The PHPA content should be run at 0.25-1 ppb throughout the section. PHPA will be added to the mud on the rig prior to drilling ahead. The success of this system is to achieve and maintain the required concentration of KCL – PHPA and GLYCOL in the system as early as possible in the drilling operation. The main function for the KCL is the primary source of the inhibitive K+ cation the inhibitive property of K+ while drilling reactive shale and clay stone is commonly known and understood. The K+ level derived from the KCL is determined from either historical date or shale characterization and is usually between 3% to 5% KCl in this specific case we would recommend a concentration of ± 11-18 ppb.

Product Concentration and Function

Salt / KCL / Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5 - 1	Alkalinity Control
Soda Ash	1 – 2	Hardness Remover

Salt	Up to 98	non - solid weighting agent
KCL	10.7 - 18	Shale Inhibitor
Limestone Powder	Up to 113	weighting agent
Starch LV	12 – 14	Fluid loss controller
Xanthan gum	1-1.5	Viscosifier
Biocide	As Requirement	Anti-bacteria
Glycol **(Optional)**	1 – 2% Vol	Shale inhibitor
PHPA **(Optional)**	0.25	Encapsulate

MIXING PROCEDURE

1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer
2. Fill the mud tank with Drill water
3. add KCL as inhibitor
4. add Salt to reach mentioned mud weight
5. Remove Hardness by add Soda ash
6. adjust PH & alkalinity by Caustic soda
7. add Starch LV to control fluid loss
8. add Xanthan gum to reach good rheology
9. Add Limestone powder to reach Mentioned Weight
10. Check the pH and keep in range by add caustic soda
11. Mix Biocide as need

Required properties

Fluid Properties Summary		
MUD PARAMETERS		**Interval**
		IV
Bit diameter	inch	12 1/4"
Density	pcf	72 – 85

Marsh Viscosity	sec	40 – 50
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 20
Gel 10 Sec	lb/100ft²	2 – 4
Gel 10 min	lb/100ft²	3 – 6
API Filtrate	cc/30'	< 2.5
pH	-	9 – 10
Ca++	mg/l	< 800
Total Hardness	mg/l	<1200
MBT	ppb	< 5
LGS (drilled Solids)	% vol	< 7
KCl	% by wt	3 - 5
CL^-	Mg/l	180000-188000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud volume

DRILLING FLUID TYPE			
Mud Density (pcf)	72 - 85	LGS Below (%vol)	6
Bit Size (inch)	12.25	Solid Control Equipment Eff. %	80
Section TD (m)	3915	Hole Wash Out %	15
Casing OD/ID (inch)	13.375 / 12.525	Dilution Factor (bbl/m)	2
Water Depth (m)		Meters To Drilled (m)	1733

Volume information (Salt KCL Pol Mud)		
Surface System	800	bbl
Cased Hole Volume	1,091	bbl
New Hole Volume	829	bbl
Wash Out Volume	124	bbl
Sub Total	2,844	bbl
Dilution Volume	1,658	bbl
Sub Total	4,502	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	4,502	bbl
New Vol Required	4,502	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1-2	82
Caustic Soda	55	0.5 - 1	82
Salt (NaCl)	2,200	Up to 98	201
KCL	2,200	10.7 - 18	37
Starch LV	55	12-14	1,146
Limestone Powder	2,200	up to 113	231
Xanthan Gum	55	1-1.5	123
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
De foamer	De foamer	55 Gal	2
Shell fish F,M	Loss control material	55 lb	each size 60 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Water Base Mud lubricant	Torque & Drag Reducer/ROP Enhancer	55 Gal	40
Starch HT	Fluid Loss Controller	55 lb	80
Pac-LV	Fluid Loss Controller	55 lb	120
PHPA	Encapsulate	55 lb	20
Glycol	Shale Inhibitor	55 Gal	28
Glass Beads - 500/800 mic	Torque & Drag Reducer	55 lb	120
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

5.3.4 8 ½" Section from 3915 m to 4180 m

After cementing the 9 5/8" inch CSG and confirmation of its proper cementing, the 8 1/2" section will be drilled from liner shoe to an estimated section TD of 4180 m horizontally by Salt/Polymer mud system necessitating a minimum requirement of 100 pcf to 106 pcf based on nearby well data.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

In the directional and horizontal section required amount of Lubricant such as Bit Lube and Glass beds will be added to active system to minimize torque and drag.

8 1/2" HOLE SIZE **FROM : 3915 MT**	**7" Liner** **TO : 4180 MT**
265 meter to drill in Fahlian, by $CaCo_3$/Polymer mud	
DRILLING FLUID SYSTEM	**$CaCo_3$/Polymer Mud**
Key products	Limestone Powder, Hematite , Salt (NaCl), Xanthan gum , PAC LV , Starch LV , Caustic Soda, Soda ash, Biocide
Water type	Drill Water + Waste Water
Solid Control	Shale shakers , Desilters , Desander, Centrifuge

Fahlyan: Limestone interbedded with green gray, Argillaceous Limestone, sticky Clay stone beds.

Product Concentration and Function

Salt /$CaCo_3$/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5 - 1	Alkalinity Control
Soda Ash	1 – 2	Hardness Remover
Salt	Up to 109	non - solid weighting agent
KCL (Optional)	10.7 - 18	Shale Inhibitor
Limestone Powder	Up to 258	weighting agent
Starch LV	12-14	Fluid loss controller
Hematite	Up to 98	weighting agent
Xanthan gum	0.25-0.5	Viscosifier
Biocide	As Requirement	Anti-bacteria

MIXING PROCEDURE

1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer
2. Fill the mud tank with Drill water
3. add Salt to reach mentioned mud weight
4. Remove Hardness by add Soda ash
5. adjust PH & alkalinty by Casutic soda
6. add Starch LV to control fluid loss
7. add Xanthan gum to reach good rheology
8. Add Limestone powder & Hematite to reach Mentioned Weight
9. Check the pH and keep in range by add caustic soda
10. Mix Biocide as need

Required properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		V
Bit diameter	inch	8 1/2"
Density	pcf	100-106
Marsh Viscosity	sec	45 – 60
PV	cps	ALAP
Yield Point	lb/100ft²	14 – 24
Gel 10 Sec	lb/100ft²	2-4
Gel 10 min	lb/100ft²	3-7
API Filtrate	cc/30'	< 2.5
pH	-	9 – 10
Ca++	mg/l	< 800
Total Hardness	Mg/l	<1200
LGS	%	<10
MBT	ppb	< 5
CL^-	Mg/l	180000-188000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud Volume

DRILLING FLUID TYPE			
Mud Density (pcf)	100 - 106	LGS Below (%vol)	6
Bit Size (inch)	8.5	Solid Control Equipment Eff. %	80
Section TD (m)	4180	Hole Wash Out %	15
Casing OD/ID (inch)	9.625 / 8.835	Dilution Factor (bbl/m)	2
Water Depth (m)		Meters To Drilled (m)	265

Volume information (Salt Pol Mud)		
Surface System	800	bbl
Cased Hole Volume	974	bbl
New Hole Volume	61	bbl
Wash Out Volume	9	bbl
Sub Total	1,844	bbl
Dilution Volume	122	bbl
Sub Total	1,966	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	1,966	bbl
New Vol Required	1,966	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1-2	36
Caustic Soda	55	0.5 - 1	36
Salt (NaCl)	2,200	Up to 109	97
KCL (Optional)	2,200	10.7 - 18	16
Starch LV	55	12-14	500
Limestone Powder	2,200	up to 258	231
Xanthan Gum	55	0.25-0.5	18
Ferobar	3,300	Up to 98	58
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
De foamer	De foamer	55 Gal	2
Pac LV	Fluid Loss Controller	55 lb	120
Starch HT	Fluid Loss Controller	55 lb	80
Shell fish F,M	Loss control material	55 lb	each size 60 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Water Base Mud lubricant	Torque & Drag Reducer/ROP Enhancer	55 Gal	40
Glass Beads - 500/800 mic	Torque & Drag Reducer	55 lb	120
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

5.4 Technical Appendix

5.4.1 Loss Control Procedure

Loss of fluid returns will lower the hydrostatic head of the drilling fluid in the wellbore, thereby inducing a kick. The influx fluid will then flow to the surface or into the zone of lower pressure. Lost circulation can occur in naturally occurring fractured, vuggy, cavernous, sub-normally pressured or pressure depleted formations. Induced losses can occur from mechanical fracturing due to pressure surges while breaking circulation. In all cases of lost circulation, attempts should be made to keep the hole full. The hole can be filled with either light drilling fluid or water. A record of the amount of fluid pumped should be made.

The concentration of LCM will vary according to losses. In the event of seepage losses, treat system with a finer grade of LCM at 20 – 30 Kg/m³. In the event of moderate to high rate of well bore loss, prepare pills at different graded concentrations and spot over thief zone. The concentration may vary between 18 to 70 lb/bbl. In order to avoid LCM plugging MWD tools a circulation sub should be installed in the drill string. The MWD engineer is to advise on the size and concentration of LCM that can be pumped through MWD Tools. Ensure that the drilling displace the LCM pill out of the string prior to shutting down the mud pumps.

Important Note: Please note below concentration and recommended materials may be change with same materials on rig site.

In case of having circulation losses:

Losses less than 10 bbls/hr:

- Active system to be loaded with 10 lb/bbl of Fiber Fine.

Losses 10- 40 bbls/hr:

- Active mud to be loaded with 15- 20 lb/bbl of Fiber Medium.
- Sweep the hole frequently with 50 bbls high viscous pill contains;
- 15 lb/bbl Mica Medium - Corse.
- 10 lb/bbl Walnut Shells Medium -Fine.
- 10 lb/bbl Oyster Shell Fine.

Losses 40 – 100 bbl/hr:

- Stop Drilling.
- Pull out string above the loss zone
- Spot 100 bbls pill contains;
- 30 lb/bbl Fiber coarse.
- 30 lb/bbl Walnut Fine.
- 20 lb/bbl Mica Med.
- Repeat the previous pill minimum of two times.

Losses more than 100 bbls:

- Up to company procedure, may be squeeze 50 bbls of Cement squeeze.

Mixing LCM Pills

The amount of LCM that can be added to a mud depends very much on the mud properties. High gels and YP will help keep the material in suspension. Since LCM is generally of low gradient, excess material tends to float on top if too much is added. The following procedure may prove useful for mixing LCM pills and avoiding the problem of flotation.

1. ENSURE the pill has enough viscosity to keep the low gradient material in suspension before addition of LCM. ADD appropriate viscosifier if required.

2. ADD the material in stages e g. 5, 10, 15, 20 etc. ppb.

3. STOP after desired concentration has been reached, or when it is observed that the material cannot be held in suspension

4. If a higher LCM content is desired, LEAVE the pill to soak for a period of time and then TRY ADDING more material.

- LCM Pill Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close the BOP and squeeze 10 bbls of LCM pill into the formation without exceeding 200 psi.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50 bbls of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20 bbls of LCM pill into the formation without exceeding 200 psi. If a squeeze pressure develops, hold the pressure for 1 hour before staging the drill string back to bottom.

Notes:

- All time hole must be full, transfer mud to trip tank and asked driller to fill hole from trip tank.

- All LCM pills must be thoroughly mixed and homogenised before being pumped.

- **When pumping an LCM pill through a bit with nozzle smaller than 18/64, DO NOT USED LCM Medium and Coarse sizes.**

- Remove filter screens from the drill pipe and if necessary from the mud pump suction lines.
- This is particularly true if the mud density is high, 14 lb/gal or greater. Fine loss circulation material which does not adsorb a lot of water should be used in weighted mud.

GUIDELINE FOR FIELD ENGINEERS FOR SHALE INHIBITION

Shale Hydration takes two forms in water-sensitive shales: surface and osmotic adsorption. Surface hydration occurs when a small volume of water is strongly adsorbed onto the planar surfaces of the clays which causes little softening or swelling, but can lead to excessive stresses if the swelling is confined.

Osmotic swelling occurs when a large volume of weakly held water is attracted to the clay surfaces by electrostatic forces. Osmotic swelling causes softening and significant swelling as the adjacent clay layers hydrate water and expand. Osmostic swelling does not generate excessive stresses, even when confined, and can be substantially reduced if Extremely difficult drilling conditions have led to more systematic approaches to analysing wellbore instabilities.

One such approach that can be used as a guideline for the mud engineer involves three basic steps:
1. Monitor and analyse the fundamental failure mechanism.
2. Determine and understand the factors governing this failure mechanism.
3. Model the problem considering impact of proposed changes.

MONITOR AND ANALYZE THE FAILURE MECHANISM

Monitoring of data and events on the rig is the first step of the process and involves:
• **Collection, organization and processing of drilling data.** Drilling data such as Rate of Penetration (ROP), Revolutions per Minute (RPM), Weight-on-Bit (WOB), Bottom-Hole Assembly (BHA), torque, over pull Trip In Hole (TIH) and Pull Out Of Hole (POOH), swab and surge, flow rate, mud weight, mud Rehology , etc. should be collected, organized and analyzed on a continual basis and compared to the symptoms of wellbore stability being observed. This will help evaluate performance and establish trends or any deviations and measure the effectiveness of treatments and changes.
• **Formation characterization.** The standard procedure in drilling operations is to apply the knowledge gained from previous wells to subsequent wells drilled in similar geologic regions. Wellbore-instability investigation requires evaluation of shales during the drilling process. Since rig site shale testing is limited due to equipment and time limitations, most evaluations are performed in a properly equipped laboratory.
Laboratory tests are conducted to evaluate shale swelling, dispersibility, cation exchange capacity, hydration capacity, colloidal content, capillary suction time, shale hardness tests and mud compatibility properties.
A number of methods have been used to perform rig site tests. These include tests as simple as the methylene blue test (CEC) measurement of shale; shale hardness measurements; hot-rolled dispersion testing with various mud formulations and additives; and sophisticated analytical measurements of various clay properties. It is particularly important to try and determine the failure mechanism: swelling and softening or brittle spalling. Information from these tests, current shaker observations and mud logger
data are used to evaluate drilling fluid performance, adjust mechanical drilling parameters and making casing design decisions. The primary options to counter shale problems at the rig site are:
• Change the drilling fluid density.
• Change to an oil- or synthetic-base system.
• Change to a more inhibited water-base mud.
• Reduce fluid loss to very low levels and increase filtrate viscosity.
• Utilize pore-plugging additives, emulsified insoluble materials or plastering agents.
• Adjust mechanical drilling parameters.
• **Evaluation and analysis of instability symptoms.**
Instability warning signs such as the presence of excessive cavings on the shakers for a given ROP, fill on connections, tripping difficulties, torque, stuck pipe, etc. indicate that something is wrong down hole. The magnitude or volume of failed rock should be estimated and recorded with the other drilling parameters. The number and location of specific incidents should be recorded, such as pipe
sticking, fill or packing-off, reaming and re-drilling. All of this information can then be organized and correlated with time and depth to detect the location and quantify the severity of critical sections in the well.

2. DETERMINE AND UNDERSTAND FAILURE MECHANISM
All of the monitored and plotted parameters are then evaluated to identify the most probable mechanism causing the instability. The most useful symptoms used to indicate the mechanism responsible for wellbore instability are:
• Cavings analysis.
• Tripping difficulties.

• Drilling conditions.
• Mud system analysis.
• In-situ conditions.

These symptoms should be evaluated based on an interdisciplinary knowledge of the causes of wellbore instability to determine the most likely failure mechanism.

3. MODEL AND CONSIDERING PROPOSED CHANGES

Once the monitoring and determination steps have produced parameters, organize and analyze this data to obtain qualitative solutions and evaluate proposed changes. One good way to do this is through the use of a decision tree or flowchart approach to evaluate the possible failure mechanism and to identify the most promising remedies. After arriving at a proposed solution, a corrective measure should be applied and
the results used as feedback into the method and decision tree method.

A decision tree or flowchart, such as shown in Figure 9 for evaluating the cause of excessive cavings, provides a logical path for identification of the mechanism and a qualitative solution. Maintaining hole stability while drilling shale sections can be particularly troublesome. Simple solutions do not always exist, but good drilling practices combined with good mud practices are most often successful. Analysis of the mechanical stress/strain relationships has helped to develop the proper density requirements for hole stability.

5.5 Directional Drilling

DRILLING FLUID PROPERTIES AND SELECTION

It is common knowledge among drilling engineers and drillers that borehole problems can be considerably more severe in a directional hole than in a vertical well. The problems are magnified as the hole angle and horizontal displacement increases. Although most of these reactions are mechanical in nature, much can be done to alleviate them by proper selection and maintenance of the drilling fluid. For example, if a water-base fluid is being used, lubricants can be added to lower the coefficient of friction between the drill string and borehole or casing. Inhibiting salts can be added to reduce the interaction of the drilling fluid with the exposed formations, thereby improving borehole stability.

There are a number of drilling factors influenced by the drilling fluid, but the following are of primary concern in drilling a horizontal well.

• Borehole instability
• Hole cleaning
• Torque and drag
• Formation damage
• Barite sag
• Hydraulics

Borehole Instability

Borehole instability is one of the more costly, but relatively common problem associated with drilling oil and gas wells. Not only can it impede drilling, it can result in loss of the hole. The term normally conjures thoughts of borehole collapse with accompanying stuck pipe. This is one of the

more critical reactions of borehole instability, but it can manifest itself in more subtle ways such as hole-fill, bridges, hole enlargement, or hole size reduction.

These reactions lead to other associated problems including increased torque and drag, logging difficulties, poor cement jobs, and increased drilling days, all of which add significantly to the drilling cost. It should be pointed out that the discussions to follow do not cover hole enlargement problems from weakly consolidated sands, or rubble zones that erode from the hydraulic action of the drilling fluid.

Stabilization

The swelling activity is controlled by adjusting the electrolyte content of the fluid to minimize expansion of the clays. Lime and gypsum-treated fluids have long been used for this purpose, but basically do not satisfy the inhibition needs of the more troublesome shales. The potassium ion is an effective swelling inhibitor and is used extensively for this purpose through the addition of potassium salts. The concentration of potassium utilized ranges from several hundred mg/L to over 60,000 mg/L. The most cost-effective potassium concentration, however, appears to be in the 3000 to 7000 mg/L range. In marine environments, seawater or seawater with additional sodium chloride, are good alternates to potassium. The tendency for the shale to disperse or subdivide into discrete particles appears to be best controlled by the addition of an adsorptive, high molecular weight water-soluble polymer.

One of the more effective polymers for this purpose is the partially hydrolyzed poly acrylamide with molecular weight in excess of 15 million. These anionic polymers adsorb on to the surface of the cuttings and borehole wall to act as a binder or elastic coating to retard dispersion. Another approach to stabilizing the hard brittle shales that contain natural or induced micro fractures is the use of water-dispersible asphalt or asphaltines. The addition of these type materials appear to plug the micro-fractures, or perhaps even surface pore throats to retard further fluid invasion. This action has proven very effective in certain shales and can be applied in most water base fluids.

Hole Cleaning

Cleaning cuttings from a vertical hole has been covered extensively by numerous investigators. Hole cleaning in a vertical hole is essentially a matter of overcoming the slip velocity of the cuttings with sufficient annular velocity and carrying capacity of the drilling fluid. Although any velocity greater than the settling velocity of the largest cutting will theoretically lift all the cuttings to surface eventually, too low an annular velocity will lead to an undesirable high concentration of cuttings in the annulus. Experience has shown that cuttings concentrations more than 5% by volume cause tight hole or stuck pipe when circulation is stopped.

Cuttings Bed Formation

Radial slip velocity increases with increasing angle of inclination, forcing more cuttings to the low side of the annulus, which eventually results in the formation of a cuttings bed. reported that bed formation was dramatic in the 40° to 50° angle with the bed sliding downward against the flow, resulting in a very high cuttings concentration.

The fact that a cutting bed forms is a serious problem, but the instability of the bed in this critical angle of the well makes the problem even more severe. High torque and drag resulting from the high cutting concentration at the lower part of the angle can obviously result in stuck pipe or loss of circulation.

Hole Cleaning Sweeps

In static drill string concluded that high-density sweeps were more effective than high-density high-viscosity sweeps based on field measurements. The higher viscosity of the latter caused flow diversion to the high side of the wellbore. Not surprisingly, high-viscosity sweeps also performed poorly. Also concluded that the best hole cleaning sweeps were high-density low viscosity sweeps, with a density 3 – 4 ppg higher than that of the drilling fluid. the following procedure is recommended:

• Calculate sweep height in the vertical part of the annulus, and added hydrostatic head in the annulus.
• Rotate at 80+ rpm while pumping the sweep, 180+ rpm is preferred. It is not recommended to pump a weighted sweep while sliding.
• Keep pumping until the sweep is out of the hole. If pumping stops while the sweep is in the deviated hole, it falls out of the "jet stream" and re-deposits captured fines.
• Avoid increasing the viscosity of the weighted sweep.
• Plan to run 3 – 4 sweeps to achieve desired results when hole cleaning problems are indicated. One sweep will probably not be sufficient to remedy an existing situation.
• Avoid checking for flow or closing the annulus with the weighted sweep in the drill pipe.
• If desired for preventative maintenance: 1) schedule sweeps every 6 – 8 hours or 500 – 1000 feet (whichever comes first, unless the hole dictates greater frequency); barite/silt build- ups are dependent on time and footage: and 2) pump a sweep while circulating bottom up before tripping, and another once back on bottom.

concluded that high-density sweeps were more effective than high-viscosity sweeps.
They strongly advocated the use of sweep reports, which should be standard practice in any drilling fluids operation.

Short Tripping

Another technique that has proven helpful in controlling problems created by the cuttings bed in the high-angle portion of the well is to make periodic short trips. Tripping the bit through the bed not only assures that the bottom hole assembly and bit can be pulled through it, but the action should agitate the bed to the point that a portion will slide down the hole and subsequently be picked up when circulation is commenced. The time intervals for these short trips will be determined by the torque and drag conditions that exist in any particular well. Lacking specific well history, a general rule of thumb is to short trip at least once for every 500 feet of new hole.

Drill String Rotation

The industry is in total agreement that drill string rotation improves hole cleaning, especially in hole sizes less than 17 ½". The general opinion in the industry is that the optimum drill string rotation rate is approximately 120 rpm. It is believed that the drill string rotation improves hole cleaning by mechanically disturbing the cuttings bed on the low side of the hole. There could also be effects due to skewing the velocity profile in the wellbore and the creation of Taylor Vortices. Encouraged the use of rotary steerable systems.

Turbulence

Bottom hole assembly design becomes important because the flow restrictions of some of the components, such as the measurement-while-drilling tools or positive displacement motors, may not allow sufficient flow to reach turbulence in the annulus. Turbulence can, of course, be induced at some constant flow rate by decreasing the annular area. This may be more difficult to accomplish in a horizontal well because of limitations on the design of drill strings and tool assemblies. An increase in the effective viscosity of the drilling fluid will decrease the Reynolds Number, and thus increase the flow rate required to obtain turbulent flow. Thus, a lower effective viscosity fluid may be desirable in the horizontal well, but attention must also be given to cleaning the larger diameter portion of the upper hole or possible washed-out sections where it is impractical to maintain turbulent flow.

It is interesting to point out a basic fact between static fluid pressure and dynamic fluid pressure. Static fluid pressure is a function of *true vertical depth*, while dynamic fluid pressure, or ECD, is a function of *measured depth*. Thus, the ECD will continue to increase, albeit slowly, even after the hole becomes essentially horizontal. The fracture pressure could be reached if the hole was extended to sufficient depth and the fluid density is close to the fracture gradient.

Hole Cleaning Guidelines

The single most important factor related to hole cleaning in deviated wells is flow rate. During directional drilling operations drilled cuttings will settle on the low side of the hole and form a stationary bed if insufficient annular fluid velocity is used. The critical flow rate required to prevent cuttings bed formation must be determined .

Typically few hole cleaning problems exist in vertical or horizontal sections. Most problems

associated with hole cleaning are seen on deviated wells and occur in the 40 – 60 degree section where gravity effects can cause cuttings beds to slump down the hole.Typical flow rates to aim for in ERD wells:

Hole Size	Typical Flow Rates
17 ½”	800 gpm minimum. Some rigs achieve 850 – 1000 gpm.
12 ¼”	Aim for 800 gpm (although 800 – 1000 gpm is typically achieved). If 800 gpm is not achievable, ensure tripping procedures are in place for poorly cleaned hole.
8 ½”	Aim for 400 gpm.

When correctly designed, both laminar and turbulent flow regimes will effectively clean a deviated well. In general increasing the viscosity of a fluid in laminar flow will improve hole cleaning as will a reduction of the viscosity of a fluid in turbulent flow. It is important that one or other regime is selected and that the transition zone between the two is avoided.

Generally viscous fluids in laminar flow are preferred because:

• It is possible to achieve higher cleaning capacity.
• Viscous fluids give better transport in the near-vertical sections.
• Viscous fluids have better suspension characteristics when circulation is stopped.
• It is difficult to achieve “turbulent flow” except in small hole sizes.

Experience has shown that good drilling fluid rheological properties are extremely important to hole cleaning when drilling a high angle well. Studies show that the effects of drilling fluid

rheological properties and annular flow regime are mutually dependent. In the laminar flow regime, increasing fluid viscosity will improve hole cleaning and this is particularly effective if the Yield Stress is high. In the turbulent regime, however, reducing fluid viscosity will help removing cuttings. Therefore the drilling fluid rheological properties should be designed to avoid the transitional flow regime. For hole sizes above 8 ½", the annular flow is laminar under most circumstances. Therefore it is necessary to specify a minimum Yield Stress. The optimum level required is best established based on field data and experience.

A higher ROP requires a higher flow rate to clean the hole. It is good practice to drill the hole with a steady ROP and select the required flow rate for hole cleaning accordingly. Before making a connection, the hole should be circulated at the normal flow rate to clear the cuttings from around the BHA. Depending upon the hole angle and length of BHA, a circulation time of 5 to 10 minutes is often required.

Before tripping out, the hole should be circulated at the normal flow rate until the shakers are clean, and the drill pipe be rotated at maximum speed/reciprocated in the meantime. This may require up to 3 bottoms-ups, depending on the hole angle and hole size. Below are listed the recommended number of bottoms-up prior to tripping.

Hole Angle, deg	8 ½"	12 ¼"	17 ½"
0 – 10	1.3	1.3	1.5
10 – 30	1.4	1.4	1.7
30 – 60	1.6	1.8	2.5
60+	1.7	2.0	3.0

Torque and Drag

Overall friction can become so great in an extended-reach well or horizontal section that the drill string can only be rotated with great difficulty. Needless to say, this condition is conducive to stuck pipe. Obviously, this coefficient of friction will be significantly affected by the lubricating properties of the drilling fluid.

Many of the causes of torque and drag are mechanical, while others are related to hole geometry. Several, however, can be directly attributed to the drilling fluid. Since wall contact with the drill string will be extremely high in a horizontal hole, it is imperative that the lubricating characteristics
of the fluid be considered.

Formation Damage

The obvious reason for drilling a horizontal well is to expose more of the producing zone to the wellbore for increased production. Thus, the importance of preventing formation damage during drilling and completion of the horizontal well is just as important as it is in its vertical counterpart.

In fact, it may be more important since the increased cost of drilling the horizontal well must show additional production for its justification. A 50% damage or permeability loss in a potential 500 bbl/day horizontal well represents a considerably greater loss in production than a 50% damage in a 100 bbl/day vertical hole through the same reservoir.

Sandstone production is by far the most susceptible reservoir rock to damage or permeability loss during drilling and completion operations. Though formation damage may be caused by poor drilling and completion practices, one of the primary causes is due to exposure to the drilling and completion fluid.

This loss, attributable to the fluid, can be caused by one or more of the following occurrences:
• Particle transport from the drilling fluid into the production zone, subsequently plugging pore throats.
• Fluid filtrate reacting with expandable clays in the rock to decrease pore throat diameters.
• Particle movement within the permeable rock due to dispersion of clays and other minerals from quartz surfaces.
• Wettability changes of the formation from exposure to the drilling fluid filtrate.
• Interaction of fluid filtrate with formation fluids to form water - insoluble precipitants.
The effect of these actions on the permeability of sandstones has been the subject of a considerable investigation. In almost all cases, some damage is caused by the drilling and completion fluid. The primary objective is to limit this damage as much as possible. Consequently, selection of the drilling and completion fluid becomes an extremely important function in drilling the horizontal well.

Recommended Fluid Parameters

• *Velocity* – One of the major obstacles to overcome in horizontal and ERD drilling is hole cleaning. In vertical holes, cuttings have room to settle without causing problems. In horizontal wells, cuttings only have to settle 4+ inches to form a cuttings bed. Research and practical experience has shown that cuttings bed formation is analogous to sand dunes on an ocean beach. The bed moves in a wave fashion up the hole. If sufficient annular velocity is
not present, the bed thickens until it reaches equilibrium, which usually occurs when turbulent flow conditions exist. With water, turbulent flow is easily obtained since viscosity is in the denominator of the turbulent flow equation, and water has a viscosity of 1. Fluids other than water require more pump output to approach turbulent flow conditions. Practical
experience in other industries has proven that materials can be transported in water slurries if the fluid velocity is between 3 and 7 ft/sec.

• *Rheological Properties* – Ideally one would want a true thixotropic fluid to drill horizontally.
In effect, two different conditions exist simultaneously. One occurs in the vertical portion of
the hole where larger diameter casing may be set, or wash-outs in the open hole occur. In that case, a fluid having a high Yield Stress, or if the fluid is visco-elastic, the initial gel strength needs to be elevated. The opposite is needed in the horizontal section from a hole-cleaning point of view. Here one wants water for the reasons pointed out earlier. Compromises are the result. If casing is set into the zone of interest, or if formations up-hole are not a problem, then water may be possibly combined with occasional sweeps.

• *Surface Apparatus* – Special attention must be placed on the solids removal equipment at the surface. Solids removal efficiency must be continually monitored so that unwanted particulate material is not retained in the fluid system that may negatively affect fluid weight
or filter cake quality, resulting in lost circulation or increased torque and drag. The cost of problems that may occur due to not doing this correctly may far outweigh the cost of equipment needed.

5.6 Hydrogen Sulphide Gas in Drilling Fluid

Description of Hydrogen Sulphide Gas
Hydrogen sulphide is a colourless gas, with an offensive odour and a sweetish taste. It is soluble in water, alcohol, oils, and many other solvents. It has a specific gravity of 1.1895 with reference to air. It is considered a weak acid, it is toxic to humans and corrosive to metals. Hydrogen sulphide can be dangerous to personnel on the surface as it is extremely toxic to human and even animal life, and is extremely corrosive to most metals as it can cause cracking of drill pipe and tubular goods, and destruction of testing tools and wire lines. When hydrogen sulphide enters the borehole during

drilling, completion, or testing for hydrocarbons, it creates several very detrimental problems. These problems are encountered regardless of the source of the hydrogen sulphide. Hydrogen sulphide has created unsolved problems in the oil field.

Origin of Hydrogen Sulphide

During oil and gas operations, hydrogen sulphide can be found as a component of formation gases, dissolved in water, hydrocarbons, or even liquid sulphur. Thermal degradation of organic materials and sulphate reducing bacteria (SRB) can create hydrogen sulphide along with other gases.

Problems Caused By Hydrogen Sulphide Gas Health and Safety of the Personnel

One of the main problems hydrogen sulphide causes is in respect to the health and safety and safety of the personnel. A maximum of eight hours exposure to concentrations greater than 100 ppm H2S will cause hemorrhage and death. Concentrations above 600 ppm can be fatal in three to five minutes. Highly stressed, high strength steel can fail in a matter of minutes in the presence of 50 ppm H_2S. At high pressure, as little as 0.1 ppm H2S can greatly reduce the time to failure of highly stressed, high strength steel. Relying solely on its odour is not a good idea because at concentrations above 100 ppm it deadens a person's sense of smell within a few minutes. The pure gas is heavier than air and can collect in low areas such as pit rooms and accommodation. below Table shows the potential hazards hydrogen sulphide can cause to the personnel from short-term exposures. There are many historical cases to document the ill effects of hydrogen sulphide exposure.
The documented cases reported in Table 2 indicate that hydrogen sulphide is very toxic and it takes only a matter of minutes to affect unprotected workers.

Health effects from short-term exposure to hydrogen sulphide	
Concentration (ppm)	**Health Effect**
0.01 – 0.3	Odour threshold
1 – 20	Offensive odour, possible nausea, tearing of the eyes or headaches with prolonged exposure
20 – 50	Nose, throat and long irritation; digestive upset and loss of appetite; sense of smell starts to become fatigued; acute conjunctivitis may occur (pain, tearing and light sensitivity)
100 – 200	Severe nose, throat and long irritation; ability to smell odour completely disappears.
500	Severe lung irritation, excitement, headache, dizziness, staggering, sudden collapse knockdown), unconsciousness and death within a few hours, loss of memory for the period of exposure (results in permanent brain damage if not rescued immediately).
500 – 1000	Respiratory paralysis, irregular heart beat, collapse and

	death without rescue.
> 1000	Rapid collapse and death

Effects of hydrogen sulphide on Metals

Hydrogen sulphide mainly cause corrosion problems to drill strings, transport pipes, storage tanks etc. Hydrogen sulphide causes sulphide stress cracking, hydrogen embrittlement and pitting corrosion in oil and gas operations. The corrosion of iron in the presence of hydrogen sulphide and water is dependent upon the dissociation of the hydrogen-sulphide molecule. Iron is oxidized to the ferrous form at the anode and hydrogen sulphide under- goes a two-step dissociation at the cathode. Tung et al[22]propose the following equilibrium reactions below.

At the anode:

$Fe \longrightarrow Fe^{2+} + 2e^-$

And at the cathode hydrogen is produced and either enters the steel or forms hydrogen gas and bubbles off:

If H_2S is present:

$2H^+ + 2e^- \; H° \rightarrow H°$

If H2S is not present:

$H° + H° \longrightarrow H_2$

Hydrogen enters the steel first by adsorption onto the water to steel interface and then by being absorbed into the steel as hydrogen. The amount of hydrogen entry depends on the corrosion rate of the steel surface and the concentration of anions such as HS- that reduce the tendency to produce hydrogen gas (as shown in second equation above) and promotes the hydrogen (Ho) to enter the steel. pH also has an influence on amount of corrosion damage. Basically, the term "*sulphides*" used in oil and gas operations includes all three water-soluble species, H2S, HS- and S2-, which coexist in a sulphide-water system. Shown in the ionization chart for hydrogen sulphide in figure 1, it can be observed that molecular H2S predominates in the acidic range, where the pH is below 7. In the range of 7 to 12, monovalent HS- predominates and above pH 12, divalent S2- predominates.

The illustration in figure 1 indicates that the most important practice is that when H2S enters in alkaline mud, it reacts to form an alkaline sulphide, most often sodium sulphide.17 While elevated pH is a good way to keep H2S from causing problems under certain conditions, it does not remove it from the fluid and any drop in pH can create a significant hazard. This then brought about the advent of sulphide scavengers in the history of oil and gas operations. The only safe method for the total removal of hydrogen sulphide or soluble sulphides is with a sulphide scavenger.

Effects of hydrogen sulphide on Drilling fluid

The first positive identification of soluble sulphide in the drilling fluids must be taken as a possible indication of H2S entering the fluid from some source. In the simplest case an increase in mud weight may be sufficient to control minor influxes. If levels persist chemical treatment will be required. This can take two forms; neutralization and scavenging. The two terms must not be confused. Neutralization of H2S in drilling fluid by caustic soda or lime is a common field approach. This alkalinity maintenance method has proved successful when small amounts of H2S enter the drilling fluid, for example, from drilling up pore spaces containing the gas.

This is not a suitable approach to the control on influxes of Hydrogen Sulphide. The alkalinity neutralization of H2S is instantly reversible by reduction in pH. Should the pH of the fluid decrease due to further influxes of H2S, Co2 or brine flows the sulphids may come out of solution as hydrogen sulphide gas. To ensure that such influxes are handled safely, it is essential to employ a sulphide scavenger. This term refers to any additive which is capable of reacting with one or more of the sulphide species and as a result, converting them to a more inert form. To be truly effective an irreversible and complete reaction must occur. As a mutual equilibrium exist between the three sulphide species in solution (H2S, HS- and S2-) irreversible and complete removal of one serves to removal all three.

To remove the problem it is necessary to precipitate the sulphide as an insoluble inorganic sulphide. Zinc or iron compounds are usually chosen because of the relative inertness of the sulphids formed. Zinc source (Zinc oxide or carbonate) has in the past been the preferred treatment. However, continuous treatment with Zinc carbonate will produce significant undesirable carbonate concentration (Carbonate contamination) which can adversely effect on drilling fluid rehology and fluid loss. See the below table:

Contaminant	FV	PV	YP	Gels	FL	pH	Pm	Pf	Mf	Cl^-	Ca^{2+}
Carbonate	↑	–	↑	↑	↗	↓	↓	↗→	↓	–	↓
H_2S	↑	–	↑	↑	↑	↓	↓	↓	↓	–	↗

Quick reference for recognizing carbonate and H_2S contamination

As a precaution against Hydrogen Sulphide influxes into the drilling fluid system should be pre- treated with Zinc Source. Zinc oxide can be used as an alternative, provided that the possible environmental payloads associated with its use are appreciated. pH and alkalinity tests (Pf/Mf) are a good first indicator of Hydrogen Sulphide influx. However, the use of scavenger is used up. It can be seen from this that while a scavenger increase safety levels, it also makes detection of small Hydrogen Sulphide influxes difficult by conventional methods. Measurement of the concentration of scavenger in the mud system is possible but the analysis method is not safe for use on the rig.

The most effective environmentally acceptable scavengers for Hydrogen Sulphide in water base muds are Zinc sources like Zinc Carbonate or Oxide, primarily because of their solubility which leads to very fast reaction rates compared to other scavengers.

Zinc Sources clearly reacts very much faster than the other scavengers. This is due to the fact that the iron in solution ensures a fast reaction with the sulphides and the chemistry in this case produces elemental sulphur as the reaction product. Elemental sulphur is completely insoluble in water. The iron is reduced from Fe^{+3} to Fe^{+2} in the process and some of this will be regenerated to produce Fe^{+3} when the mud comes in to contact with the air. Any regeneration of Fe^{+3} will aid in reducing product consumption without affecting is scavenging properties.
Recommended treatment levels of Zinc sources are as follows:
pre- emptive treatment is recommended at 1.0 lb/bbl treatment of a mud system in the event of gas influx should be made on the basis that 2.0 lb/bb; will treat approximately 100 ppm Hydrogen Sulphide.

Testing / Reporting/ Treatment for Hydrogen Sulphide

The active mud system must be regularly analyzed and monitored for the presence of active soluble Sulphide with the Garrett Gas train. This test must be carried out at the same time as each complete mud check, i.e.: three times per day while drilling.
Alkalinity checks should be carried out intermittently in between these full checks and any significant reduction in the pH and Mf/Pf should be followed up by testing for the presence of active soluble sulphides with the Garrett Gas Train.
For routine checks lead acetate paper can be used to give a qualitative check. Should this test be positive then it must be followed by a qualitative analysis with a Dragger tube.
The following actions must be taken after the initial positive test result:

1. Inform the Drilling Supervisor
2. Start adding H2S Scavenger to the active mud system. Active soluble sulphides can be analysed and monitored because of their characteristic of readily regenerating hydrogen Sulphide gas in reaction with mild acid. By adding citric acid to whole mud sample in a Garrett gas train detection of Hydrogen Sulphide on lead acetate paper indicates the presence of active Sulphides in the mud. When this qualitative test proves positive a dragger tube should be used to give qualitative results.
However it must be remembered that pre-treating the mud with MILGARD Fe means that no
Sulphides will be delectable until all the scavenger has reacted, i.e.: it is impossible to detect small influxes of Hydrogen Sulphide in a system pre- treating with MILGARD Fe .

Contingency stock to be kept on hand on the rig

The following Contingency stock of treatment chemicals must be kept on hand on the rig over and above the normal mud maintenance material requirements.
Zinc Carbonate or Oxide 120 Sxs 25 kg
Lime 200 Sxs 25 kg
As above mentioned Hydrogen Sulphide is a poisonous and dangerous acidic gas encountered in many formations. It can quickly deaden senses and can be fatal, even at low concentration. Hydrogen Sulphide is characterized by its typical “rotten egg” smell. For safety reasons, it should be neutralized immediately with caustic soda or lime to increase pH to >11.5 to form sulphide (S2-), then precipitated with a source of Zinc.
Neutralization with caustic soda:
2 NaOH + H2S 2Na+ + S2- + 2H2O (at pH>11.5)
Neutralization with maintaining excess lime:
2 Ca(OH)2 + H2S Ca2+ + S2- + 2H2O (at pH>11.5)
Converting Hydrogen Sulphide into just sulphide by increasing pH is not a permanent reaction. If the pH were drop into the acid region, sulphide will be converting back into the poisonous H2S form. For this reasons H2S should always be precipitated by adding a source of Zinc such as Zinc Oxide.
Removal by precipitation with treatments of (ZnO):
H2S + ZnO ZnS + H2O
if H2S gas detected in mud following action should be taken to:
4. Immediately rise up the pH to at least 11.5 to 12 with caustic soda.
5. Buffer the pH with lime.
6. Begin treatment with Zinc oxide to remove sulphide (S2-) from the system. *(1 lb/bbl treatment of Zinc oxide removes approximately 1100 mg/lit of sulphide. No more than 2 lb/bbl of Zinc oxide should be used to pre-treat the system for H2S)*
If it appears that H2S bleeding in (flowing) form the formation, the mud density should be
increased to shut off the flow of gas into the well-bore.

5.7 Health, Safety and Environment

Drilling fluid engineer - roles and responsibilities

- Attend Operators/Contractors safety meetings and advise on all HSE matters pertaining to our products. The requirements for Drilling Fluids Engineers is not only to attend but to contribute to Safety meetings onsite is also important, including giving
- Presentations on the fluids and chemicals being used.

- Give toolbox / pre-tour talks on chemical safety. ƒ Take part in risk assessments relevant to fluids, in particular for the first use of new systems e.g. SBM / OBM etc.,
- Follow all procedures related to HSE and follow all well site directives issued by the operator and / or drilling contractor.
- Ensure correct and updated safety posters are in place in the mud and sack rooms or mixing areas for land rigs.
- Ensure MSDS are up to date and easy to locate.
- Use engineering controls and personal protective equipment, as
- Appropriate.
- Use rig specific HSE Observation system e.g. STOP.

Hazard and risk assessment

The use of chemicals in the workplace presents hazards and risks to personnel involved in their handling and application. In order to minimize these hazards risk assessments are performed and HSE control measures and management systems are established to achieve the following:-

- Identify hazards
- Safely manage those hazards.

- Identify risks
- Where possible, eliminate those risks through control / engineering measures e.g. ventilation and collection of dust.

- Where not possible, manage those risks through processes including the use of PPE.

- Provide training and awareness systems designed to achieve the above and promote continuous improvement.

Hazardous materials

- The effect on a person of a hazardous material depends on:
- The nature of the hazardous material.
- The site of the action.
- The amount of the hazardous material involved (“dose”).
- The reaction of the individual (“susceptibility”).

Hazardous Material Effects

Local effects:

- Skin and eye irritation & burns.
- Skin defatting leading to dermatitis.

Systemic effects:

- Central nervous system (headaches, nausea, dizziness).
- Cardiovascular system (CO poisoning).
- Sensitisation (allergy) and asthma.
- Teratogenic and carcinogenic.

Chemical hazardous effects may be :

- Acute - effects lasting minutes, hours or days e.g. irritation i.e. generally short term recoverable effects.
- Chronic - effects lasting weeks, months or years e.g. occupational asthma – generally long and possibly permanent effects.
- Effects may be reversible or irreversible.



Material safety data sheet – MSDS

An assessment is made of the physical and health hazards of each chemical supplied by our company. This information is included in a Material Safety Data Sheet (MSDS) and, in part, on container labels.

Material Safety Data Sheets contain the following information:
1. Product Identification
2. Composition, information on ingredients
3. Hazard identification
4. First aid measures
5. Fire fighting measures
6. Accidental release measures
7. Handling and storage measures
8. Exposure controls, personal protection
9. Physical and chemical properties
10. Stability and reactivity

Ensure that the following practices are followed regarding MSDS information at the workplace:

- Provide active up-to-date MSDS files covering all drilling fluid chemicals on location either on CD or paper copies. This must also include the laboratory testing chemicals.
- Distribute the MSDS files to the Well site Manager/ OIM, Medic (or designate) and sack storage / mixing areas.
- Update the MSDS file list whenever a new item is received.

Mixing guidelines

Pre-Job

- Select the chemicals to be mixed.
- Review MSDS (Material Safety Data Sheets) .
- Review the wall charts.
- Obtain appropriate tools, e.g. barrel pump.
- Inspect the condition of the chemicals to be mixed.
- Obtain appropriate PPE and WEAR IT.
- Ensure mixing personnel have clear written instructions.
- Perform a Job Hazard Analysis for any new chemicals or personnel.

Mixing

- Check that hopper is running and that the correct lines valves and pits have been selected.
- Ensure sufficient extraction and ventilation in hopper area.
- Ensure that sacks and drums are conveniently positioned and use correct lifting procedures.
- Be aware of any forklift operations.
- Clean up spills as soon as possible.
- Close the hopper any time chemicals are not being mixed.

Cleaning up

- Inform derrick man, pump man or supervisor that job is complete.

- Clean up mixing area.
- Dispose correctly of empty sacks, drums and pallet waste, e.g. Banding, wood & plastic wrapping in the correct manner.
- Ensure forklift is parked in designated area with forks lowered.

6. Waste Management Program

6.1 Drilling Waste Management Design

The contractor is responsible for designing the optimum practices for treating the produced waste during the operation. The initial technical waste management proposal should be included the practices for minimizing, treating and recovering of the wastes.
According to the below categories:

- Water based mud containing minerals and natural synthetic polymers, and
- Cuttings contaminated with WBM.

The contractor is encouraged to offer possible practices to minimize the drilling fluids waste and its regeneration.

6.1.1 Water Based mud containing minerals, naturals and synthetic polymers

- **Primary operation**

  The proposed primary solids control package should be able to control drilling fluid properties, reducing of mud materials usage against conventional solid control equipment, and significantly reduce the number of excess fluids generated throughout drilling each phase of the well.

- **Secondary Operation**

  All excess mud and spillage (water based) should be collected in rig site pits and free surface fluid should be treated using a built for propose flocculation station (has to join with centrifuge) consisting of integral dosing and pumps along with a centrifuge for efficient, quick and controlled separation of the solids before the cleaned fluids is returned to active system or buffer tanks.

  This system should be able to remove colloidal sized particles that cannot be removed with primary solids control system.

  This system may use to clean up the remaining water-based mud or decrease weight of oil base mud at the end of the drilling fluids programmed.

6.1.2 Cuttings Contaminated with WBM

Spud mud and water-based mud cuttings should be fixed and be ready for transport to disposal zone. For fixing the cutting, 100-150 kg cement shall be used for each cubic meters of cutting as well as 3-4-liter Sodium Silicate for each cubic meter of cutting. (Consumption of cement and sodium silicate varies in different formations). Gypsum is not allowed for cutting fixation. This method is physically Bonds Contaminants to Prevent Leaching to Environment. Final product will pass can test and in case of approval by company man it'll send to landfill.

6.2 Waste Discharge and Handling

The Company is responsible for designing of required pits and The Contractor is responsible for procedure for burring of waste cuttings in disposal zone; supervision and managing of requirement operation for deposit oil waste cutting.
The contractor is responsible for treating all the generated waste in site without any discharges in environment or damages. It is also the responsibility of the contractor for handling/discharge of generated waste after treatment.

The contractor manages and supervises the solids and fluid waste streams loading and off-loading on/from the designated areas and responsible for the loader along with the responsibility of the storage pits, tanks, etc.

6.3 Laboratory equipment and analysis

The contractor should prepare suitable laboratory analysis equipment to perform complete analysis services of waste matter. The provided equipment shall make this opportunity to study the mentioned environmental policy of waste analysis.

6.4 Report and Comments

The contractor provides the following reports and documentations on contract duration:

- Daly and monthly waste management report.
- Final waste management report, including:
  - Summarized amount of generated waste on each section,
  - Total treated waste,
  - Total chemical consumption,
  - Cost monitoring report, and
  - Special activities.

6.5 The Equipment Installation and Commission

- **First location**

The Maximum days for installation and commissioning of all equipment and necessary piping will be 7 days after rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

- **Other locations**

The Installation of all equipment should be completed during rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

6.6 Clean Up Process

After drilling operations and drilling rig removal, the well area, waste pit, and corral should be completely cleaned. The contractor shall remove the fixed cuttings to pre-determine locations (disposal) which can protect the environment is very effective.

6.7 Supplying Services

Waste treatment package

- **Dewatering unit**
- **Filter press**

Transfer pump package

- **Air diaphragm pump**

- **Submersible pump**
- **Centrifugal pump**

Storage tank

- **Waste water storage tank**

Cutting treatment

- **Loader**

Laboratory equipment's

- **Photo meter**
- **Jar Test**
- **pH /EC meter**
- **Retort kit**
- **Solutions**
- **Hot plate magnet**

Other

- **Wash gun**
- **Laboratory and cabin**

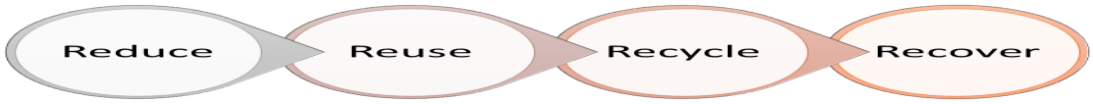
6.8 Waste Management Policy

Waste management services should be utilized to minimize generated waste and contaminated discharged cuttings during drilling operations. This helps keep the environment clean and ensures that our operations are in compliance with DOE (Department of Environment) legislations.

We provide cost-effective and optimum solutions from environmentally acceptable procedures to achieve zero-discharge plans and improve fluids recovery. The outcome reduces operational costs and enables recycling.

The waste is analysed to study various parameters such as chloride, nitrate, and phosphate. Other analysis is carried out to monitor treatment processes on a regular basis during waste generation. Our waste management personnel are fully trained experienced and have the competencies in place to provide the best reliable services with concentrated support and the least liabilities.

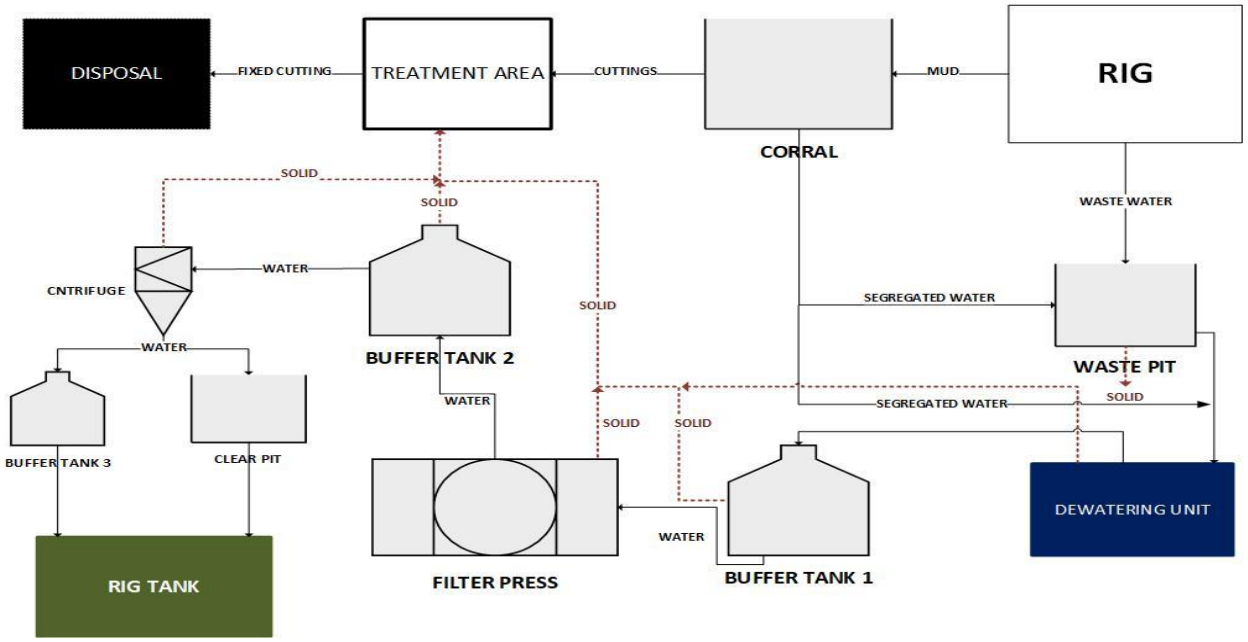
6.9 Drilling Waste Management Steps



6.10 Calculation

6.10.1 Waste Water Treatment and Disposal

6.10.1.1 Dewatering System



6.10.1.2 Estimated Volume

	unit					Total
Hole Size	Inch	26	17.5	12.25	8.5	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	150	2182	3915	4180	-
Mud system	-	Bentonite Mud	Salt saturated KCl, etc. mud	Salt saturated KCl, etc. mud	OBM	-

Mud weight	PCF	63-70	75-95	105-125	55.5-57	-
Interval	Meter	150	2032	1733	265	-
Operating duration	Day(s)					
Estimated mud volume	M^3	221	1095	971	530	2817
	bbl	1370	6790	6018	3288	17466
Estimated Treating waste water	M^3	133	657	3147	2447	6384
	bbl	822	4074	19511	15173	39579

6.10.1.3 Dewatering Chemical

	Application	**Required concentration**
Poly Aluminum chloride	Coagulant	0.5~1.5 Kg/m^3
Flocculent polymer	Flocculant	0.1~0.3 Kg/m^3
Hydrated lime	Coagulant	0.25~0.5 Kg/m^3
Soda-ash	Harness control	0.5 ~ 0.6 Kg/m^3 decreasing 1000 mg/Lit of Ca, at average
Caustic soda	pH control	0.5 ~ 0.6 Kg/m^3 Depended on hardness
Biocide	Microbial activity control	20 ml/ m^3

6.10.1.4 Treated Water Properties

Item	**Amount**
pH	7~9
Total hardness	< 800 mg/liter
Turbidity	< 50 NTU
TSS	< 80 mg/liter
Nitrate	< 50 mg/liter
Aluminum	< 3 mg/Lit
Sulfate	< 50 mg/liter
Ca^{2+}	< 600 mg/liter

6.10.1.5 Material Consumption Estimation

	unit					**Total**
Hole Size	Inch	26	17.5	12.25	8.5	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	150	2182	3915	4180	-
Interval	Meter	150	2032	1733	265	-
Operating	Day(s)	-	-	-	-	-

duration						
Poly alum	Dosage of kg/M^3			0.5~1.5		
	25 kg sxs	5	26	126	98	255
Poly electrolyte	Dosage of kg/M^3			0.1~0.3		
	25 kg sxs	1	5	25	20	51
Soda ash	Dosage of kg/M^3			1		
	25 kg sxs	5	26	126	98	255
Caustic soda	Dosage of kg/M^3			0.5~0.6		
	25 kg sxs	4	20	94	73	192
Biocide	Dosage of ml/M^3			20		
	25 kg sxs	27	131	629	489	1277
Lime	Dosage of kg/M^3			0.5~1		
	25 kg sxs	5	26	126	98	255
Citric acid	Dosage of kg/M^3			0.3~0.5		
	25 kg sxs	2	8	38	29	77

6.10.2 Waste Management Equipment

6.10.2.1 Flocculation Unit

This unit, has the bellow specifications:

- Fully automated system,
- Full 316 stainless steel material,
- Eqipped with high pressure level and pH sensors,
- Equipped with digital flow meter and totalizer to monitor flow rate, and
- Touch panel technology to control prose.







6.10.2.2 400 bbl Dewatering Unit

This tank provides more than 400 bbls capacity to store contaminated water for using in dewatering process and ability to adding chemicals such lime from hopper. Two Mixer 25 hp and Jet System give ability to holding any type of mud for hours without any stealing.



6.10.2.3 400 bbl Storage Tank

Each tank provides more than 400 bbls capacity to store clear water for recycling.

6.10.2.4 High Performance Filter Unit

Filtration plays an important role in the natural treatment of waste water as it percolates through the soil. It is also a major part of most water treatment. Groundwater that has been softened, or treated through iron and manganese oxidation, requires filtration to remove floc created by coagulation or oxidation processes. Since surface water is subject to run-off and does not undergo natural filtration.

it must be filtered to remove particles and impurities. Filtration can be compared to a sieve or micro- strainer that traps suspended material between the grains of filter media. However, since most suspended particles can easily pass through the spaces between grains of the filter media, straining is the least important process in filtration. Filtration primarily depends on a combination of complex physical and chemical mechanisms, the most important being adsorption. Adsorption is the process of particles sticking onto the surface of the individual filter grains or onto the previously deposited materials. Forces that

attract and hold particles to the grains are the same as those that work in coagulation and flocculation. In fact, coagulation and flocculation may occur in the filter bed, especially if coagulation and flocculation before filtration was not properly controlled. Incomplete coagulation can cause serious problems in filter operation.



6.10.2.5 Centrifugal pump

This pump can be utilized in different situation. 3 x 4 pump can be used for transferring processed water back to rig water tanks for reuse.



6.10.2.6 Submersible Pump

These pumps designed to clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.



6.10.2.7 200 bar wash down guns

These wash guns minimized water consumption and can be used on the shakers to unplug screens or washing al the area on the rig from rig floor to mud pits.



6.10.2.8 Fluid transfer diaphragm pump

This pump works with air pressure and has the responsibility of transferring mud and contaminated water.
Three inches' diaphragm pump transfers mud and waste from cellar to coral to keep cellar empty.



Two inches' diaphragm pumps transfer waste water and mud between different spots on location. They also can be utilizing to transfer mud from mud tanks dead volume to treatment tanks.



These pumps designed to transfer clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.

6.10.2.9 DWM laboratory cabin

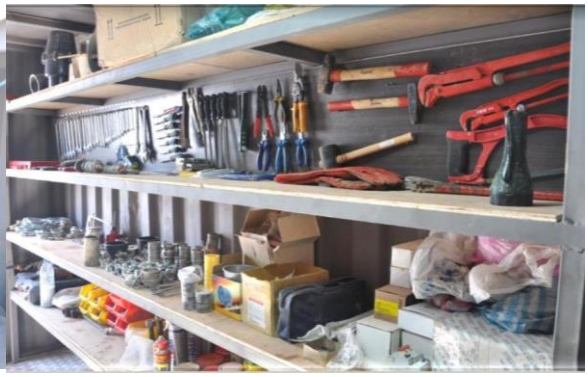
Lab equipment for DWM is consisting of:

- Retort kit,
- API Filtrate test kit,
- Testing reagents and equipment for Chloride, Calcium, Hardness and PH, and
- Hatch Water Test kit




This kits analyst all concerned parameters, including aluminum, iron, nitrate, nitrite, zinc, phosphate, sulphate, sulphide, copper and cyanide concentration.

6.10.2.10 Storage Container

6.10.2.11 Loader





7. Cement Program

Summary:

- Cement of the well is part of the well integrity barrier envelope. In addition, cement protects the casing from corrosion due to potential formation fluids on the outside well.
- Cementing in the 17 ½" section returns to surface has historically been a challenge. This is because of the formation strength below the 20" surface casing is not sufficient to bring cement all them way to the surface.
- Cementing the 12 ¼" section is as equally challenging as for the 17 ½" section, due to small limit between pore pressure and fracture pressure.
- 20" surface casing will be set @±150m. Cemented up to surface till return to surface (cellar floor).
- 13 ⅜" intermediate casing will be set @±2182m. Cemented up to cellar floor. 100 % excess on slurries for open hole-Casing volume & 20% excess on slurries for casing-casing volume.
- 9 ⅝" production casing will be set @±3915m. It will be cemented up to cellar floor. 100% excess on slurries for open hole-Casing volume & 20% excess on slurries for casing-casing volume.
- 7" liner will be set @ ±4199 mMD. It will be cemented up to liner lap @ ±3862.5 m (80 m overlap on production CSG). 100% excess on slurries for open Hole-Casing volume & 20% excess on slurries for casing-casing volume.

General Procedure:

Prior to each cement job, the unit will be checked accurately. Fluid ends, engines, etc.

The lines will have to be flushed with the mixing water. Cementing heads, plugs will be inspected before the job.

The plugs will be inserted into the cementing head in presence of the Drilling Operation Representative.

Bumping the plug is MUST.

The cement density will be checked, using pressurized balance.

Volume calculations & displacement will be done by & cement engineer.

Rig pump should be used to displace the cement.

Important: The following program is a sample; therefore, it should be tested using the field drilling water, cement sample and additives.

Cement table:

Casing/liner		20" Surface Casing	13-3/8" Intermediate Casing in Block I	13-3/8" Intermediate Casing in Block II & III	9-5/8" Production Casing in Block I	9-5/8" Production Casing in Block II & III	7" Production Liner in Block I	7" Production Liner in Block II & III
Cementing Method		Stab-in cementing	Double Plugs	Double Plugs	Double Plugs	Double plugs	Rotating/Non-Rotating Liner	Rotating/Non-Rotating Liner
Top of Cement		Surface	Surface				Top of Liner	
Slurry Weight (sg)	Lead	NA	1.30-1.50		1.9-2.08	1.30-1.35	1.35	1.9
	Tail	1.85-1.89	1.9		1.9		1.9	
Slurry Excess		80%-100% OH volume or return to surface	50%-100% OH volume depending on hole condition & 20% for cased hole annular volume				30% over caliper or 60% OH for tail and 40% for lead slurry & 20% for cased hole annular volume	
Thickening time (min)		≥Cementing time plus 40min	Lead ≥Cementing time plus 60min Tail ≥Pumping tail time+Displacement time plus 60min				Cementing time plus 90min	
Fluid loss(ml)		NA	LEAD/TAIL≤150		LEAD ≤80 TAIL≤50		LEAD/TAIL≤50	
Free water (ml/250ml)		NA	LEAD/TAIL : 0					
Initial consistency (BC)		≤30	LEAD/TAIL : ≤30					
Compressing strength (MPa/24h)		≥3.5	LEAD : ≥7 TAIL : ≥14					≥14

Casing/Liner String	Lead Cement (PCF)	Tail Cement (PCF)	Interval (mt.)	Length (m
20"	-	-	0-150	149
13-3/8"	-	-	0-2182	2181
9-5/8"	-	-	0-3915	3914
7"	-	-	3815-4180	364

Note: Lead Cement and Tail Cement Weight will be to update and and issue in due time for each Section.

Centralizer placement based on FEED:

(Base on Cementing simulation prog. number of Centrlizer might be change)

Cement program and centralization pattern

Casing/Liner	20"	13-3/8"	9-5/8"	7"
CMT Class	"G"	"G"	"G"	"G"
Excess (OH-CSG)	50-100%	50-100%	50-100%	40-60%
Excess (CSG-CSG)	20%	20%	20%	20%+7Std Wet
Lead Cement (PCF)	117.5	90.5	85-90	-
Tail Cement (PCF)		115.5	115-118	115-118
CMT Additives	It will be issued in due time			
Thickening Time (min)	It will be issued in due time			

"Cement program and centralization pattern will be update and issued in due time For Each Section"

8. Bit Hydraulic program

8.1 26” hole Section

26 " KHD115GC (660.4mm)

KINGDREAM江钻股份

- **Rubber O-ring sealed roller bearing**

Sealed roller bearing structure. Surface of the thrust bearing couple are specially treated for decreasing friction and increasing rigidness, load capacity of the bit and anti-galling ability of the bearing are consequently improved.

- **Cutting structure**

Hardfaced gage and all tooth surfaces, and the gage fitted with compacts for enhanced gage protection.

- **Integrated style**

The bit's structure is integrated style as same as small and middle size bits. The leg is integrally forged, so as to increase strength and reliability of bit effectively.

- **Special gage protection for cone**

Add one row of trimmers between heel row and gage row, which can repair wall of well as well as protect cone shell.

- **Reinforced bit head OD**

Cutter arrangement on head OD area is utilized to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.

- **Hardfacing on shirttail**

This feature can provide better protection for seal and bearing to extend the bit's service life.

- **Center jet (nozzle)**

Center jet can avoid bit balling, eliminate resorting area of fluid at bottom hole, expedite upward flow of cuttings and improve ROP. It is suitable for drilling in soft to medium soft formations with low compressive strength and high drillability.

Technical specifications

IADC	1-1-5
Bearing and seal	"O" ring sealed roller bearing
Number of rows	3—3—3
Number of cutters	86
Main cutter shape / Cutter exposure	Wedge teeth //68mm
Gage row cutter shape	Wedge teeth
Type of nozzle	S5
Type of center nozzle	SC4
Shank	7 5/8 REG
Recommended makeup torque	46.1~54.2 (KN.m)

Recommended operating parameters

WOB：165 ～ 460（KN）
Ideal WOB range：140～280（KN）
Rotary speed： 200 ～ 80 （rpm）
Ideal Rotary speed range：190～100（rpm）
The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

8.2 17 ½" Hole Section

17 1/2 " KGA115GCT
(444.5mm)

ULTRATEC

- **Roller bearing and rubber seal**

Roller bearing design suitable for high WOB and high RPM is used. Conduct friction reducing treatment and hardening treatment on secondary surface of thrust bearing. Bearing O ring seal adopts high saturation nitrile butadiene rubber(NBR) which can bear high abrasiveness and high temperature, larger seal ring has decreased sealing force, and protruding part on the outside of seal has increased the reliability of sealing.

- **Head OD enhancement**

Adopt teeth layout on head OD to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.

- **Hardfacing on shirttail**

This feature can provide better protection for seal and bearing to extend the bit's service life.

- **Special gage protection for cone**

Add one row of trimmers between heel row and gage row, which can repair wall of well as well as protect cone shell.

- **Center nozzle**

Center nozzle can avoid bit balling, eliminate fluid stagnant area at bottom hole, increase rising velocity of drilling cuttings, and improve ROP, which is suitable for drilling formations from soft to middle soft with low compressive strength and high drillability.

Technical specifications

IADC	1-1-5
Bearing and seal	Roller bearing and rubber seal
Number of rows	3—2—2
Number of cutters	70
Main cutter shape / Cutter exposure	Hardfaced tooth surface
Gage row cutter shape	
Layout of diamond enhanced inserts	/
Type of nozzle	S5
Type of center nozzle	S4
Shank	7 5/8 REG
Recommended makeup torque	46.1～54.2(KN.m)

Recommended operating parameters

WOB： 110～310 (KN)
Ideal WOB range：120～270 (KN)
Rotary speed： 80～200 (rpm)
Ideal rotary speed range：100～180 (rpm)
The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

Applicable formations

Very soft with low compressive strength and high drillability, such as soft shale, clay and salt bed, etc.

17 1/2 " KGA115GCT
(444.5mm)

ULTRATEC

- **Roller bearing and rubber seal**

Roller bearing design suitable for high WOB and high RPM is used. Conduct friction reducing treatment and hardening treatment on secondary surface of thrust bearing. Bearing O ring seal adopts high saturation nitrile butadiene rubber(NBR) which can bear high abrasiveness and high temperature, larger seal ring has decreased sealing force, and protruding part on the outside of seal has increased the reliability of sealing.

- **Head OD enhancement**

Adopt teeth layout on head OD to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.

- **Hardfacing on shirttail**

This feature can provide better protection for seal and bearing to extend the bit's service life.

- **Special gage protection for cone**

Add one row of trimmers between heel row and gage row, which can repair wall of well as well as protect cone shell.

- **Center nozzle**

Center nozzle can avoid bit balling, eliminate fluid stagnant area at bottom hole, increase rising velocity of drilling cuttings, and improve ROP, which is suitable for drilling formations from soft to middle soft with low compressive strength and high drillability.

Technical specifications

IADC	1-1-5
Bearing and seal	Roller bearing and rubber seal
Number of rows	3—2—2
Number of cutters	70
Main cutter shape / Cutter exposure	Hardfaced tooth surface
Gage row cutter shape	
Layout of diamond enhanced inserts	/
Type of nozzle	S5
Type of center nozzle	S4
Shank	7 5/8 REG
Recommended makeup torque	46.1～54.2(KN.m)

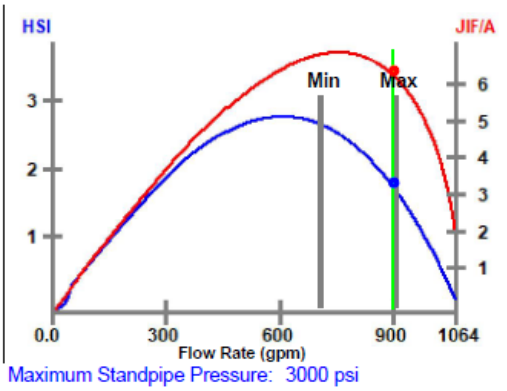
Recommended operating parameters

WOB： 110～310（KN）
Ideal WOB range：120～270（KN）
Rotary speed： 80～200（rpm）
Ideal rotary speed range：100～180（rpm）
The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

Applicable formations

Very soft with low compressive strength and high drillability, such as soft shale, clay and salt bed, etc.

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 3000 psi

Hydraulic Details @ 7159 ft

Flow Rate / Hole Diameter	50.8 gpm/in
System Pressure Loss	2132 psi
Bit Nozzle Pressure Drop	862 psi
Annular Pressure Drop	38 psi
Pump Input Hydraulic Power (M.E.=85%)	1553 hp
Bit Hydraulic Horsepower	447 hp
Jet Impact Force	1544 lbs
Impact Force/ Hole Area	6.42 lbs/in²
Hydrostatic Head	4347 psi
Equivalent Circulating Density	11.8 ppg
Cuttings Slip Velocity (Chip Size = .30 in)	43.56 ft/min
Surface Equipment Length	150 ft X 3.0 in
Buoyant Collar Weight	78679 lbs

Well Information

Well Name:
F20- 17.5" PDC

Drilling Contractor:

Operator:

Prepared For:

Location:

Prepared By:

Bit Run Data

Bit Number	0
Bit Size	17 1/2 in
Bit Type	PDC

Recommended Hydraulics

Nozzles	12\12\12\12\12\12\12\12\12 32nds
Total Flow Area	0.99 in²

	IN	OUT
BIT HSI	1.41	1.86
JIF/Area	4.89	6.42 lbs/in²
Nozzle Velocity	288	288 ft/sec
Flow Rate	889	889 gpm
SPP	1635	2994 psi
% Pressure @ Bit	40.1%	28.8%
Measured Depth	2788	7159 ft
TVD	2788	7159 ft
Mud Weight	8.9	11.7 ppg
Plastic Viscosity	10	15 cp
Yield Point	24	40 lbs/100 ft²

Drill String And Annular Velocity Data @ 7159 ft

Depth ft	OD X ID in	Length ft	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. ft/min	Est. Chip Vel. ft/min	Crit Vel. ft/min	Flow Type
0-492	DP 5 x 4.276	492	19.124	94.8	2.0	64	20	370	LAM
492-6522	DP 5 x 4.276	6030	17.500	1162.4	30.6	77	34	370	LAM
6522-6704	HWDP 5 x 3	182	17.500	193.6	.9	77	34	370	LAM
6704-6977	DC 8.5 x 3	273	17.500	290.4	2.3	93	50	373	LAM
6977-7159	DC 9.75 x 3	182	17.500	193.6	1.9	103	60	375	LAM

8.3 12 ¼" Hole Section

12 14 HAT117G.pdf

12 1/4 " KHJ517GT
(311.1mm)

ULTRATEC

- **Metal face sealed journal bearing**

HJT series bit uses journal bearing and metal face seal; it can perform consistently in high speed drilling application.



- **Head OD reinforcement**

Head OD are designed with carbide inserts to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.

- **Hardfacing on shirttail**

This feature can provide better protection for seal and bearing to extend the bit's service life.

- **Special gage protection on cone**

A row of trimmers is added between heel row and gage row compacts. As a result, not only cone shell is protected, but also the borehole wall is properly trimmed.

Technical specification

IADC	5-1-7
Bit diameter（in）	12 1/4
Shank	6 5/8 REG
Recommended makeup torque	37.9~43.3 (KN.m)
Type of nozzle	S4
Type of center nozzle	N/A
Journal Angle	33°
Offset	9.53mm
Seal and bearing	Metal face sealed journal bearing
Number of rows	4-4-4
Inner row	Offset crested scoop compacts
Outer row	Wedge compacts

Recommended operating parameters

WOB： 110 ～ 310（**KN**）

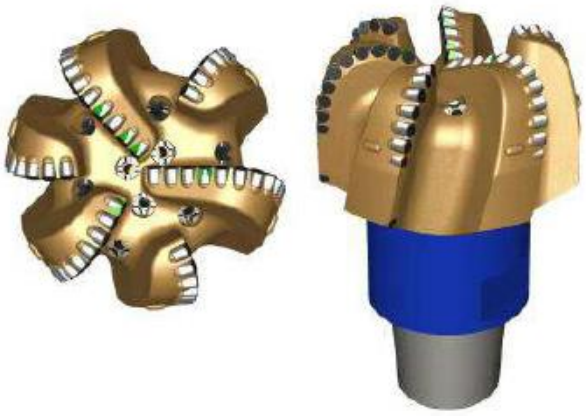
Rotary speed： 50 ～ 240 （**RPM**）

The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

Applicable formations

Soft with low compressive strength and high drill-ability, such as shale, clay ,sandstone and soft limestone, etc .

12.25"M1664 Hybrid Bit Specification



Product Specifications	
Size:	12.25"
Type:	M1664
Part ID:	038709-045C
Body Material:	Matrix
Blade Qty:	6
Active Gauge Length ("):	1.6
Passive Gauge Length("):	4.5"
Connection:	6-5/8"API REG
Junk Slot Area (sq.in):	31.9
Make up Length ("):	16.4
IADC Code:	M323

Nozzles & Ports	
Type	Quantity
Variable Nozzles	9
Ports	0

Cutting Structure		
Type	Diameter	Quantity
Sharper Cutter	16	5
Primary Normal Cutter	16	39
Back-up Cutter		0
Active Gauge Cutter	16,13	13
Total		**57**

Operating Parameters	
WOB (klbs):	7-32
RPM(rpm):	60-260
Flow Rate (gpm):	444-1110
Make up Torque (kftlb):	38-42

Design Features & Application

Feature:

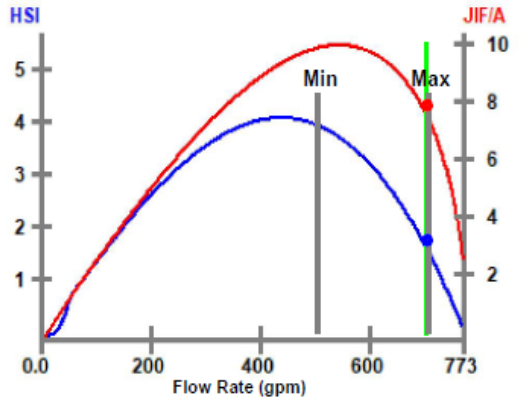
It is a matrix body PDC bit with 6 curved blades. Force-balance design and Low-friction Gauge improve anti-whirl ability .CFD technology is applied in optimizing hydraulic efficiency, enhance the anti-balling ability.

Application:

Application formation: Soft to Medium formation with low compressive strength and good drillability . such as Clay stone , Marl, Lignite, Sandstone, Tuff.

All information on this specification sheet is confidential.

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 3000 psi

Hydraulic Details @ 12845 ft

Flow Rate / Hole Diameter	56.9 gpm/in
System Pressure Loss	2462 psi
Bit Nozzle Pressure Drop	538 psi
Annular Pressure Drop	61 psi
Pump Input Hydraulic Power (M.E.=85%)	1220 hp
Bit Hydraulic Horsepower	219 hp
Jet Impact Force	940 lbs
Impact Force/ Hole Area	7.98 lbs/in²
Hydrostatic Head	7533 psi
Equivalent Circulating Density	11.39 ppg
Cuttings Slip Velocity (Chip Size = .30 in)	41.94 ft/min
Surface Equipment Length	150 ft X 3.0 in
Buoyant Collar Weight	99907 lbs

Well Information

Well Name:
F20-12.25
Drilling Contractor:
MSA/OEOC OEOC208
Operator:
MSA-OEOC
Prepared For:
PPL
Location:
South Azadegan
Prepared By:
Engineering Team/MSA

Bit Run Data

Bit Number	0
Bit Size	12 1/4 in
Bit Type	PDC

Recommended Hydraulics

Nozzles	14\14\14\13\13\13\13 32nds
Total Flow Area	0.96 in²

	IN	OUT
BIT HSI	1.58	1.86
JIF/Area	6.78	7.98 lbs/in²
Nozzle Velocity	233	233 ft/sec
Flow Rate	697	697 gpm
SPP	1672	3000 psi
% Pressure @ Bit	27.3%	17.9%
Measured Depth	7159	12845 ft
TVD	7159	12845 ft
Mud Weight	9.6	11.3 ppg
Plastic Viscosity	10	25 cp
Yield Point	10	18 lbs/100 ft²

Drill String And Annular Velocity Data @ 12845 ft

Depth ft	OD X ID in	Length ft	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. ft/min	Est. Chip Vel. ft/min	Crit Vel. ft/min	Flow Type
0-492	DP 5 x 4.19	492	19.124	71.6	.4	50	8	258	LAM
492-7159	DP 5 x 4.19	6667	12.415	969.9	27.8	132	90	268	LAM
7159-12118	DP 5 x 4.19	4959	12.250	721.4	21.9	137	95	269	LAM
12118-12300	HWDP 5 x 3	182	12.250	132.5	.8	137	95	269	LAM
12300-12845	DC 8.5 x 3	545	12.250	396.8	9.9	220	178	289	LAM

8.4 8 ½" Hole Section

8 1/2 KGA215G(215.9mm)

ULTRATEC

- **Roller bearing and rubber seal**



Roller bearing design suitable for high WOB and high RPM is used. Conduct friction reducing treatment and hardening treatment on secondary surface of thrust bearing. Bearing O ring seal adopts high saturation nitrile butadiene rubber(NBR) which can bear high abrasiveness and high temperature, larger seal ring has decreased sealing force, and protruding part on the outside of seal has increased the reliability of sealing.

- **Head OD reinforcement**

Head OD are designed with carbide inserts to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.

- **Hardfacing on shirttail**

This feature can provide better protection for seal and bearing to extend the bit's service life.

Technical specifications

IADC	2-1-5
Bearing and seal	Roller bearing and rubber seal
Number of rows	4—3—3
Number of cutters	136
Main cutter shape / Cutter exposure	Hardfaced tooth surfaces
Gage row cutter shape	Hardfaced tooth surfaces
Layout of diamond enhanced inserts	/
Type of nozzle	S4
Type of center nozzle	None
Shank	4 1/2 REG
Recommended makeup torque	16.3~21.7 (KN.m)

Recommended operating parameters

WOB: 64 ~ 150 (KN) Ideal WOB range: 85 ~130 (**KN**)

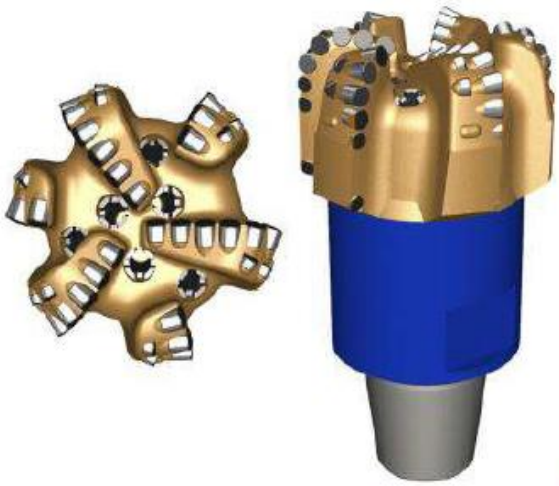
Rotary speed: 180 ~80 (**rpm**) Ideal rotary speed range: 140 ~100 (**rpm**)

The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

Applicable formations

Medium hard with hard stringers, such as hard shale, sandstone and limestone, etc.

8 1/2" M1665 Bit Specification



Product Specifications	
Size:	8 1/2"
Type:	M1665
Body Material:	Matrix
Blade Qty:	6
Active Gauge Length ("):	1.6
Passive Gauge Length("):	2
Connection:	4-1/2"API REG
Junk Slot Area (sq.in):	14.5
Make up Length ("):	11.7
IADC Code:	M323

Nozzles & Ports	
Type	Quantity
Variable Nozzles	6
Ports	0

Cutting Structure		
Type	Diameter	Quantity
Primary Cutter	16	24
Back-up Cutter	13	12
Active Gauge Cutter	16,13	12
Total		48

Operating Parameters	
WOB (klbs):	5-23
RPM(rpm):	60-260
Flow Rate (gpm):	350-555
Make up Torque (kftlb):	18-21

Design Features & Application

Feature:

It is a matrix body PDC bit with 6 curved blades. Force-balance design and Low-friction Gauge improve anti-whirl ability .CFD technology is applied in optimizing hydraulic efficiency, enhance the anti-balling ability.

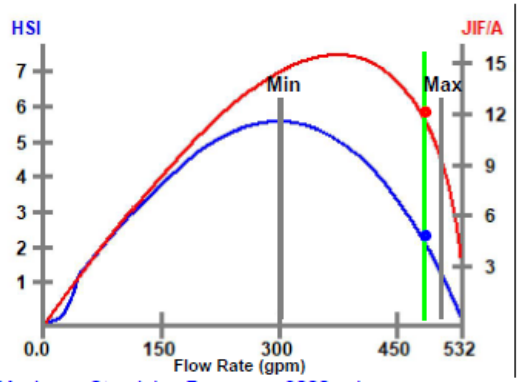
Application:

Application formation: soft to medium hard formation.

BESTE 百施特 BESTEBIT

All information on this specification sheet is confidential.

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 3000 psi

Well Information

Well Name:
F20-8.5
Drilling Contractor:
MSA/OEOC OEOC208
Operator:
MSA/OEOC
Prepared For:
PPL
Location:
South Azadegan
Prepared By:
Engineering Team/MSA

Bit Run Data

Bit Number	0
Bit Size	8 1/2 in
Bit Type	PDC

Hydraulic Details @ 13714 ft

Flow Rate / Hole Diameter	56.5 gpm/in
System Pressure Loss	2503 psi
Bit Nozzle Pressure Drop	495 psi
Annular Pressure Drop	309 psi
Pump Input Hydraulic Power (M.E.=85%)	840 hp
Bit Hydraulic Horsepower	138 hp
Jet Impact Force	694 lbs
Impact Force/ Hole Area	12.23 lbs/in²
Hydrostatic Head	10036 psi
Equivalent Circulating Density	14.53 ppg
Cuttings Slip Velocity (Chip Size = .30 in)	29.94 ft/min
Surface Equipment Length	150 ft X 3.0 in
Buoyant Collar Weight	69068 lbs

Recommended Hydraulics

Nozzles	13\13\13\13\13\13 32nds
Total Flow Area	0.77 in²

	IN	OUT
BIT HSI	2.3	2.44
JIF/Area	11.53	12.23 lbs/in²
Nozzle Velocity	200	200 ft/sec
Flow Rate	480	480 gpm
SPP	2437	2998 psi
% Pressure @ Bit	19.1%	16.5%
Measured Depth	12845	13714 ft
TVD	12845	13714 ft
Mud Weight	13.3	14.1 ppg
Plastic Viscosity	26	46 cp
Yield Point	12	26 lbs/100 ft²

Drill String And Annular Velocity Data @ 13714 ft

Depth ft	OD X ID in	Length ft	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. ft/min	Est. Chip Vel. ft/min	Crit Vel. ft/min	Flow Type
0-492	DP 5 x 4.19	492	19.124	48.6	.4	35	5	282	LAM
492-7159	DP 5 x 4.19	6667	12.415	658.2	31.0	91	61	297	LAM
7159-12845	DP 5 x 4.19	5686	8.535	561.4	190.4	246	216	333	LAM
12845-12895	DP 5 x 4.19	50	8.500	4.9	1.7	249	219	334	LAM
12895-13077	HWDP 5 x 3	182	8.500	89.9	6.3	249	219	334	LAM
13077-13714	DC 6.5 x 2.5	637	8.500	757.9	78.9	392	362	392	LAM

*****Note: Bit Specification in each Hole Section will be Finalized in due time.*****

9. Casing Program

9.1 Casing Stress check & Casing Specification

Casing Size (in)	Depth (m) from	Depth (m) to	ppf	Grade	Connection	Collapse Pressure(psi)	Burst (psi)	ID (in)	Drift ID (in)	CSF	BSF	TSF
20"[1]	0	150	94	K-55	BTC	520	2110	19.124	18.936	2.11	1.23	1.6
13-3/8"[2]	0	2182	68	N-80	BTC	2260	5020	12.415	12.259	1.34	1.65	1.2
9-5/8"[3]	0	3915	53.5	110SS	Premium Gas Tight Connection	7,950	10,900	8.535	S.D: 8.5	1.39	1.34	1.45
7"	3815	4180	32	110SS	Premium Gas Tight Connection	10,780	12,460	6.094	5.969	1.61	6.184	S.D.

[1] Considering full casing volume evacuation for collapse design and full column SW (normal gradient) with MAASP for burst design.
[2] Considering 2/3 casing volume evacuation for collapse design and full column SW (normal gradient) with MAASP for burst design.
[3] Considering 2/3 casing volume evacuation for collapse design and full column oil with MAASP for burst design.
[4] Considering full casing volume evacuation for collapse design and full column oil with MAASP for burst design.

9.2 26" Hole Section, 20 " Casing

150m in Aghajari FM

Important Note:

1. The casing shoe must be duplex, stab in, NON-DIFFERENTIAL type.
2. Fill casing by non-LCM fresh mud. Consult about casing filling program by operation manager.
3. Tighten casing with chain before cementing.

String Design

- 20" stab-in float Shoe, 94 ppf, K-55, BTC.
- ±150 m, 20" Casing, 94 ppf, K-55, BTC.
- Centralizer: Shall be mentioned after simulated and programed in CMT plan.

Burst design criteria: Full column SW with MASP

Collapse design criteria: Full casing evacuation

Casing Specifications

Pipe OD (in)	20
Weight (lb/ft)	94
Grade	K-55
Connection	BTC
Pipe ID (in)	19.124
Collapse Resistance (psi)	520
Pipe Body Internal Yield (psi)	2110
Pipe Body Yield (KLbs)	1480
CSF: 2.11, **BSF:** 1.23, **TSF:**1.6	
Make Up Torque (ft-lb), optimum	Based on triangle

9.3 17 ½" Hole Section, 13 3/8" Casing

2182m (+/-5m TVD below top of Gurpi FM)

String Design

- 13 3/8" Float Shoe, 68 ppf, N-80, BTC.
- Two Joints (Approx.±24m) 13 3/8" Casing, 68 ppf, N-80, BTC.
- 13 3/8" Float Collar, 68 ppf, N-80, BTC.
- 13 3/8" Casing, 68 ppf, N-80, BTC, **(0-2182m)**
- Centralizer: Shall be mentioned after simulated and programed in CMT plan.

Burst design criteria: Full column salt water (normal gradient) with MASP
Collapse design criteria: 2/3 casing volume evacuation

Casing Specifications

Pipe OD (in)	13 3/8
Weight (lb/ft)	68
Grade	N-80
Connection	BTC
Pipe ID (in)	12.415
Collapse Resistance (psi)	2260
Pipe Body Internal Yield (psi)	5020
Pipe Body Yield (kips)	1556
CSF: 1.34, **BSF:** 1.65, **TSF:1.68**	
Make Up Torque (ft-lb), optimum	Based on triangle

9.4 12 ¼" Hole Section, 9 5/8" Casing

3915m (top of Fahliyan)

String Design

- 9-5/8" Float shoe, 53.5 ppf, 110S, Premium Gas tight connection
- Two Joints 9-5/8" Casing 53.5 ppf (Approx.±24m), 110S, Premium Gas tight connection
- 9-5/8" float collar, 53.5 ppf, 110S, Premium Gas tight connection
- 9-5/8" Casing 53.5 ppf, 110S, Premium Gas tight connection

Burst design criteria: Full column oil with MAASP
Collapse design criteria: 2/3 casing evacuation

- Centralizer: Shall be mentioned after simulated and programed in CMT plan.
  - During Casing running TMS (Torque Monitoring System) shall be used.

Casing Specifications

Pipe OD (in)	9 5/8
Weight (lb/ft)	53.5
Grade	110S
Connection	Premium Gas tight connection
Pipe ID (in)	8.535
Collapse Resistance (psi)	7,950
Pipe Body Internal Yield (psi)	10,900
Pipe Body Yield (KLbs)	1,710
CSF: 1.39, BSF:1.34, TSF:1.45	
Make Up Torque (ft-lb), optimum (for V. Top Conn)	19,080

9.5 8 ½" Hole Section, 7" Liner

4180m (Fahliyan)

String Design

7" Float Shoe, 32ppf, C-110, Premium gas tight connection, 6.094".

- Two Joints 7" Liner, 32ppf, 110SS (Approx.±24m), Premium gas tight connection, 6.125" S.D.
- 7" LDC Float Type, 32ppf, C-110, Premium gas tight connection, 6.094".
- 7" Liner, 32ppf, 110SS, Premium gas tight connection 6.094".

Burst design criteria: Full column oil with MAASP
Collapse design criteria: Full casing evacuation

- Hydraulic set liner hanger assembly

Liner Specifications

Pipe OD (in)	7"

Weight (lb/ft)	32
Grade	110SS
Connection	Premium gas tight Conn
Pipe ID (in)	6.094
Wall Thickness (in)	0.453
Collapse Resistance (psi)	10,780
Pipe Body Internal Yield (psi)	12,460
Pipe Body Yield (KLbs)	1,025
CSF: 1.35**, BSF:** 1.61,	
Make Up Torque (ft-lb), optimum (for V. Top Conn)	13,860

9.6 Material Selection

Casing

Corrosive Environment Analysis and Strength Check

Based on the data of offset wells in this oilfield, there is a certain degree of corrosive problems in the 4 layers to be produced. Corrosive medium is principally composed of 2 parts: one is the corrosive gases like H_2S and CO_2 existing in the produced fluid; and the other is the high salinity formation water. The contents of main corrosive fluids in offset wells are shown in Table.

H2S and CO2 Content

Well No	*Reservoir formation*	*H ₂S from PVT*	*CO ₂ from PVT (mol%)*	*H ₂S from Seperator*	*CO ₂ from Separator*
AZN2	Sarvak	0.13	6.71	7.93/150psi	0.06/150psi
AZN4	Sarvak	0.06	5.44	0.02/150	6.32
	Kazhdumi	0	2.62	0	2.99/150psi
	Gadvan		2.64		
AZN6	Gadvan		2.47		2.566/400psi

1) Corrosion in this oilfield is a result of combined action of H_2S, CO_2 and high salinity formation water. The corrosive mechanism is complicated, and the corrosive rate on the casing is difficult to determine. Based on the NACE standard and the data from the Table, the producers result in severe corrosive environment. However, by the research on development conditions and technical indices, it can be found that although the corrosive fluid contents are high in the oilfield, the corrosive environment is not extremely serious as the following reasons:

2) In general, the temperature that cause the SSC/SCC is concentrated in the range from 20°C to 30°C. The average depth of the horizontal section is 2,730m with the reservoir temperature of 96°C, and tensional stress on the casing does not exist in the horizontal section. Therefore, the influence of SSC/SCC may not be considered for the casing/liner.

3) Based on the RMDP, the production GOR in the first development phase should not be higher than the initial GOR, therefore, the flowing pressure is higher than the bubble point pressure all the time, that is, degassing problems do not exist in the horizontal section and some vertical section, the H2S and CO2 have no chance of separation out of oil phase, and it is lack of prerequisite for casing corrosion.

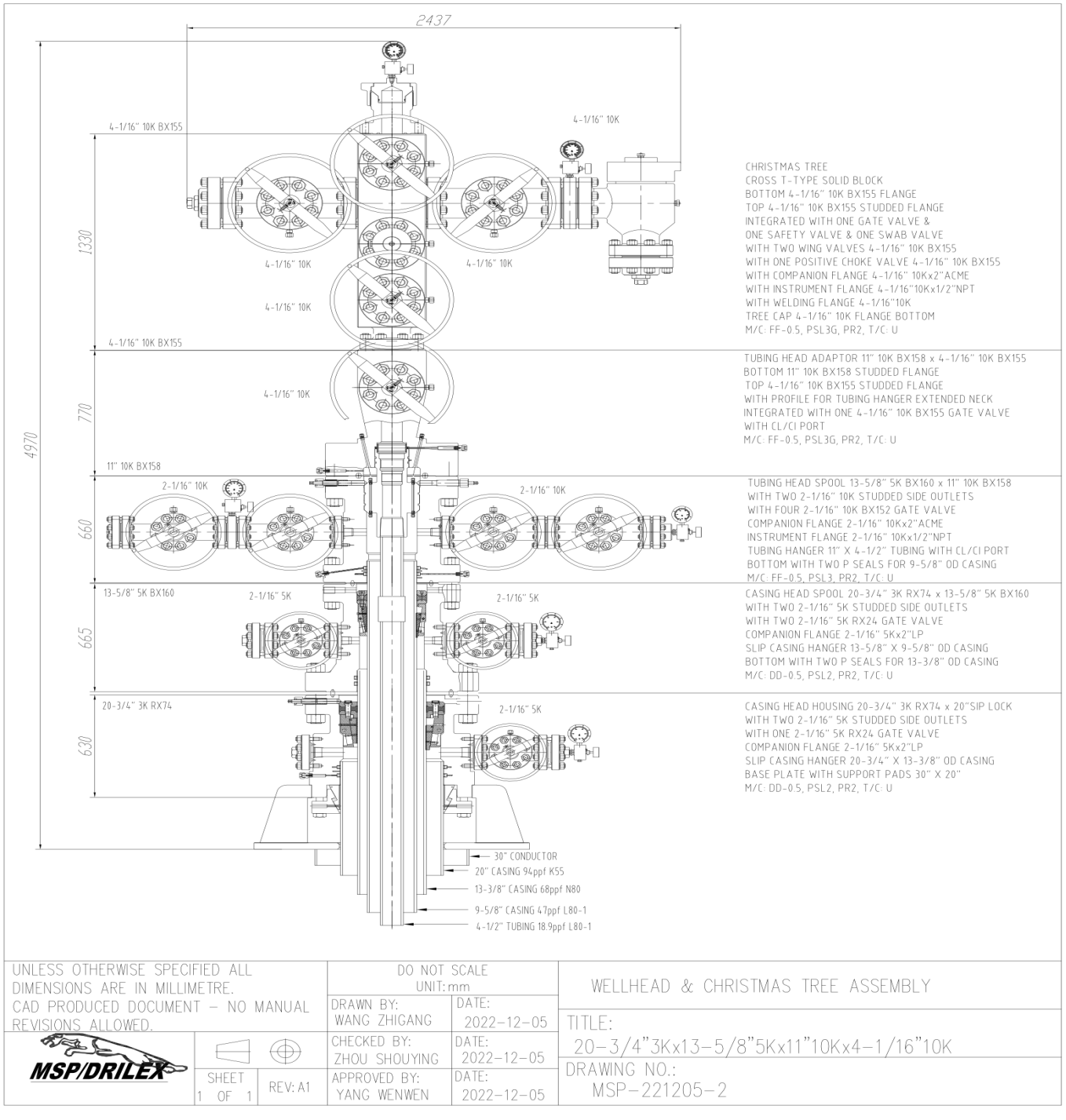
4) Based on above all, the corrosive environment of the casing string is not severe. The optimal selection for the casing string can be stated briefly as follows

Material Selection of Casing String

2) In this oilfield, the corrosion of production string obviously occurs in 2 regions. One is the region where the pressure is above the bubble point; in which degassing problems do not exist, and the corrosion environment is relatively weak. Another one is the region where the pressure is below the bubble point; because of the decline of pressure, the concentration of corrosive gas will increase and the corrosion is relatively serious. Considering the corrosive environment is not serious in the low water cut production stage and the production practice in this oilfield has proved the L80 is fit for it, So, the L80 should be taken as the material of 9-5/8" casing, 7", 5 1/2" and 4-1/2" liner. In order to prevent the hazard of gas leakage after production degassing occurs, 9-5/8" casing, 7" and 4-1/2" liner should be Gas tight thread. 2) The 110S is selected as the material of 9-5/8"/7"/4-1/2" casing rather than L-80 due to the high strength requirement in Fahliyan wells, what's more the 110S is fit for sour gas environment.

10. Wellhead & BOP Specification

10.1 Wellhead & X.M.T Schematic and Specification



20 -3/4" x 13- 5/8" x 11" x 4- 1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm** Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe （specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

10.2 Wellhead Material

1) **Selection of material class of Wellhead & X-mas Tree**:

H_2S & CO_2 in Offset Wells

Well No	*Reservoir formation*	*H2S from PVT (mol%)*	*CO2 from PVT (mol%)*
AZN2	*Sarvak*	*0.13*	*6.71*
AZN4	*Sarvak*	*0.06*	*5.44*
	Kuzhdumi	*0*	*2.62*
	Gadvan		*2.64*
AZN6	*Gadvan*		*2.47*

Considering the information above, API 6A 19th & NACE MR-0175 (2003) /ISO 1516, and Cl- content which is being most concerned, the South AZADEGA X-mas tree material and the tubing hanger should be FF0.5.

class.As to the casing head, it's barely exposed to resident fluid, or only for quite a short time, when it comes to desert working condition, according to API 6A19th & NACE-MR-0175 (2003) /ISO 1516, the tubing spool should be EE0.5, the casing head material class should be DD0.5 (ANSI 4130) , stainless steel cladding in ring groove

Selection of PSL (product specification level)

According to API 6A 19th, The final PSL for wellhead and X-mas tree are as below:

PSL for wellhead and X-mas tree

Item	*Specification*
Casing Head Housing assembly	*PSL-2*
Casing spool assembly	*PSL-2*
Tubing spool assembly	*PSL-3*
Tubing hanger	*PSL-3G*
X-Tree assembly	*PSL-3G*

Selection of PR (Performance Requirement)

According to API Specification 6A 19th edition, performance requirements are specific nd unique to the product in the as-shipped condition. All products shall be designed to perform according to the requirements of API 6A clauses 4 and clauses 10 while in the pressure and temperature ranges and corrosion envirement. Other requirements include load capability, cycles, and operating force or torque. There are two performance requirement levels: PR1 and PR2. Valves operating as safety valves shall be of the PR2 performance level and meet the requirements of Annex I in API 6A For SA wells, the guideline for PR selection is that all parts not contacting well fluids in nomal production period are PR2, all parts contacting well fluids are PR2 as follow, Casing head housing and casing head spool assembly – PR2. Tubing head spool, tubing hanger and X-mas tree - PR2.

Temperature Class

Environment temp:
Well temp: 0.0244×4100+27.384=127°C
According to API 6A 19th, temp should be U (-18°C-121°C)

API TEMP. CLASSIFICATION	*OPERATING RANGE*
U	*-18 TO 121*

Designed Pressure

Base on wells data of this project, and API spec 6A 19TH, considering the planning casing program, Wellhead and X-mas tree system for the project is designed as followed:

Wellhead and X-mas tree

Item Well type	*Casing Head Housing*	*Casing Spool*	*Tubing Spool*	*X-mas Tree*
Savark	*20-3/4" 3K*	*13-5/8" 5 K × 20 3/4" 3K*	*11" 5K × 13 5/8" 5 K*	*5K*
Gadvan / Fahliyan	*20-3/4" 3K*	*13-5/8" 5 K × 20 3/4" 3K*	*11" 10K × 13 5/8" 5 K*	*10K*
Water disposal wells	*20-3/4" 3K*	*13-5/8" 5 K × 20 3/4" 3K*	*11" 5K × 13 5/8" 5K*	*5K*
Item	***Specification***			
Casing Head Housing assembly	*MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U*			
Casing spool assembly	*MC:DD-0.5 5000PSI PSL-2 PR2, TC: U*			
Tubing spool assembly	*MC:* FF-05 *5000PSI PSL-3, PR-2, TC: U (Savark&Water disposal wells)*			
	MC: FF-0.5 10000PSI PSL-3, PR-2, TC: U (K&G&F wells)			
Tubing hanger	*MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Savark&water)*			
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)			
X-Tree assembly	*MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Savark&Water disposal wells)*			
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)			

10.3 BOP Specification and Schematic

Casing String	Wellhead	BOP	Hole Size
30” conductor	N/A	No BOP required while drilling 26” hole (only 30” bell nipple welded on	26" hole

		30" conductor)	
20" Surface Casing	Section A Casing Head Housing 20" Slip lock on casing x 20 3/4" 3K	21-1/4" 3K Annular BOP+ 21-1/4 3K Double Rams (Pipe Rams & Shear Rams) + 20-3/4 3K Drilling Spool+ 20-3/4 3K Spacer Spool	17-1/2" hole
13-3/8" intermediate casing	Section B Casing head spool 20 3/4" 3K BTM x 13 5/8"5K TOP	13-5/8" 5K Annular BOP+ 13-5/8" 10K Double Rams (Pipe Rams & Shear Rams) + 13-5/8" 10K Single Ram+13-5/8" 10K Drilling Spool+13-5/8" 10K Spacer Spool + Adapter flange 13-5/8" BTM 5K-Top 10K	12-1/4" hole
9-5/8" production casing	Section C Tubing head spool 13 5/8"5K BTM x 11"10K TOP (Kazhdumi/Gadvan/Fahliyan well)	13-5/8" 5K Annular BOP+ 13-5/8" 10K Double Rams (Pipe Rams & Shear Rams) +13-5/8" 10K Single Ram+13-5/8" 10K Drilling Spool+ Adaptor flange BTM 11" 10K-TOP 13-5/8" 10K	8-1/2" hole

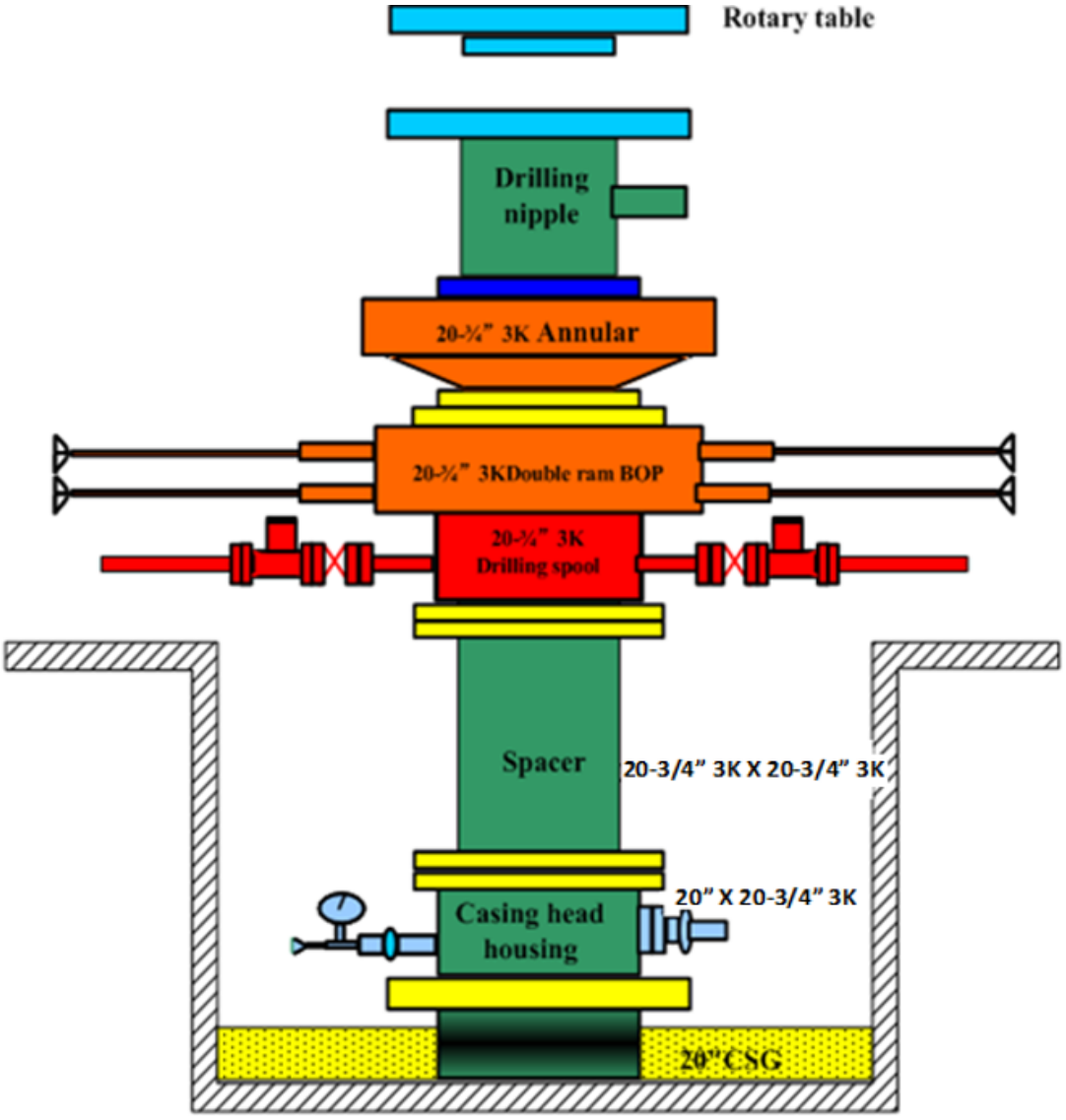
10.3.126" Hole Section

No well control equipment is required in drilling the section.

10.3.217 ½" Hole Section

The BOP stack in this section consists of an annular preventer (as diverter) orderly from top to bottom as follows:

- 20 3/4" -3000 annular preventer.
- 20-3/4" – 3k Double Rams



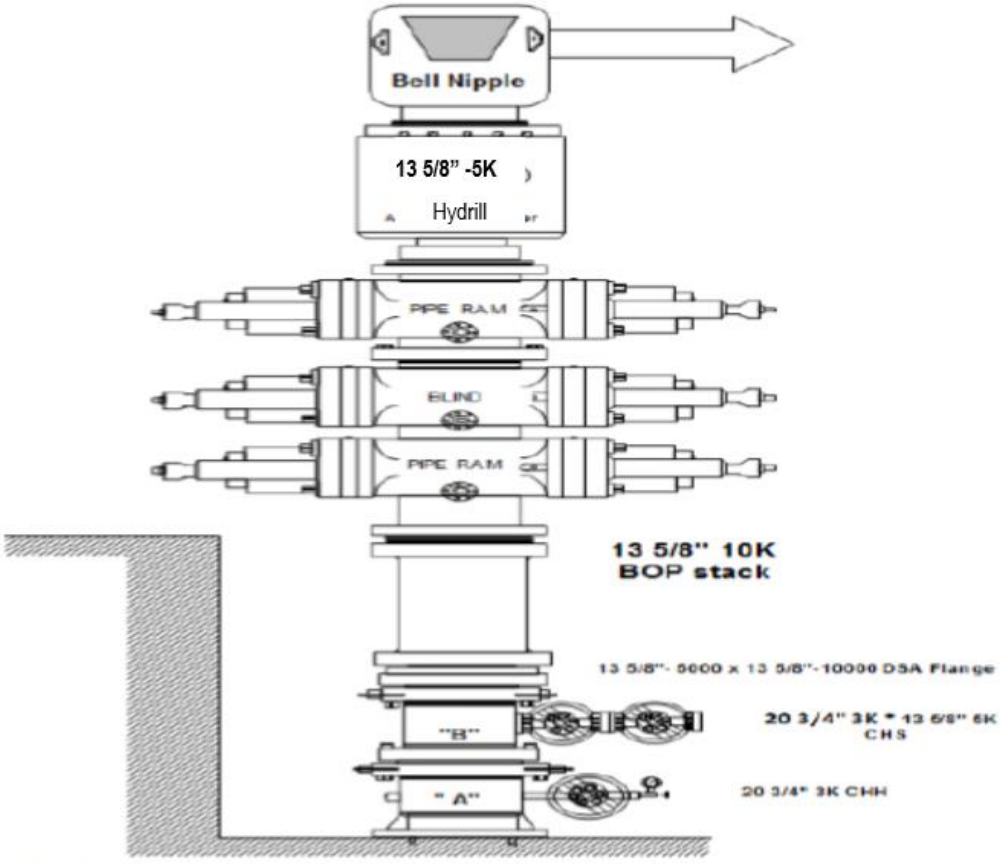
10.3.3 12-1/4" hole section

An annular BOP and three ram type Bops (a double stack on top and a single stack at bottom) represent the BOP configuration in drilling the section. All ram type BOPs shall be fitted with manual locks, to hold the rams closed in the events of hydraulic pressure loss.)

- 13 5/8"- 5000psi annular preventer
- 13 5/8"-10,000psi single stack ram type BOP (equipped with pipe rams)
- 13 5/8"-10,000psi double stack ram type BOP (equipped with pipe rams at top and shear/blind rams) Required ram sizes for 13 5/8"-5000 Ram type BOP.
- 1 set blind
- 3 set 5"
- 1 set 3 VV"

> 1 set Fix (5")
> 1 set variable (4 1/2"-7")

1 set ea. For 9 5/8" and 7" casings 3" nominal bore choke and kill lines shall be provided. Stab type connectors shall be fitted to allow quick connection of kill and choke lines to the BOP stack.



Well control Equipment Schematic in 12 1/4" Section

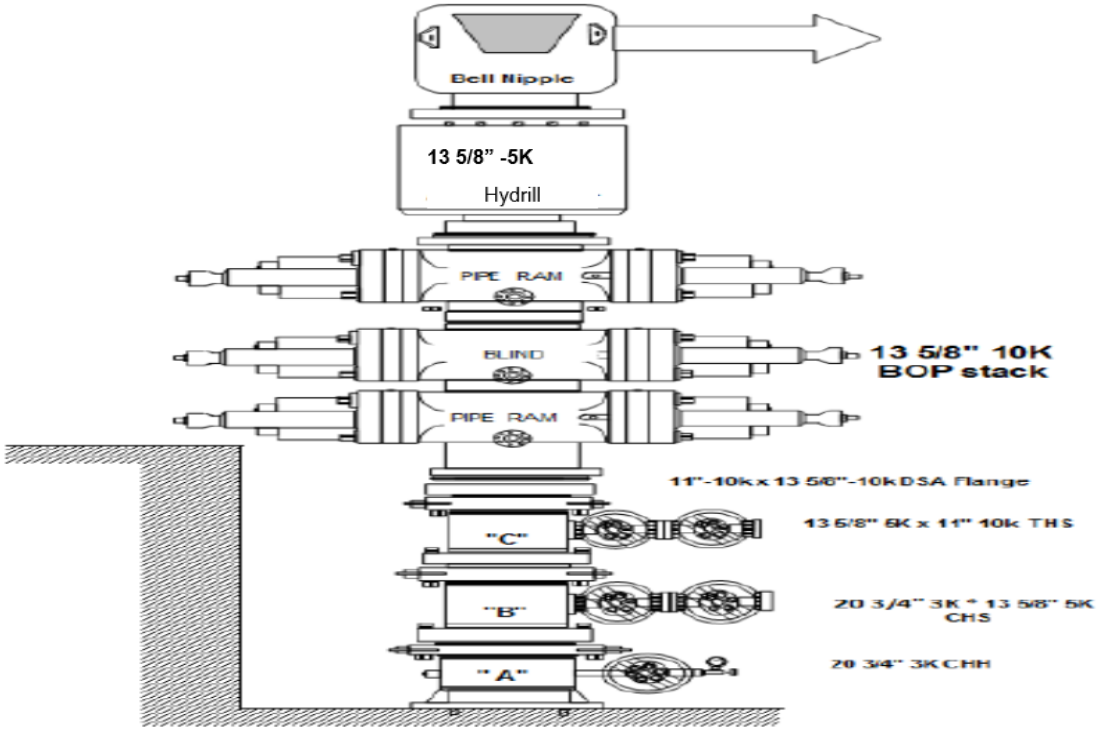
10.3.48 ½" Vertical hole section

An annular BOP and three ram type Bops (a double stack on top and a single stack at bottom) represent the BOP configuration in drilling the section. All ram type BOPs shall be fitted with manual locks, to hold the rams closed in the events of hydraulic pressure loss.)

> 13 5/8"- 5000psi annular preventer
> 13 5/8"-10000psi single stack ram type BOP (equipped with pipe rams)
> 13 5/8"-10000psi double stack ram type BOP (equipped with pipe rams at top and shear/blind rams) required ram sizes for 13 5/8"-10000 Ram type BOP.
> 1 set blind
> 3 set 5"
> 1 set 3 VV"

- 1 set Fix (5")
- 1 set variable (4 1/2"-7")

1 set ea. For 9 5/8" and 7" casings 3" nominal bore choke and kill lines shall be provided. Stab type connectors shall be fitted to allow quick connection of kill and choke lines to the BOP stack.



Well control Equipment Schematic in 8 1/2" Section

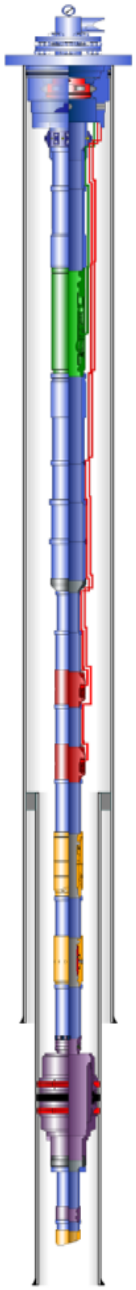
11. Completion

*** *completion program to Will be update and issued in due time* ***

No.	Fahliyan wells (10,000 psi)	
1	1/4" Control Line	Thickness: 0.049", INC 825, 120m/Roll, 15,000Psi WP, type: Seamless
2	3/8" Injection Line. encapsolated	Thickness: 0.049", SAF2507, 4000m/Reel, 15,000Psi WP
3	32# Hydraulic Oil	200 Liter Per Tank
4	Fluted Adaptor	4-1/2" 18.9ppf, PREMIUM Pin X Pin, S13CR-110, 10,000 Psi WP
5	4-1/2" Flow Coupling	4-1/2" 18.9ppf, PREMIUM Box X Pin, S13CR-110, 6 FT, 10,000 Psi WP
6	4-/2" Tubing Retrievable Safety Valve. self-equalizing, flapper type	4-1/2" 18.9ppf PREMIUM Box x Pin, S13CR-110, Min ID: 3.437", 10,000 Psi WP, 325 F WT
7	4-1/2" Flow Coupling	4-1/2" 18.9 ppf, PREMIUM Box X Pin, S13CR-110, Min ID: 3.4", Length: 4 FT, WP: 10,000 Psi
8	Cross-over	4-1/2" 18.9#, premium Box X 3-1/2" 10.2# PREMIUM Pin, S13CR-110, Length:24"
9	3-1/2" Travel Joint	3-1/2" 10.2ppf PREMIUM B x P, Stroke Length: 10ft, S13CR-110, WP: 10,000 Psi, WT: 300 F
10	3-1/2" Chemical Injection Nipple, with permanent dual check valve	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110 , WP: 15,000Psi
11	Chemical Injection dual check Valve	S13CR-110, WP: 15,000Psi, WT: 325 F
12	3-1/2" Sliding side door, with integral profile	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, Min.I. D: 2.75" Selective Profile, Open Down, WP:10,000 Psi, WT:300 F
13	7" x 3 ½" Hydraulic set Permanent Packer. With anchor latch and MOE	7" 26-29 PPF casing, 3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, WP:10,000 Psi, WT:300 F
14	2.75"No-Go Landing nipple	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, 2.75" No-Goe Profile, WP:10,000 Psi
15	3-1/2" Re-entry guide	3-1/2" 10.2ppf PREMIUM Box Up, S13CR-110, WP 10,000 psi
16	Fitting for safety valve	Size 1/4" male × 1/4" Tube OD, WP 10000Psi, Material: INC 825
17	Fitting for injection nipple	Size 3/8" male × 3/8" Tube, WP 15,000Psi, Material: SAF2507
18	4-1/2" Cross-coupling tubing protector	Matched 4-1/2" 18.9ppf tubing, fit for 9-5/8" 47 to 53.5lb/ft Casing, including 4 slots, two for 1/4" control line, two for 3/8" injection line, Carbon steel.
19	3-1/2" Cross-coupling tubing protector	Matched 3-1/2" 10.2ppf PREMIUM, fit for 7" 29 to 35lb/ft Casing, including 4 slots, two for 1/4" control line, two for 3/8" injection line, Carbon steel.

*** *completion program to Will be update and issued in due time* ***

No.	Fahliyan wells (10,000 psi)	
1	1/4" Control Line	Thickness: 0.049", INC 825, 120m/Roll, 15,000Psi WP, type: Seamless
2	3/8" Injection Line. encapsolated	Thickness: 0.049", SAF2507, 4000m/Reel, 15,000Psi WP
3	32# Hydraulic Oil	200 Liter Per Tank
4	Fluted Adaptor	4-1/2" 18.9ppf, PREMIUM Pin X Pin, S13CR-110, 10,000 Psi WP
5	4-1/2" Flow Coupling	4-1/2" 18.9ppf, PREMIUM Box X Pin, S13CR-110, 6 FT, 10,000 Psi WP
6	4-/2" Tubing Retrievable Safety Valve. self-equalizing, flapper type	4-1/2" 18.9ppf PREMIUM Box x Pin, S13CR-110, Min ID: 3.437", 10,000 Psi WP, 325 F WT
7	4-1/2" Flow Coupling	4-1/2" 18.9 ppf, PREMIUM Box X Pin, S13CR-110, Min ID: 3.4", Length: 4 FT, WP: 10,000 Psi
8	Cross-over	4-1/2" 18.9#, premium Box X 3-1/2" 10.2# PREMIUM Pin, S13CR-110, Length:24"
9	3-1/2" Travel Joint	3-1/2" 10.2ppf PREMIUM B x P, Stroke Length: 10ft, S13CR-110, WP: 10,000 Psi, WT: 300 F
10	3-1/2" Chemical Injection Nipple, with permanent dual check valve	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110 , WP: 15,000Psi
11	Chemical Injection dual check Valve	S13CR-110, WP: 15,000Psi, WT: 325 F
12	3-1/2" Sliding side door, with integral profile	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, Min.I. D: 2.75" Selective Profile, Open Down, WP:10,000 Psi, WT:300 F
13	7" x 3 ½" Hydraulic set Permanent Packer. With anchor latch and MOE	7" 26-29 PPF casing, 3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, WP:10,000 Psi, WT:300 F
14	2.75"No-Go Landing nipple	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, 2.75" No-Goe Profile, WP:10,000 Psi
15	3-1/2" Re-entry guide	3-1/2" 10.2ppf PREMIUM Box Up, S13CR-110, WP 10,000 psi
16	Fitting for safety valve	Size 1/4" male × 1/4" Tube OD, WP 10000Psi, Material: INC 825
17	Fitting for injection nipple	Size 3/8" male × 3/8" Tube, WP 15,000Psi, Material: SAF2507
18	4-1/2" Cross-coupling tubing protector	Matched 4-1/2" 18.9ppf tubing, fit for 9-5/8" 47 to 53.5lb/ft Casing, including 4 slots, two for 1/4" control line, two for 3/8" injection line, Carbon steel.
19	3-1/2" Cross-coupling tubing protector	Matched 3-1/2" 10.2ppf PREMIUM, fit for 7" 29 to 35lb/ft Casing, including 4 slots, two for 1/4" control line, two for 3/8" injection line, Carbon steel.

Schematic		Description	Max OD	Min ID
	1	Elevation (floor - TBG Head Flange)		
	2	TBG HANGER 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	3/68
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI	5/5	3/64
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5/118	3/64
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5/118	3/64
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT	5/5	3/64
	7	4-1/2" X 3.437" TRSSV-10 TUBING RETRIEVABLE SURFACE CONTROLLED SAFETY VALVE WITH SELF-EQUALIZING FEATURE, WITH AF PROFILE	7/5	3/437
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT	5/5	3/64
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5/118	3/64
	10	293 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5/118	3/64
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5/118	2/922
	12	5 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4/25	2/922
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2/922
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4/25	2/922
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2/922
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4/25	2/922
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	5/315	2/922
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4/25	2/922
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4/551	2/75
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4/25	2/922
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD	4/00	2/922
	31	7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4"	5/91	2/922
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4/25	2/922
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4/25	2/922
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN	4/5	2/635
	37	3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI	4/25	2/922
		END OF COMPLETION STRING		

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12. HSE & H2S Safety plan & Well Control

12.1 General Quality, Health, Safety & Environmental Requirements

- PPL/PEDEC's committed to conducting operations in a manner which minimizes the risks to the health and safety of those involved in, and affected by, such operations. PPL/PEDEC's also committed to conducting its operations in a manner which minimizes the impact on natural and 'built' environments in the area of operations.
- This section sets out minimum Quality, Health, Safety and Environmental (QHSE) requirements for the WORK that MSA and subcontractors will be required to comply with whilst conducting operations for PPL/PEDEC.
- PPL/PEDEC may perform audits and assessments of MSA's and its subcontractors' performance under the CONTRACT. PPL/PEDEC reserves the right without prior notice to conduct quality, health, safety, security and environmental audits of the operations.
- MSA shall at all times be responsible for performing the WORK in a manner so as to protect against loss of life, injury to people, damage to property and the environment. The detailed manner means and methods of performing the WORK shall be under the control and direction of MSA.
- Prior to the start of Operations, certain QHSE processes must be completed. These QHSE processes include developing the final QHSE plan format and frequency of any QHSE audits, permitting and licensing and PEDEC's approvals. MSA shall ensure that sufficient resources are dedicated to this task to ensure completion prior to mobilization.
- PPL/PEDEC may at any time stop the WORK or any portion of the WORK without penalty to PPL/PEDE Cif the WORK is not conducted in accordance with the Quality, Health, Safety or Environmental guidelines specified in this CONTRACT or by the laws of the appropriate jurisdiction. PPL/PEDEC reserves the right to require MSA to remove any person from the operations in the case of non-compliance with QHSE procedures. PPL/PEDEC may also prohibit the use of equipment and plant in cases where that equipment or plant does not meet the required standards or is deemed unsafe.
- MSA shall ensure that activities associated with this CONTRACT are conducted in compliance with applicable national and local HSE laws and regulations, MSA's standards and PEDEC's requirements and IADC documents.
- Limits of Authority
- PEDEC's Representative(s) do not have the authority to relax health, safety or environment standards. If they judge the operation to be inadequate with regards to any aspect of quality, health, safety, security or the environment, then the operation may be immediately suspended.

- MSA shall be required to develop a project-specific QHSE plan to manage the WORK and associated risks. This plan shall, as a minimum, address topics included in the list detailed in Item 7. Where existing MSA QHSE publications detail the topics listed, it is sufficient to either include them verbatim, or provide reference to where they may be found in MSA's QHSE Manuals. The final project-specific QHSE plan shall satisfy MSA standards, PPL/PEDEC requirements and applicable HSE laws and regulations.
- MSA should establish an effective HSE system and implement it strictly.
- MSA must adhere to the standards and regulations of Health, Safe and Environment constituted by the local government.
- MSA must strictly adhere to the procedures / standards and regulations of Health, Safety and Environment promulgated by NIOC.
- MSA should comply with all applicable local government regulations including guidelines and recommended practices and any other Iranian authorities having jurisdiction.
- All personnel should respect the local culture, traditions and social values of the indigenous population of the Iran and South Azadegan region.
- MSA shall hire qualified and experienced worker and familiar his personnel with hazards especially with H2S and Mine and UXO (unexploded ordinance) according to related procedure.
- MSA shall conduct proper training about UXO (unexploded ordinance) and mine awareness to all personnel

12.2 Health & Safety Management

As per national/local and client (PPL/PEDEC) standards and MSA/OEOC HSE plan. Following are minimum requirement for the AZNS project:

- MSA/OEOC shall take all necessary safety measures in relation to the WORK to be provided and shall conduct itself and its work force in such a way as to comply at all times with the provisions of the national and local health and safety regulations pertinent to the contracted Scope of WORK.
- MSA/OEOC shall also comply with all and any of PPL/PEDEC'S safety standards and requirements as particularly applicable to the Operating Area and/or as may be imposed by PPL/PEDEC from time to time.
- MSA/OEOC and its subcontractors shall take all necessary steps to provide a safe and healthy working environment for employees, PPL/PEDEC's personnel and third parties in the performance of the WORK.
- MSA/OEOC shall operate a management system that ensures:
- MSA/OEOC's and sub contractors' personnel operate safe systems of work at all times for the performance of the WORK.

- Hazards associated with the work are identified and appropriate actions are taken to prevent accidents.
- Risks to safety and health of personnel involved in the WORK are identified and assessed, and any actions and measurements are implemented to either eliminate or reduce these risks to a level that is considered as low as reasonably practicable, given the nature of the WORK.
- Personnel are aware of the identified hazards and risks to their safety and health and to the safety and health of others. The detailed safe working procedures established by MSA/OEOC to ensure that the WORK will be conducted with due regard to the QHSE aspects, shall be detailed or referenced in the QHSE plan. Copies of the source documents shall be provided to PPL/PEDEC if the procedures are referenced only.

QHSE Representative:

- MSA/OEOC shall provide a dedicated, on-crew QHSE Representative. The primary responsibility of the QHSE Representative shall include, but not be limited to, the maintenance and monitoring of the implementation of HSE guidelines and procedures and where required provide HSE awareness training for the crew. Such activities shall be approved and actively supported by MSA/OEOC at corporate management level.

Training and Competence:

- MSA/OEOC shall ensure that its employees and those of its subcontractors are trained and competent to perform the WORK in a safe and environmentally responsible manner. Such training and demonstration of competence will include but not be limited to vocational, safety, emergency, environmental and other training required by regulatory agencies and IADC/OGP guidelines. MSA/OEOC's senior staff shall have received training in safety management.

Accident/Incident Reporting:

- MSA/OEOC shall report all incidents. Any hazardous incident/near miss involving PPL/PEDEC, MSA/OEOC's, or any third-party personnel, plant or equipment, shall be reported to PPL/PEDEC, whether or not injury to personnel or damage/loss to plant or equipment resulted. The following incidents shall be verbally reported immediately to the on-site PPL/PEDEC Representative(s): damage to property or equipment belonging to PPL/PEDEC, third parties, MSA/OEOC or its subcontractors; injury to personnel of PPL/PEDEC, third parties, subcontractors or MSA/OEOC; near misses which may have had serious potential consequences; environmental damage; media attention. An initial written report, containing factual information only, shall be prepared by MSA/OEOC and delivered to the PPL/PEDEC Representative within twenty-four (24) hours of the incident.
- PPL/PEDEC reserves the right to participate in the investigation of any incident arising out of performance of the WORK.

- MSA/OEOC and all subcontractors shall have an accident and incident reporting system, details of which shall be included in the QHSE plan.
- MSA/OEOC shall maintain and forward to PPL/PEDEC, a monthly summary of its safety performance.
- MSA/OEOC shall prepare and forward to PPL/PEDEC, a weekly report stating all incidents and near misses that occurred during that week.
- All MSA/OEOC and subcontractor personnel and PPL/PEDEC Representative shall attend safety meetings on a regular basis. PPL/PEDEC Representative shall be given a copy of the minutes of the meetings.

Safety Equipment & Personal Protective Equipment (PPE) requirements:

- MSA/OEOC shall ensure that appropriate PPE compliant to national standards is supplied to MSA/OEOC's personnel involved in the WORK. PPL/PEDEC intends to use water-based mud and hence MSA/OEOC's Personnel that may be exposed to this must be provided with appropriate PPE.
- MSA/OEOC shall be solely liable for all costs in equipping its own and all other personnel on contract to, or under the control of, MSA/OEOC with the required safety equipment. MSA/OEOC will provide to all MSA/OEOC's Personnel appropriate PPE throughout the WORK. Any defective PPE will be replaced without delay. Personnel without the correct PPE will not be allowed to continue work, until replacement PPE has been issued. If any person is disallowed to work owing to inadequate PPE, then PPL/PEDEC will deduct his day- rate from MSA/OEOC's invoice.
- MSA/OEOC shall detail in the QHSE plan the PPE proposed for issue to the crew. This proposal should include the arrangements for replacement (due to wear and tear and damage) of this equipment.
- This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.
- A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.
- An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centers manned by qualified doctor 24-hours per day in Tehran or Ahwaz city.
- Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.
- MSA/OEOC shall, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with good oil field practices and OGP Guidelines. An assessment of the Contract Area with regard to "Environmental and Occupational Health Risks" and "Local Medical Support" shall be provided by MSA/OEOC in the QHSE plan. This

assessment and the resulting "Health Risk Assessment Score" will be discussed with PPL/PEDEC, and a final assessment will be agreed.

- To ensure the fitness of those personnel engaged to undertake the WORK, MSA/OEOC shall conduct periodic medical examinations of its staff and ensure that its subcontractors' personnel are also subject to periodic medical examination. As a minimum this shall include pre-hire medical examinations of temporary local personnel and bi-annual medical examinations of its permanent and expatriate staff. Evidence should be made available to PPL/PEDEC to verify an appropriate program of immunizations consistent with the recommendations of the World Health Organization for the area of operations.
- MSA/OEOC shall be responsible for the cost of medical examinations as required to provide PPL/PEDEC with medical certificates which confirm the fitness of MSA/OEOC's assigned personnel for the WORK. Each drilling rig shall be equipped with a hospital, doctor and an ambulance type-2 level.

Drug Abuse and Control:

- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to a search of property of such Personnel when PPL/PEDEC suspects that such Personnel have in their possession drugs and/or other prohibited substances or any other item likely to prejudice safety or security. PPL/PEDE will provide a representative to attend the search.
- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to medical testing for the presence of drugs or other prohibited substances.
- MSA/OEOC further agrees that any Personnel who refuse to provide such consent or consents or withdraws such consent or consents shall at PPL/PEDEC's option be removed from the WORK and shall not, at PPL/PEDEC's option, be assigned to any future services to be provided by MSA/OEOC for PPL/PEDEC.
- MSA/OEOC further agrees that it shall immediately notify PPL/PEDEC in the event that MSA/OEOC becomes aware that any of its Personnel has been found in possession of drugs and/or any other prohibited substance. MSA/OEOC is required to notify PPL/PEDEC in the event that any Personnel provided for the WORK have previously been found in possession, or have abused drugs and/or other substances prohibited at Location. The acceptance of such Personnel for the WORK shall be at the sole discretion of PPL/PEDEC.

Camp Standards:

- MSA/OEOC shall abide by all statutory Iranian governmental health and safety regulations, and maintain acceptable standards for cleanliness and hygiene for quarters, camps and the Operating Area(s) in general. MSA/OEOC shall detail the standards and facilities proposed for its camp in the QHSE plan.

Emergency Preparedness:

- MSA/OEOC and its subcontractors shall have emergency preparedness and response plans plus procedures available at all times throughout the duration of the CONTRACT.
- MSA/OEOC shall take all necessary measures to ensure that:

I. Potential emergency situations have been identified and assessed;

II. An emergency response plan, including clear responsibilities, is established;

III. MSA/OEOC's management, employees and third parties are aware of response procedures and responsibilities;

IV. Regular drills and exercises are carried out to test the emergency response organization and procedures and that any resultant issues are properly addressed.

MSA/OEOC shall notify PPL/PEDEC immediately with respect to any emergency situation and shall provide regular updates on emergency planning procedures, tests etc.

MSA/OEOC shall detail in the QHSE plan a list of potential contingencies for which emergency response procedures are planned. The QHSE plan should also detail a schedule of proposed exercises and drills to verify the effectiveness of the proposed arrangements in place. Development and implementation of the above plan shall incorporate any PPL/PEDEC assigned personnel to the project.

12.3 Environmental Management

- COMPANY shall comply with any and all laws, rules and regulations of governmental agencies having jurisdiction, which now exist or may be promulgated during the term of the CONTRACT, relating to the control and prevention of damage to the environment.
- Without prejudice to above, COMPANY shall ensure that all personnel use the highest prevailing standards of care and diligence to prevent unlawful disposal of all waste oil, water and other waste materials or fluids which may accumulate or be caused by COMPANY's performance of the CONTRACT. COMPANY shall not dispose of any objects, scrap, waste, rubbish, equipment, fluids or fuel into the surrounding area or allow anything to be stored loose where it can be driven into surrounding areas by wind. COMPANY shall take all reasonable precautions to prevent pollution of any nature or kind from its performance of the WORK. COMPANY shall have contingency plans and procedures in place for such events.
- In the event of any disposal prohibited by Item 3.2 above, whether by dumping, spilling, emission or otherwise arising from COMPANY's performance or non-performance of the CONTRACT, COMPANY shall be fully liable for and bear the costs of removal of any such item(s) dumped, spilled, emitted or otherwise disposed regardless of the sole, concurrent or contributory negligence or breach of duty (statutory or otherwise) of PPL/PEDEC. In the event of any such disposal, COMPANY shall immediately undertake all reasonable remedial and protective measures required in such circumstances.
- COMPANY shall notify PPL/PEDEC immediately with respect to any pollution, loss, damage, claim or demand (or occurrence which may give rise to the same) resulting from the CONTRACT. COMPANY shall report to PPL/PEDEC any incidents of non-compliance with

legislative and regulatory environmental requirements that occur during the performance of the WORK. Without relieving COMPANY of any of its obligations under Item 3.2, if COMPANY does not promptly proceed to perform its obligations under this Item 3, PPL/PEDEC may take part to any degree it deems necessary in the control and removal of any pollution, dumping, spillage or contamination which is the responsibility of COMPANY under this Item 3.2, or which is due to any act or omission of COMPANY. COMPANY shall reimburse PPL/PEDEC for costs arising from such actions of PPL/PEDEC upon receipt of invoice from PPL/PEDEC. PPL/PEDEC shall have the right to deduct such costs from any sum due or becoming due to COMPANY under the CONTRACT.

- COMPANY shall operate a management system that is able to demonstrate compliance with environmental laws, rules and regulations applicable to its activities, products and services.
- MSA/OEOC and its subcontractors shall take all necessary steps to ensure that the environmental impacts of their work are responsibly managed and shall take all necessary measures to ensure that:
- COMPANY's personnel are aware of the environmental impact of the COMPANY's operations and are sufficiently trained and competent to perform the WORK.
- COMPANY's personnel operate systems, which will ensure that significant environmental impacts of WORK are identified and managed in accordance with a commitment to legal compliance and continual improvement.
- The environmental risks of operations associated with the WORK are evaluated and appropriate actions are taken to prevent accidents and reduce pollution, and that contingency plans are in place in the event of an incident.
- COMPANY will have adequate processes for monitoring the environmental performance of their WORK.
- COMPANY will, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with IADC Environmental Guidelines. COMPANY's proposals for environmental protection should be detailed in the QHSE Plan. This should include, but not be limited to:

- procedures and precautions to minimize damage to crops and vegetation, the local ecology and disturbance of wildlife and farm animals;
- recycling and waste disposal;
- Clean-up and restoration of seismic lines, Rig sites access roads and trails, fixed sites occupied during the survey.

- An ecological team shall be set up by COMPANY, with the purpose of collecting all waste, rubbish, food scraps, office waste and other discarded material from camps, created by the WORK. Such activities shall be performed on a daily basis. Such material shall be appropriately disposed of, according to PPL/PEDEC approved methods or existing methods established by COMPANY, which are proven to be acceptable to PPL/PEDEC. Well location and camp site shall be delivered to PPL/PEDEC without any waste, rubbish, food scraps, office waste and other discarded material. PPL/PEDEC has the right to hold Completion certificate till cleaning the well location and camp site by COMPANY.
- The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting

and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

- Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.
- Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.
- Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.
- COMPANY shall perform on site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation COMPANY in accordance with the following waste hazard classification.
- Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)
- Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass
- Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.
- The COMPANY/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.
- It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to
- local community are recommended.

Hazardous Wastes:

- Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site:

- Mud purifying circulation system
- Cooling water spraying and circulating system for drilling pump
- Special recovery tank for deposed oil
- Reserve house for valuable mud chemicals
- Automatic mud weighting supply system
- BOP stack is equipped according to drilling design
- Define a separation area for gathering all waste according to waste management procedure.
- Requirement of environment protection on location Sewage pit Area of the sewage pit: 1500 m2
- Prior to spud mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.
- Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.
- Install oil supply lines and check whether there is any leakage.

Management of environment protection on location:

- In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

- Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.
- Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.
- Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.
- After the well is completed, reinforce the sewage pit， solidify the disposed mud and have environment procedure fulfilled.
- Install fuel supply lines and keep them in good condition.
- All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.
- The environmental protection personnel from COMPANY and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.
- Transition Area for Hazardous Waste shall also be provided by each COMPANY in his allocated area,
- PPL/PEDEC shall approve each area before use as well. The transition area of hazardous waste shall be:

- Paved, fenced, marked and illuminated
- Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g., Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).
- In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).
- Configured to accommodate a safe physical separation of incompatible waste materials (e.g., acids, and alkalis, organic materials and strong oxidizing agents).
- Capable enough to provide storage for the entire range of hazardous wastes generated during the work.
- The area shall be provided with Spill Containment and means of preventing run off to and from the area.
- All waste containers to be stored shall be appropriately labeled.
- Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from PPL/PEDEC.

- Standard of utilization and discharging of wastes on location
- Conduct drilling operation according to the design, drilling mud is needed before spud.
- Sewage that cannot be recovered shall be put into the pit, do not discharge
- Make full use of waste water. Water in mud pump and water brake must be utilized by circulating. Water that is used to wash drill floor, shall be recovered after deposition. The recovery rate of waste water shall reach 85%.
- Disposed oil shall be recovered entirely. Recovery rate shall be 100%. Dumping of waste oil shall be prohibited.
- In drilling operation, non-chrome thinner high temperature mud system shall be used often if engineering is qualified.
- While drilling in special area adopt mud creation inhabitation techniques to decrease the waste mud creation.

- After completion, all the mud chemicals, bentonite, clays shall be recovered and shall not be left at the well site. Mud in purifying tank shall also be recovered after completion.
- In case of any flammable or noxious gas encountered, steps shall be taken at once to prevent kicks and blowout. Gas shall be guided into ignition equipment to burn it away.
- Burning material that may result in serious smog or horrible smell at the well site is not allowed. For cement or barite that may result in granular dust pollution, operation shall be performed in a closed system.
- For abandoned well, filled with cement plugs to seal the oil, gas or water formation.
- COMPANY/Generators are required to transport all generated wastes to the location/facilities designated, and licensed by authorities and/or approved/verified by the PPL/PEDEC (as appropriate).
- All wastes shall be transferred by approved Sub- COMPANY (which could be a 3rd party Sub-COMPANY of the COMPANY/Generator itself). The Waste Transporter and any other waste management service COMPANY shall be licensed in accordance with local regulations and are in compliance with such regulations. PPL/PEDEC shall verify and approve the Waste Transportation COMPANY by checking licenses and qualifications, audit and inspections, verification of human resources and received trainings etc.
- The transportation COMPANY shall ensure that all wastes are transported in a safe and responsible manner.

12.4 Security

As per PPL/PEDEC as main client and MSA/OEOC agreed plan.

- Security for the WORK shall be arranged and administered by the MSA/OEOC. MSA/OEOC shall be fully advised in writing covering all aspects of security upon. Company shall liaise with PPL/PEDEC QHSE to develop operational and contingency plans prior to Mobilization.
- The Security guards shall be appropriately housed and provided for. Transport for the Security guards to perform field activities shall be made available according to standard Iranian Oilfield procedures.
- All MSA/OEOC's personnel involved in the WORK shall possess the necessary security clearance documentation required by the Iranian authorities to work within the Operating Area.

12.5 Quality

1. Ensure that relevant experiences from the appraisal and production wells (if any) are captured and included in the operation programs /procedures.
2. Check lists to be prepared for all equipment, available for PPL/PEDEC prior to start of operations.
3. Procedures planned for all operations, available for PPL/PEDEC prior to start operations.
4. High focus and follow up of KPIs (Key Performance Indicators) by NIOC procedures.
5. Drilling risk assessment.
6. All goods and equipment shall be checked as per engineering design and inspection certificates, PPL/PEDEC approval shall be obtained before running any goods in hole.
7. Routine reports shall be submitted to PPL/PEDEC and any commented shall be corrected and informed engineering team.

- Rig Specification shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
- Rig Layout shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
- Rig certification (COF) and record shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
- Apply PPL/PEDEC inspection work instructions.
- Drilling String inspection must be done before and during operation based on client request and its certificates and record shall be submitted to PPL/PEDEC.
- Pre-Spud check list must be verified by PPL/PEDEC before operation.
- NDT certificates and record shall be submitted to PPL/PEDEC.

12.6 Meetings

HSE meetings shall be convened on a monthly basis with an agenda to discuss the general issues about HSE, any change in the contractor, in the nature and risk of the activities, HSE training program, safety analysis of the activities, as well as investigation and analysis of the accidents. The Contractor's site manager and HSE Representative shall attend such meetings and report to the Client's site manager and HSE representative.

Weekly HSE meeting shall be conducted with the presence of MSA/OEOC HSE & Rig Contractor HSE Supervisor.

12.7 List of Minimum Required Contents

As per below tables:

Minimum Onshore Rig HSE Requirements	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Muster station sign board	2
Warning Sign (H2S Danger, Danger zone...)	As needed.
MSDS For all Chemicals	As needed.
T- Card Board	1 Panel in entrance of Rig & 1 Panel in Camp

Wind Socks	3
Fall Arrestor	Minimum 6 sets with different length Install in (Monkey board, 2 for Monkey Board ladders, Crown, In mast for hanging block, Under Rig floor)
Bug Blower (Explosion proof)	Minimum 3 no. Set in (Rig Floor, Cellar area, Spare)
Explosion Proof Torches	Minimum 6 (Company man, Tool pusher, HSE, Medic, Maintenance crew, Mud Crew)
Rescue Equipment	As needed.
Security Well Marker	By MSA/OEOC HSE Concern
Derrick man's escape system I.e. REM chute	By MSA/OEOC HSE Concern
Emergency shower with eye wash to be situated at assigned positions	Minimum 3 sets (Rig floor, chemical matters storage and handling areas)
Portable eye wash set	Minimum 5 sets
Life Ring	2
Spill Kit	3
EX Radio	At least 7
First Aid Kits	Rig Floor, Medic room, Camp Boss, Client Office, Store, Camp Galley
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall+ Safety Helmet + Safety boots for all personnel on location and for visitors.	As needed.
Safety Eye protection & Ear Protections (Ear muff & Ear Plug)	As needed.
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	As needed.
Fire Fighting Equipment	
Fire extinguishers: "Powder-50 kg"	7(In Rig site) 4(In Camp site)
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	By MSA/OEOC HSE Concern
Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	By MSA/OEOC HSE Concern
Fire extinguishers "Foam-45 Liter" For fire potential areas where required.	4 No. wheeled (Set in Diesel tanks, Engine generators in Rig site and Camp site)

Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	By MSA/OEOC HSE Concern
Mobile Foam Package (Fiberglass Body-Z4 Inductor) + 1000 Liter FFFP 3% Foam + S4 Low expansion Nuzzle (B.S)	By MSA/OEOC HSE Concern
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	By MSA/OEOC HSE Concern
Fire blankets for galley, Welding & Cutting operations	As needed.
Fire Fighting Suits	Minimum 4 sets in each Rig site and Camp site
Water pump diesel, capable of charging the fire main in case of deep well failure.	1(Rig site)1(Camp site)
Medical Equipment	
Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Basket Stretcher with lifting belts and hooks) (1No.) Long Back Board with spider straps and immobilizer) (1No.) Army Stretcher(2No.)
Ambulance	1No. (B Type)

Minimum Rig-less HSE Requirements for South Azadegan Project	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Eye wash & safety Shower to be situated at assigned position	(1) chemical storage and handling area
Muster station Sign board	2No.
Warning Sign (H2S Danger, Danger zone, ...)	Required
MSDS For all Chemicals	Required
T- Card Board	Required

First Aid Kits	Hospital, All offices
Wind Socks	2No.
Air Fans (Explosion proof)	1No.
Explosion Proof Torches	Required
Rescue Equipment	Required
Spill kit	3
EX Radio	At least 6
Security Well Marker	Required
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall + Safety Helmet + boots for all personnel on location and for visitors.	Required
Safety Eye protection & Ear Protections (Ear Plug)	Required
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	Required
Fire Fighting Equipment	
fire extinguishers: "Powder-50 kg"	At least 3No.
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	Required
Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	Required
Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	Required
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	Required
Fire blankets for galley, Welding & Cutting operations	Required
Fire truck (10000 Liter Water-1000 Liter Foam) Or (Firefighting systems)	1No.
Fire Fighting Suits	As per fire fighters' number
***The following equipment recommended meanwhile the risk can be retained in case that the company risk assessment considering the risk is so far as reasonably foreseeable. The measures and precautions also need to be as low as reasonably practicable.**	
Medical Equipment	

Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Back Board (1No.) Army Stretcher (2No.)
Ambulance	1(B Type)

12.8 Well Control

1. Installation Requirements of BOP Stack and Control System:

- The distance between BOP control lines and relief lines shall be not less than 1m. Steel Gap Bridge should be placed in such a way to protect lines from damage by vehicle crossing. Sundries shall not be packed on the control line shield, on which welding operations are forbidden.
- The air supply shall be separated from that of driller's console and directly connected to the main air supply system, and shall be equipped with liquid/gas separators. Excessive bending and pinching on air hoses are forbidden.
- Electric power to BOP control system shall be directly from the main switch of the power board, and controlled by independent switches.
- Accumulators shall be in good condition, charged at its rated value, and ready for work at any time.

2. Installation Requirements of Choke Lines and Kill Lines, and Relief Lines:

a) Beside the drilling spool, one manual and one HCR gate valves on the side of chock manifold and two manual gate valves on the side of kill manifold shall be equipped, and the manual gate valve close to the spool shall be always open.

b) The high-pressure co-flex hose shall be installed as straight as possible and fixed by anchor after installation.

c) Relief lines (burning line and bleed line):

- No field welding is permitted on relief lines and the connecting style of relief lines must be flanged.
- The arrangement of relief lines shall take wind directions, dwellings, roads, oil tank areas, power lines and other facilities into consideration.

- While the two lines go in the same trend, the distance between the lines shall be at least 0.3m, and the two lines shall be fixed independently.
- The outlets of bleed line connecting with choke manifold shall be in a safe area which is at least 300m. The bleed line on the kill manifold side connecting with drilling spool shall be prepared and connected when necessary (about 70m from the wellhead).
- At every 10m~15m, turning points, and outlets, the lines shall be supported and fixed by cement foundations with anchor bolts or ground anchors. Hanging portions shall be supported firmly. If crossing the stumbling blocks such as brooks or ponds is needed, metal bridging supports are necessary.
- Pre-buried anchor bolts for cement foundations shall not be less than 20mm or 3/4" in diameter, and greater than 0.5m in length.

3. Pressure test value and requirements:

a) A pressure test shall be conducted successfully where there is no pressure decline in 5 minutes for a low-pressure test and a minimum of 10 minutes for a high-pressure test.

b) Prior to each high-pressure test, a low-pressure test with 200-300 psi shall be carried out to verify sealing capability at the circumstance of low pressure and avoid washing out sealing elements.

c) High pressure test should be to the rated working pressure of ram preventers or the rated working pressure of the wellhead, whichever is the lowest, or to the rating required in DRILLING ROGRAM. Variable bore rams should be initially pressure tested on the largest and smallest OD pipe sizes that may be used during the OPERATION. Annular preventers, with a joint of drill pipe installed, should be tested to 70% of their working pressure. The high test for Kelly cocks, kill lines and choke lines should be to their rated working pressures, each part in the choke manifold is tested respectively to its rated working pressures, the relief lines be pressure tested to no less than 1,450psi (10Mpa).

d) When casing cup tester is used, the above test pressure should not exceed 80% of the internal yield pressure (burst pressure) of the exposed casing.

e) The control system of BOP shall be pressure tested to 3,000psi (21Mpa) for its reliability before putting into use.

f) Except that the BOP control system is tested with hydraulic oil, other well control equipment shall be pressure tested by clear water.

g) Testing tools

- All required sizes of the plug type testers, and cup type testers should be available for pressure testing purpose.
- When the cup type tester(s) is used for pressure test, the connection of the cup testers has to be checked, to ensure it will stand the load during the high pressures test. For surface casing, the cup tester is to be set in middle of the first casing joint in the top, so the first casing connection (between wellhead and casing head housing) below the well head is to be tested too while testing BOP and wellhead.

4. Pressure test frequency:

All ram type preventers, annular BOP, valves, choke and kill lines, choke manifold, Kelly valves and safety valves should be pressure tested at the following frequencies:

- Prior to installation where possible.

- After installation of wellhead and BOP assembly and prior to drilling.
- After the disconnection or repair of any pressure containment seal in the BOP stack, choke line, or choke manifold, but limited to the affected component.
- Prior to drilling into a confirmed or suspected high-pressure zone.
- At any time requested by the COMPANY REPRESENTATIVE.
- In any case, every one month or during the first trip after the one-month interval with a maximum interval of 45 days. This frequency is not required for shear rams. As a minimum, the shear rams are to be tested prior to resume drilling operation after 9-5/8" casing string has been set and adequate WOC.

5. Work Requirement of Well Control Equipment:

- With shut-in casing pressure lower than 1000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

6. Management of Well Control Equipment:

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

7. Killing Method and Shut-in Method:

- Driller's method is recommended for well killing.
- The hard Shut-in method is recommended for shutting in the well in Fahliyan reservoir.

8. Well Control during drilling:

a) Shut in Procedure in While on Bottom

- Position Kelly (if in use) and tool joint so that tool joints are clear of sealing elements.
- Shut off pumps, check for flow.
- If flow is noted, close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. toll joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (if possible).
- Record shut-in drill pipe and casing pressures.
- Record volume gain and mark pits.
- Weigh mud in suction pits.
- Check blowout preventer and manifold for leaks.
- Check flowline and choke exhaust for flow.
- Check accumulator pressure.

b) Shut in Procedure in While Tripping

- If a flow is detected, install full opening safety valve and close same.
- Position tool joints clear of the sealing elements and close annular preventer and open remote-controlled choke line valve (HCR).
- Make up grey valve (dart type inside BOP on full open safety valve). Then Make up Kelly or top drive (insert a pup joint or single between safety valve and top drive) and open safety valve.
- Record casing pressure.
- Check blowout preventer, stack, and manifold for leaks.
- Check valves for correct position.
- Check accumulator pressure.
- Begin stripping to bottom.

c) Shut in Procedure for Empty Wells

- Try to RIH with drill pipe as more as possible if hole condition permits, and the drill pipe must be installed a Kelly cock or safety valve.
- Close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. tool joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (jf possible).
- Record casing pressure

9. Well Control During Running Casing:

The well control while running casing is important and critical, best practices shall be followed, including but not limited to:

- Change out pipe rams to casing rams prior to running casing into hydrocarbon bearing formation, if possible.
- The crossover between casing and drill pipe has to be available, and good practice is to make it up with a joint of DP and keep on V-door.

10. Shut in Procedure During Logging:

Cable shall be pull out of the hole immediately when overflow is found during logging. If pit gain exceeds the specified amount, the cable shall be cut immediately and treat the overflow in accordance with the shut-in procedure for empty wells. Continuing to pull the cable through the closed annular preventer is not recommended.

11. Well Control During Testing:

- No difference to while DRILLING OPERATION, since either there is testing string in the well (same as Shut-in procedure while on bottom), or tripping the string. (Same as shut-in procedure while tripping).
- Likely there is some crude in the testing string, and precaution has to be taken while circulating the crude out of hole during killing operations to ensure it is properly handled without causing environmental impact.
- To minimize the formation damage, the killing fluids used shall be weighted brine if possible, however in well control case when it occurs, killing well safely will take priority rather than protecting formation from damage on selection of killing fluids.

12. Well Control During Stimulation:

- Similar to during testing.
- For any residual or alive acid to be circulated out during killing operations, measures have to be taken to treat the returned acid properly to avoid personnel from getting hurt, and without causing any environmental damage.

13. Well Control Drills:

Drills shall be conducted only when hole conditions and other rig activities permit. Drills should be scheduled so that crews can practice drills while on bottom, while tripping in and while out of the hole. Each operating shift shall perform at least one BOP drill under different situations per month.

14. Basic Requirement of Well Control:

- For 13-5/8”, 10K BOP, BOP should be schemed as follows: upper rams 5”, middle rams are shear rams, lower rams is variable rams (2-7/8” to 5”). For easily connecting, armored fire-resistant 10000psi and ID 4-1/16" flexible hoses are recommended as kill/choke lines which connected with BOP directly.
- The BOP stack and choke manifold, kill line and choke line, HCR and manual gate valves, inside BOP (including kelly cock) have to be H2S trimmed and stand the working temperature from -10 degrees C (in case of doing N2 for DST) to the maximum up to 100 deg C (estimated highest temperature condition).
- The BOP equipment shall be visually inspected, function tested, and hydrostatically pressure tested during commissioning and Acceptance of Drilling Unit prior to Commencement Date, then upon installation on a wellhead, prior to drilling any hole section with a potential hydrocarbon bearing reservoir, after breaking any BOP connection (such as after installation of an additional wellhead section after running casing), and at least every 14-21 days while the BOP is installed on the Well.
- CONTRACTOR shall submit the current well control procedures manual of CONTRACTOR to COMPANY for review and approval by COMPANY. Valid IWCF Well Control Certificates (or IADC equivalent) must be held by all tool pushers, Drillers (optional) at Drilling Supervisor level and Drillers and Assistant Drillers at Driller level, as further specified in CONTRACTOR Personnel HSE Requirements.
- The accumulator must be of sufficient capacity so that it is possible to close one (1) ram and the bag, open a ram and the bag, re-close a ram and the bag and retain 1200psi accumulator pressure.

- The "OPEN" and "CLOSED" operating positions of BOP activators of the rams, annular, and remote actuated valves must be clearly identified on the BOP master control panel and remote panels, in English in addition to the native language of the majority of the senior drilling crew.
- Shear-ram BOP activator must have a flap-cover to ensure only pre-meditated and deliberated activation and to prevent accidental WORK.
- All lines and connections in the choke and kill manifold, standpipe manifold, choke and kill lines and high-pressure mud lines should be welded hard-piping with weld-on FMC Weco "hammer-lug" 1502 unions (or Western equivalent, such as Anson) or weld-on API flanges. The well control system should not include any threaded screw-on hammer-lug unions, flanges, or other threaded screw-on connections.
- All adapter or spacer spools between the wellhead and BOP must be manufactured in accordance with API 6A, and must be complete with all required studs, nuts, and ring gasket seals for both lower and upper connections; minimum spare parts should include two (2) OEM spare studs complete with nuts and two spare OEM ring gasket seals for both bottom and top connections.
- There should be a wide-bore gas vent line from the "poor-boy" degasser; the line and line outlet must be securely anchored, the line outlet must be a safe distance from the well cellar, rig floor, choke manifold, and mud logging unit.
- Well Control Equipment Welding: All welds in the well control system must be NDT inspected and records maintained of all welding and weld inspections, including choke and kill manifold, standpipe manifold, mud pump discharge manifold, choke and kill lines.
- Well Control Rig Floor Information: The following must be posted on the rig floor (in the Driller's shack / "dog-house"): Well Shut-In Procedure, Well Information Sheet (showing casing and hole sizes, depths, and such like), Kill Sheet (including most recent slow circulation rate pressures, formation integrity leak-off test information and MAASP, present mud weight, present drill string and well volumes, and such like).
- The derrick substructure clear height (rotary table to ground level) needs to be enough for fitting in above BOP configuration (including a tubing head and a bell nipple). This need to be double checked in rig tender process. According to the calculation of height of BOP stack, the substructure clear height is at least 9 m.
- In the case if drilling 6-1/8" hole becomes necessary, or using 3-1/2" drill pipe for the cleanup run (including drill 7" shoe track), then taper drilling string (5" DP + 3-1/2" DP) is to be run, so a pair variable ram be installed in the single ram preventer. The defect installing variable ram in single ram preventer is that the ram may be damaged and the string fall in hole bottom due to the ram hanging up too much weight when the drill pipe be cut. If the pipe ram be installed in the single ram preventer and the variable ram be installed in the double ram preventer, it would be very dangerous due to the variable ram failure in frequently using.

12.9 H2S Monitoring and Preventing Measure

As per PPL/PEDEC SOW South Azadegan oilfield contains H_2S bearing formations, precautions shall be taken in place especially in 12-1/4" & 8-1/2" hole section (for typical Fahliyan vertical well) while drilling, completion and well testing operation.

H_2S can be released any time while drilling, completion and well testing operation, therefore:

- The Rig shall be prepared for H_2S (Training, procedures and protective gear).
- Make sure everyone on the Rig should be made aware of the hazard presented by H_2S during their induction when they first arrive on the rig.
- The mud shall be treated to protect the drill pipe and the casing.

- All training program shall be recorded and reported.
- H_2S emergency drill must be conducted frequently.
- Adequate sensors and alarms must be installed for the scope of monitoring the H_2S releases.

13. H2S Monitoring & Control

NOTE: Final plan for H2S Monitoring and Control will be issued in due time by Service company instruction and Rig Contractor standards based on Oilfield safety and NIOC standards.
Minimum H2S Requirements for Rig site will be as following:

Table Error! No text of specified style in document.**-1: Minimum Rig H2S Requirements for South Azadegan Project**

Minimum Rig H_2S Requirements for South Azadegan Project	
Gas Detection Equipment	
H_2S Service cabin	1 No.
H_2S Unit location point	Comply with MSA/OEOC HSE
Main Monitoring Panel	1 No. - 8 channels
Max. number of channels	8 No.
Number of H_2S channels	4 No.
Measuring range	0 - 100
Adjustable alarm siren	Minimum 4 set (Min 120db voice load)
Measuring range	0-25 PPM
Adjustable alarm setting levels	2 No. (low 10 PPM and high 15 PPM)
Gas concentration reading displays	As sensors head number
Electrical Classification	Required
Certification	Required
Monitor panel installation site	H2S Unit
Automatic switch-on back up battery (min. 12 hr. power supply)	1
SO_2 Equipment	
Portable Single Detector	2 No.
H_2S Sensors	
H_2S Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required

Spare sensor head	2 No.
Response gas concentration	10 ppm low & 15 ppm High
LEL Sensors	
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required
Response gas concentration	20 % low & 40% High
Spare sensor head	Required
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Cables (connecting sensors with monitor panels)	
Suitable for hazardous area	as much as required
Electrical Classification	as much as required
Visual-Acoustic H_2S Alarms	
Low level visual alarms	Yellow flashing lights
High level visual alarms	Steady red lights
Low level acoustic alarms	Intermittent horn
High level acoustic alarms	Continuous horn
Above items suitable for hazardous area	Required
Number of alarm stations	5 No.
Alarm points location	comply with MSA/OEOC HSE
Calibration kit	At least 2 No.
PORTABLE H_2S MONITORING SYSTEM	
Hand pump colorimetric tubes H_2S detectors	1 No.
H_2S colorimetric tubes ranges	Req., comply with MSA/OEOC HSE
H_2S colorimetric tubes quantity	Req., comply with MSA/OEOC HSE
Personal electronic H2S detectors	Min. 6 No
Multi Gas Portable Detectors	
Quantity	4 (two main two spare)

Measuring range	0 – 100 ppm
Alarm setting levels	two
Certification	Required
Calibration kit	Required
Electrical Classification	For hazardous area
Certification	Required
Respiratory Protection Equipment	
Air Compressors	
Quantity	2 No. (Electrical- Diesel)
Approved for air breathing	Required
Gross weight	Required
Flow rate capacity	480 lit./min each
Charging pressure	3200
Filtration and purifier device	Specify approval
Certification	As per EN-132
Alarm for contaminating gases	Required
Auto shut down for contaminating gases	Required
Air compressor driving system	Electrical
Suitable for refilling bottles of air breathing apparatuses	Required
Air test kit	At least 1 No.
FIXED CASCADE AIR BREATHING SYSTEM	
The fixed cascade air breathing system complies with code type	Specify
Certification details	Required
No. of racks	At least 3*8
Dimensions (LxWxH) each rack	required
Bottles certified as pressure vessel	Required
Total air storage capacity	50 Liter
At max. working pressure of	3300 Psi

Air Pressure regulator	Req., victory Type
Safety/relief valve	Required
Low pressure alarm device	Required
Stainless steel piping/lines	Required
Air Breathing Delivery Stations	
Number of stations	At least 7 No.
Stations locations	Comply with MSA/OEOC HSE
Each Fixed Station Consisting of:	
Outlet manifold	4-man manifold 3 No & 6-man manifold 2 No
Connection outlets	Required
Hose lines	20(25 feet)20(15 feet)
PORTABLE AIR BREATHING APPARATUSES	
Self-Contained Breathing apparatuses (SCBA 45 min)	25
EEBA (20 min.)	20
Mask type	Full face (positive pressure)
Compliance with specifications	Comply w/ EN 137
Approval Authority	required
Air bottle working pressure	required
Air bottle certificate	required
Air regulator	required
Low Pressure audible alarm	Required
Suitable to be quick connected to cascade outlet manifold	Required
Stowage suitcase	Required
SCBA Radio Communication	Required
Spare Bottles for SCBA	At least 3
BREATHING APPARATUSES location points	comply with MSA/OEOC HSE

14. Requirements

14.1 26” Hole Section

Item	Description	Qty.
1	26” Bit, Code (1-1-5), 7-5/8” Reg. Pin	1 No.
2	Nozzle (16/32”, 18/32”, 20/32”)	As Req.
3	Nozzle’s installation & Removal tools, snap rings and O’rings, Centre jet installation & Removal tools, plug centre jets.	1 Set
4	Stab-in stinger w/Redress O-Ring for stinger	1 Set
5	20”, 94 (lb/ft), K-55, BTC, “Stab in” type duplex shoe	1 No.
6	20”, 94 (lb/ft), K-55, BTC	± 165 m
7	26” x 20”, Bow Centralizers	6 No.
8	20” x 5”, Bow Centralizers	2 No.
9	20” Stop Collar	16 No.
10	Thread sealing compound (Dope)	1 Dr.
11	Thread lock compound (Baker Lock)	1 Can.
12	Cement, Class “G”, 110 lb/sxs	As Req.
13	Cement Additive	As Req.
14	20-3/4" x 5000 psi Casing Head Housing	1 No.
15	RX-74 Ring Gasket (20-3/4" x 5000 psi)	1 No.
16	2” * 370 mm stud & Nut ((20-3/4" x 5000 psi))	20 No.
17	2 1/16” x 3000 psi Side outlet Valves	1 No.
18	Rx-24 Ring Gasket (2-1/16” 3000 psi)	3 No.
19	7/8” * 166 mm stud & Nut (2-1/16” 3000 psi)	24 No.
20	2 1/16” x 5000 psi Companion Flange + 2” Bull Plugs + ½” pipe plug	2+2+1 No.
21	1-1/2” VR-Plug	1 No.
22	Pressure Gauge, 0-3000 psi Range	1 No.
23	20-3/4" Wear Bushing to Fit with C.H.H W/O-Ring	1 No.
24	20-3/4" Plug Tester to Fit with C.H.H W/O-Ring	1 No.
25	½” Needle Valve	1 No.

14.2 17 ½" Hole Section

Item	Description	Qty.
1	17-1/2" Bit, Code, 1-1-5, 5-3-7, PDC Bit, Type: S223, 7 5/8" Reg. Pin	1+1+1 No
2	Nozzle (11,12,13,14/32")	As Req.
3	Pipe screen for inside 5" DP	2 No.
4	13 3/8" Float Shoe (PDC drillable), 68 ppf, N-80, BTC.	1 No.
5	13 3/8" Float Collar (PDC drillable), 68 ppf, N-80, BTC.	1 No.
6	13 3/8" Casing, 68 ppf, N-80, BTC	± 2500 m
7	13 3/8" Top & Bottom cementing plugs (non-rotating type)	1+1 No.
8	17 1/2" x 13 3/8", Bow Centralizers	80 No.
9	20" x 13 3/8", Rigid Centralizers	5 No.
10	13 3/8" Stop Collar	170 No.
11	13 3/8" Pup Joint, 6ft & 9ft 68 ppf, N-80, BTC	1+1 ea.
12	Thread sealing compound (Dope)	1 Dr.
13	Thread lock compound (Baker Lock)	1 Can.
14	XOS 13 3/8" BTC Pin x 4 1/2" IF Box	1 No.
15	Cement, Class "G", 110 lb/sxs	As Req.
16	Cement Additives	As Req.
18	20-3/4" 3000 psi x 13-5/8" 5000 psi Casing Head Spool	1 No.
19	BX-160 Ring Gasket (13-5/8" 5000 psi)	1 No.
20	1-5/8" * 320mm stud & Nut (13-5/8" 5000 psi)	16 No.
21	Slip & Seal Assembly 20" * 13-3/8" OD Casing (Non-Auto. & Auto.)	1+1 No.
22	2 1/16" x 5000 psi Side outlet Valves	2 No.
23	Rx-24 Ring Gasket (2 1/16" x 5000)	4 No.
24	7/8" * 166mm stud & Nut (2 1/16" x 5000)	32 No.
25	2 1/16" x 5000 psi Companion Flange + 2" BullPlugs + ½" pipe plug	2+2+2 No.

26	1-1/2" VR-Plug	2 No.
27	Pressure Gauge, 0-5000 psi Range	2 No.
28	13-5/8" Wear Bushing to Fit with C.H.H W/O-Ring	1 No.
29	13-5/8" Plug Tester to Fit with C.H.H W/O-Ring	1 No.
30	½" Needle Valve	2 No.
31	BX-160 Ring Gasket (13-5/8" 5000 psi)	1 No.
32	13-5/8" 5000 psi BOP Stock & Related Equipment	1 Set.

Note: Qty and spec to be conformed with Contact's SOW. In case of any shortage or discrepancy, the Contract's SOW will be reference.

14.3 12 ¼" Hole Section

Item	Description	Qty.
1	12-1/4" Bit, Code 5-3-7 & PDC Bit, Type:M323 ,6 5/8" Reg. Pin	1,1,1 no.
2	Nozzle (12,13,14,16/32")	As Req.
3	Pipe screen for inside 5" DP	3 No.
4	9-5/8" Float Shoe (PDC drillable), 53.5 ppf, 110SS, Premium gas tight conn.	1 No.
5	9-5/8" Float Collar (PDC drillable), 53.5 ppf, 110SS, Premium gas tight conn.	1 No.
6	9-5/8" Casing 53.5 ppf, 110SS, Premium gas tight conn.	4400 m
8	9-5/8" Top & Bottom cementing plugs (non-rotating type)	1+1 No.
9	12 1/4" x 9 5/8", Bow Centralizers w/nail	120 No.
10	13 3/8" x 9 5/8", Rigid Centralizers w/nail	40 No.
11	9 5/8" Stop Collar w/nail	320 No.
12	9 5/8" Pup Joint, 6ft & 9ft 53.5 ppf, 110SS, Premium gas tight conn	1 ea.
13	Thread sealing compound (Dope)	2 Dr.
14	Thread lock compound (Baker Lock)	1 Can.
15	XOS 9 5/8" BTC Pin x 4 1/2" IF Box	1 No.
16	Cement, Class "G", 110 lb/sxs	As Req.
17	Cement Additives	As Req.

20	13-5/8" 5000 psi x 11" 10,000 psi Tubing Head Spool	1 No
21	TBG Hanger: 11"BOWLx4-1/2" 18.9PPF BOX VAM TOP BOTTOM & 4-1/2" 4-2G ACME BOX	1 No.
22	Metal Seal + S-Seal	1 Set
23	BX-158 Ring Gasket (11" 10,000 psi)	1 No.
24	1-3/4" * 381 mm stud & Nut (11" 10,000 psi)	16 No
25	Slip & Seal Assembly 13-5/8" x 9-5/8" OD Casing Automatic (Non-Auto. & Auto.)	1 +1 No.
26	2-1/16" INSTRUMENT FLANGE, WP:10,000 PSI	1 No.
27	2 1/16" x 10,000 psi Side outlet Valves	4 No.
28	BX-152 Ring Gasket (2 1/16" x 10,000)	7 No.
29	3/4" * 139 mm stud & Nut (2 1/16" x 10,000)	48 No.
30	2 1/16" x 10,000 psi Companion Flange +2" Bull Plugs + ½" Pipe plug	2+2+2 No.
31	1-1/2" VR-Plug	2 No.
32	Pressure Gauge, 10,000 psi Range	2 No.
33	Angle Needle Valve, WP:10,000 psi	2 No.
34	11" Wear Bushing to Fit with T.H.S W/ O-Ring	1 set
35	11" Plug Tester to Fit With T.H.S W/ O-Ring	1 set

Note: Qty and spec to be conformed with Contact's SOW. In case of any shortage or discrepancy, the Contract's SOW will be reference.

14.4 8 1/2" Hole Section

Item	Description	Qty.
1	8-1/2" Tri Cone Bit, Code 2-1-7, 5-3-7& PDC Bit, Type: M433	1+1+1No.
2	Nozzle (11,12,13,16,18/32")	As Req.
3	Pipe screen for inside 5" DP	3 No.
4	7" Float Shoe (PDC drillable), 32ppf, 110SS, Premium gas tight connection	1 No.
5	7" LDC, 32ppf (PDC drillable-float type), 110SS, Premium gas tight connection	1 No.
6	7" Liner, 32ppf, 110SS, Premium gas tight connection, 6" S. D	± 440 m
7	X.O.S 7", 32ppf, 110SS, Premium Gas Tight Connection pin x 4-1/2" IF Box	1 No.
8	7" Hydraulic set liner hanger (tandem cone) c/w top packer, setting ball & tie back sleeve	1 Set
9	7" * 8-1/2" Bow Integral type Centralizer C/W Nail and Pins	As Reg.

10	7" * 8-1/2" Rotating Centralizer C/W Nail and Pins	30 No.
11	7" * 8-1/2" Rigid Centralizer C/W Nail and Pins	5 No.
12	7" Stop Collar	80 No.
13	Wiper plug & pump down plug	1 Set
14	S.N. Packer 7" 32ppf, 110SS	1 Set as Req.
15	Dresser Mill for liner tie back sleeve + Polish mill	Each 1 No. As Req.
16	Thread Sealing Compound (liner Dope)	2 Dr.
17	Thread Locking Compound (Baker Lock)	1 Can
18	Cement "G", 110 Lb/SX	As Req.
19	Cement Additives	As Req.
20	TUBING HEAD ADAPTOR, BOTTOM STUDDED FLANGE 11", 10000 PSI,	1 No.
21	CROSS T-TYPE SOLID BLOCK, BOTTOM FLANGED 4-1/16" API 10,000 PSI WP, TOP FLANGED STUDDED 4-1/16" API 10000 PSI WP	1 No.
22	4-1/16" POSITIVE CHOKE VALVE W/ 10,000 PSI WP W/ BEAN SIZES for POSITIVE CHOKE VALVE (8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 INCH,	1 No.
23	COMPANION FLANGE API 4-1/16", 10,000 PSI WP.	1 No.
24	Blind Flange, 4-1/16", 10,000 psi API WP, C/W ½" NPT FEMALE THREAD Body	
25	WELDING FLANGE, 4-1/16", 10,000 API FLANGE x WELDING NECK, FOR PIPE SIZE 4", MATERIAL: 410SS, M/C:FF-0.5 T/C:U PSL:3 FORGED BODY	1 No.
26	GASKET SELF SEALING RING BX-155 ,4 1/16"-10000 PSI MATERIAL: SS316	10 No.
27	INSTRUMENT FLANGE, 4-1/16", 10,000 PSI API BX155, WITH FOUR PORTS 1/2" NPT, ALIGNED INCLUDING 4 PIECES OF 1/2" NPT PLUG, M/C:FF-0.5 T/C:U PSL:3G PR:2 FORGED BODY	1 No.
28	PRESSURE GAUGE, 1/2" NPT MALE CONNECTION, 0-10,000 PSI, 4" FACE DIAMETER, STAINLESS STEEL FOR COVER AND INTERNAL PARTS	2 No.
29	X-MAS TREE CAP, 4-1/16" API 5000 PSI WP, FLANGED BOTTOM (4-1/16", BORE), 11-1/2" 4TPI ACME THREAD UNION TYPE, TAPPED WITH 1/2" NPT PORT M/C:FF-0.5 T/C:U PSL:3G PR:2 FORGED BODY	1 Set.
30	2" Bull plug & ½" PIPE PLUG NPT THREAD, FORGED BODY, STAINLESS STEEL AISI 410,HEXAGONAL 10,000 PSI WP, M/C:FF-0.5 T/C:U PSL:3G PR:2	2+2 No.
31	STUD & NUT 1 1/8" x 203 MM LONG ASTM A193 B7/A194 2H STUD/NUTS PTFE COATED FOR 4 1/16"-10,000 FLANGE	56 No.
32	ANGLE NEEDLE VALVE ½" NPT MALE * FEMALE 10,000 PSI WP M/C: FF-0.5 T/C: U PSL: 3G	2 No.
33	MANUAL GATE VALVE API 4 1/6" 10,000 WP. FLANGED ENDS AND FULL BORE M/C: FF-0.5 T/C:U PSL:3G	2 No.
34	4" BACK PRESSURE VALVE, ONE WAY, WP:10k, W/ REDRESS KIT	1 Set
35	4" BACK PRESSURE VALVE, TWO WAY, WP: 10K, W/ REDRESS KIT	1 Set

Note: Qty and spec to be conformed with Contact's SOW. In case of any shortage or discrepancy, the Contract's SOW will be reference.

15. Drilling Services

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	APD
4	Electrical Logging and Perforation	PANIZ/SEALAND
5	Mud Logging	PGD
6	Directional Drilling (MWD)	N/A
7	H2S Monitoring and Control	BONYAN IMEN
8	Casing and Tubing Running	OEOC
9	Cementing	SPAD ENERGY/MEHRAN
10	Dry Test, DST and RTTS	OEOC
11	Slick Line	OEOC
12	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
13	Surface Well Testing	OEOC
14	Liner Hanger	PARS POOLAD
15	Well Head	OEOC